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- I'm proud of my hard work. Working hard won't always lead to the exact things we desire. There are many things I've wanted that I haven't always gotten. But, I have a great satisfaction in the blessings from my mother and father, who instilled a great work ethic in me both personally and professionally.

  AlhmduLillah

Disclaimer:

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ZOO510 Economic Zoology

Final term 120 to 239

120 Poultry Feeding 5

B- Vitamin D:

- *MCQS*: Vit.D required for **bone formation, egg production, reproduction & prevention of rickets.**
- Def. symptoms:poor **growth, lameness & rickets.**
- Poultry do not exposure to **sunlight,** ration must suppl. With vit.D.

C- Vitamin E: which disease caused by vitamen E?

- Vit.E in vegetable is not readily available as in oil concentrates.
- Answer: **Vit.E essential to prevent encyphalomalacia or crazy (softening of the brain) chick disease.**



D- Vitamin K:

- **MCQs**:Def. of vit. K **delay clotting time of the blood** & produce serious hemorrhage
- All mixtures should be suppl. With vit. K
- Treatment by sulfonamide(man made synthetic drugs) vit. K req.

E- Riboflavin:

- Def. of vit.B2 curled-toe paralysis, dwarfism **(Dwarfism in chickens is an inherited condition found in chickens consisting of a significant**

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delayed growth, resulting in adult individuals with a distinctive small size in comparison with normal specimens of the same breed or population.) & degeneration of nerve trunks.

- Requirement: Broilers & breeder

4.4mg/kg

Layers 2.5 mg/kg ration



dwarfism and toe paralysis

F- **Thiamin:**

✓ Def. of thiamin nerve deg., convulsion & heart abnormalities.

121 Poultry Feeding 6

G- **Niacin:**

- Def. of niacin inflammation of tongue & mouth cavity (black tongue).
- Young chick required niacin more than adult due to less bacterial action synthesis.

H- **Vit.B12: What are affects caused by Def. of vita B12?**

- Animal proteins are good sources of vit.B12.
- Answer: **Def. of vit.B12 irritability, poor feathering & poor hatchability**.

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Feeding space:

- 1 inch feeder space /chick for 2 weeks age & 2 inches after that.

Water:

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- **mcqs:** Bird drink **about twice as much water by weight of feed consumed.**
- Water consumption increase or decrease according to the environmental temperature.
- Some medications are administered in the drinking water.

122 Feeding of Broilers 1

Age / Nutrients	Protein (%)	ME (Kcal/kg)
Starter ration (0-3weeks)	22-24	2800
Grower ration (3-5 weeks)	20-22	3000
Finisher ration (5-7 weeks)	18-20	3200

Feed conversion (FC)= $\frac{\text{Feed intake (g or kg)}}{\text{Weight gain (g or kg)}}$

Feed conversion of broilers = 2.2

Sq:lq: what are the Factors affecting feed conversion?

Factors affecting feed conversion:

1. Type of feed fed
2. Strain of the birds
3. Environmental temperature
4. Age and weight of the birds

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5. Diseases and condemnations
6. Rodent & flying bird control in feeding area
7. Antibiotics and medications iprove FC
8. Debeaking & size of baby chicks
9. Feed wastage
10. Form of the feed

123 Feeding of Broilers 2

SQ: wjhat is the feeding of Broiler breeding pullets?

- Chickens breed for meat production grow rapidly & reach sexual maturity at early age too many small egg & not good for hatching.
- Bird kept for breeding purposes, it is necessary to slow down their rate of growth & development of sexual maturity.

Methods used:

MCQs:1-Restricting feed intake to approximately **70%** (when pullets **at 7-9 weeks till 23 weeks**),or

2-A skip –a-day program involves full feeding every

other day, or 3-Feeding a diet containing 10% protein.

A 'pullet' is a young hen that is under 1 years of age, and has not started laying eggs yet

Energy feeds:

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- Grain, grain by-products and animal & vegetable fats and oils supply the most of energy in the poultry diets.
- **Mcqs: Corn is the most common grain** used in formulating poultry diets
- Other grains such as grain sorghum & wheat substituted part of corn
- **Mcqs:** Animal & vegetable fats added in limited amounts **(5-10% of the diet)**



grain sorghum

124 Feeding of Broilers 3

Protein supplements:

- Protein suppl. Added to provide the essential AAs.
- Several protein sources used to achieve a better balance of the needed AAs.
- Animal protein sources are more variable in their amino acids than plant protein
- AAs req. for poultry differ from that of other animals in that glycine & serine are dietary req. & required glycine for uric acid formation.

A-Plant protein:

- Soybean meal is most commonly plant protein source & has a better balance of AAs than other plant protein (cottonseed meal, corn gluten meal**(Corn gluten meal is the principal protein of corn (maize) endosperm consisting mainly of zein and glutelin.)** , linseed meal). **Alsi ka khana**

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Linseed meal

- **Mcqs:** Cottonseed meal used in grower poultry ration to replace **up to 50% of the soybean meal**, while linseed meal not more than 3-5% of diet.

B-Animal protein:

- The most commonly used are fish meal, meat by-products, milk byproducts, blood meal(**Blood meal is dried, ground, and flash-frozen blood and contains 12% nitrogen. While it's a great organic fertilizer,**)
, feather **meal (Feather meal is a byproduct of processing poultry; it is made from poultry feathers by partially grinding them under elevated heat and pressure, and then grinding and drying.**) & poultry by-product meal.
- Fish meal have a good balance of AAs, but must not used in large amount (used at 2-5%) to avoid fishy flavor in eggs & poultry meat.

MCQs: Zain Official

125 Feeding of Broilers 4

S.Q:WHAT ARE MINERAL SUPLLEMENTS?

Mineral supplements:

- Sources of Ca in poultry diets are ground oyster shell, limestone, bone meal, dicalcium phosphate.
- Inorganic P supplied by bone meal, di-calcium phosphate, rock phosphate.

 Na & Cl adding as common salt (**0.5-1% of diet**) Mn (Mn sulfate), Zn (Zn sulfate)

S.Q:WHAT ARE VITAMIN SUPPLEMENTS?

Vitamin supplements:

 Natural feedstuffs provide some vitamins for poultry.

 Vitamin premixes are commonly used to provide the required vitamins in poultry.

S.Q:HOW FEED PREPARATION OCCOUR?

Feed preparation:

 Commercial feeds for poultry as mash, pellets or crumbles.

 Less wastage when using pellets or crumble and poultry grow faster and more commonly used for broilers and turkey than for laying hens (laying hen tend to become too fat unless they are on the restricted feeding pr





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126 Feeding of Laying Hens 1

L.Q:WHAT ARE NEUTRIENTS REQUIREMENTS OF LAYING HENS?

Nutrient requirements of laying hens:

1-Energy requirement:

 For maintenance (2kg wt.) = 220 Kcal

MCQs:

For 70% production = 130 Kcal

For 1 g gain/day = 3 Kcal

The usual energy conc. Is 2.8 Mcal ME/kg diet

Energy conc. Than 2.3 Mcal energy intake & egg production

2-Protein requirement:

- Laying hen receiving diet containing 3.1 Mcal ME/kg DM require 16.5% protein.
- To get maximum economic return from laying hen flock, a feed efficiency of m.q:1.6-1.8 kg of feed per dozen of eggs produced is need.
- A laying ration should contain about 15% protein based on 2900 Kcal ME/kg of diet.

• Effect of environmental temperature:

 Small light body weight hens consumes: ✓ In Summer mcqs: 90g feed (19% protein ✓ 17g protein/ hen/ day). In Winter 110g feed (15.5% protein 17g protein / hen / day)





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S.Q:What are essential AAS FOR4 LAYING HENS?

Essential AAs for laying hens:

1. Leucine, isoleucine, lysine, methionine, tryptophan and arginine.
2. Methionine is first limiting AAs for egg production.
3. Mash for laying hens should contain **not less than 3-4%** animal protein supplement.
4. Feathers are high in **sulfur amino acids (required methionine).**

Fat supplement:

- Fat addition egg yield in winter
- Fat addition amount of feed required / dozen eggs.

S.Q:WHAT MINERALS ARE REQUIRES FOR LAYING HENS?

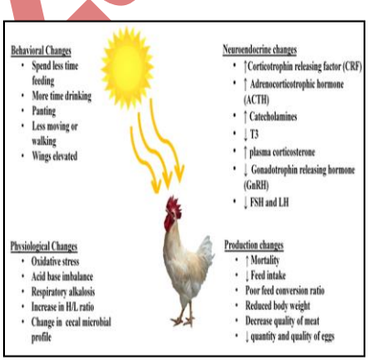
3-Mineral requirements:

A-Calcium:

- Laying birds need large amounts

Of Ca because egg shells composed entirely of CaCo3 •
Ca in laying ration egg production & egg shell weak.

- Bird stored Ca for about 10-14 days before the first egg was laid in the marrow of long bone.





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128 Feeding of Laying Hens 3

WHAT ARE PURPOSE OF B-PHOSPHOROUS?

B-Phosphorus:

- Protein supplement used in poultry rations (mat meal, tankage, fish meal & dairy byproducts) usually be sufficient in phosphorus.
- Plant protein supplement (SBOM) should supplement with P & Ca.
- Inorganic P is more available than phytate P.

C-Manganese:

Ca in laying ration egg production & egg shell weak & hatchability.

- $\frac{1}{4}$ lb Mn sulphate added to ton of mash fed without grain & $\frac{1}{2}$ lb to mash fed with grain

D- Iodine:

- Iodine in laying ration goiter

✓ Iodized salt must be used instead of common salt in the ration of poultry.

E- Selenium:

- Se in laying ration Exudative diathesis

F- Zinc:

- Zn in laying ration skeletal abnormalities, ataxia, necrotic dermatitis & thin shell & hyper keratinization of epidermis.

G- Salt:

- 0.5-1% of the total ration salt





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129 Feeding of Laying Hens 4

S.Q:WHAT VITAMINS ARE REQUIRED FOR LAYING HEN?

4-Vitamin requirements:

A-Vitamin A :

- Laying hens require higher content of vit.A in their feed in very hot weather than cold because they consume less feed.
- vit.A in laying ration Nutritional roup (sticky materials from eye & nostrils)





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130 Feeding of Laying Hens 5

S.Q:WRITE PURPOSE OF VITAMIN D?

B-Vitamin D :

- vit.D in laying ration thin shell eggs, egg production & hatchability, breast bone become soft & bones of legs & wings become fragile.

C-Riboflavin & vit.E :

- Riboflavin & vit.E in laying ration low hatchability

Phase-feeding of laying hens:

To adjust nutrient intake in accordance with the rate

of egg production

A-Phase I (most critical period):

During 20 W period (22-42 W of age) pullet :

1. egg production from zero to peak (85-90% production).
2. body weight from 1300 to 1900g.
3. egg size from 40g/egg at 22W to over 56g/egg at





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131 Feeding of Laying Hens 6

B-Phase II :

✓ Period after 42W of age when the hens attained mature body weight ✓ The period ranged from 42-72W of age.

S.Q:WHAT ARE EFFECTS OF TEMPRATURE ON EGG SHELL?

Effect of temp. on egg shell:

- Hot weather respiration rate Co2 loss blood bicarbonate level egg shell formation

Stage of egg production:

- Egg production hen usually cover a period of 15 months
- Commences at 22W of age peak at 28-30W of age
- gradually decline to 65% after 15 months of lay.

lighted period feed intake & stimulation of pituitary gland egg laid

Feeding systems:

1. Whole grain method

2. Grain & mash
3. All mash: fed at first 8W
4. Wet mash feeding (more palatable)
5. Pellets

- With grain fed must used insoluble grit
- Also fresh green feed is fed to poultry.





132 Feeding of Laying Hens 7

Nutrition and egg quality: A. Egg size (egg weight):

S.Q:WHAT FACTORS EFFECT THE SIZE OF EGG?

Factors affecting egg size:

1. Level of protein in diet:

- 14-20% CP rations balanced AAs heavier eggs
- The choice of protein level in layer diet depend on accurate evaluation of extra-cost for the additional protein compare with the income from larger eggs obtained.

2-Energy intake

3-Mineral & vitamin levels:

Ca & vit.D egg weight

4-Level of linoleic acid:

- Linoleic acid formation lipoprotein in liver ovary uptake by ova higher egg weight

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5-Strain

B-Shell quality:

- The quality of egg shells depend on the presence of adequate levels of vit.D_3 & certain minerals including Ca, P & Zn.
- Def. or imbalance of vit,D_3 , Ca & P shell thickness & misshapen eggs egg production
- Mn thin & brittle-shelled eggs

✓ The blood carbonate is the source of carbonate in the shell formation





133 Feeding of Laying Hens 8

- Very hot weather poor quality egg shells
- End of laying period falls egg shell quality due to failure in Ca metabolism & Ca of ration
- Sulphonamide **drugs thin shelled eggs**
- Insecticides & fungicides in grains malformed eggs
- Rancid cod liver oil in diet rough shells
- Diseases poor shell quality

C-Internal egg quality:

- The nutritive content of the egg depends upon the level of these nutrients in the diet of laying hen
- Suitable iodine in diet I content of eggs
- Def. of vit.B2 **slight yellowish-green tinge in albumin**

S.Q:DESCRIBE ON WHICH FACTORS YOLK COLOR DEPEND?

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D-Yolk color:

- The color of egg yolk depend upon the presence of carotenoid pigment (xanthophylls) in the ration
- (fresh & good dried green feeds & feed additives)
- When 30% yellow maize **or 5% good quality alfalfa** or up to 22mg xanthophyll/kg deep-yellow yolks
- Highly pigmented plants undesirable colored yolks

Large amount of untreated CSM brown mottled yolk & pinkish tint of albumin

Pimento pepper in **diet orange-red yolks**





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134 Feeding of Turkeys 1

- The general principles of feeding turkeys are similar to those for feeding broilers. Major differences are in the protein levels required and the importance of the vitamins biotin & pyridoxine in turkey diets
- Poults must be fed & watered as soon as possible after hatching & if feeding delayed beyond 36h after hatching difficulty learning to eat & drink.
- Vits. & minerals suppl. of the diet essential for good hatchability of turkey eggs.
- At 10-12W of age separate hens from toms

Period	Protein (%)	ME (Kcal/kg)
First 3 weeks	30-33	2930-3000
0-4 W	28	2930-3000
4-8W	26	2900
8-12W	20-22	3100
13-16W	19	3200
17-20 W	16	3275
21 W-market	13-14	3350
Laying hen	15-18	2925
Peak production	19	2755

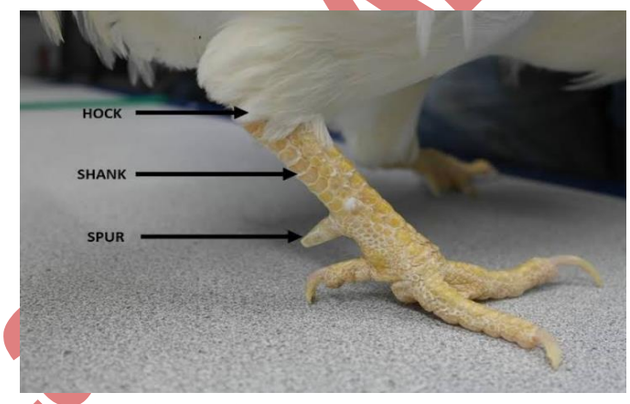
135 Feeding of Turkeys 2

Nutritional disorders of turkey:

1-Leg weakness disorders:

Cause: def. of Ca, P, vit.D, choline, biotine, folic acid, Mn & zinc.

2-Enlargment of hock joint:



Cause: def. Of niacin, biotin, vit.E & zinc.

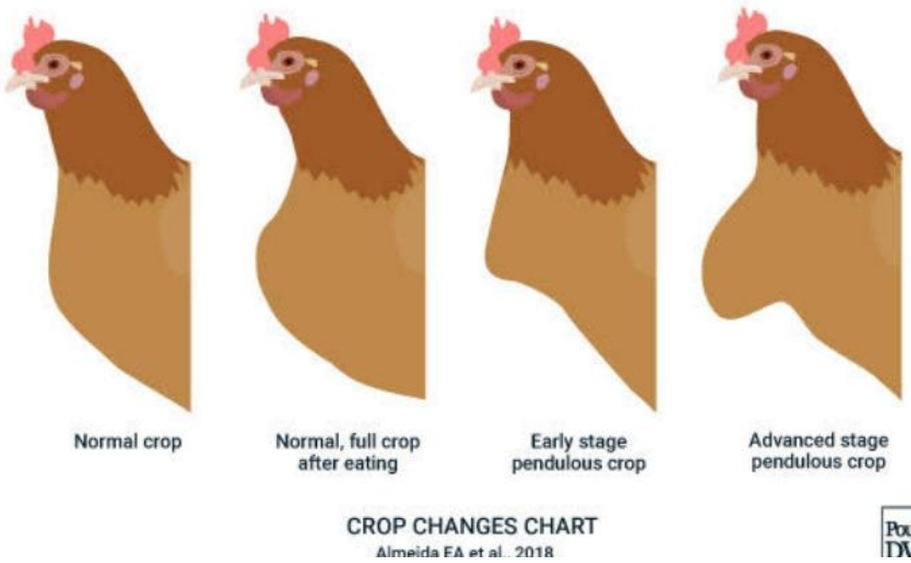
3-Footpad dermatitis:

Cause: biotin deficiency

Symptoms: sticky droppings adhere to the feet & cause dermatitis

4Pendulous crop:

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Cause: yeast proliferation in crop

Symptoms: gas production from fermentation of carbohydrate interfere with passage of ingesta from crops to proventriculus pendulous crop

Treatment: fungal inhibiting antibiotics

5-Ascitis:

Cause: high salt intake fluid accumulation in body cavities

6-Exudative diathesis:

Cause: Selenium deficiency

7Aflatoxicosis:

- Aflatoxin affect the immune system increase susceptibility to disease

- Mycotoxin hemorr luish the carcass





136 Feeding of Ducks

- Commercial feeds in mash, pelleted or crumbles form available for ducks & geese
- If a commercial **feed for ducks & geese is not available,** chicken feed may be used (not contain coccidiostat)
- Geese will start to eat pasture when they are only few days old & feed additional grain if pasture is not of good quality.

FEEDING OF GEESE

Period	Protein (%)	ME (Kcal/Kg diet)
0-4 W (starter)	20	2900
After 4 W (grower)	15	3000
Breeding	15	2900

FEEDING OF DUCKS

Period	Protein (%)	ME (Kcal/Kg diet)
0-2 W (starter)	22	2900
2-7 W (grower)	16	3000
Breeding	15-18	2900

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137 Biosecurity 1

Biosecurity for poultry

Disease

S.Q:HOW MANY TYPES OF DISEASE OCCOUR IN POLTRY PRODUCTION?

Prevention

➢ There are basically 2 types of diseases that must be considered in poultry production:

1. diseases of detriment to bird health
2. diseases of potential human health concerns

S.Q:WRITE CUSES OF HEALTH ISSUE?

Causes of Health Issues

1. Pathogens

i. bacteria, viruses, parasites, protozoa, fungi

2. Nutritional deficiencies
3. Chemical poisons
4. Overmedication
5. Poor management

Biosecurity

What is "Biosecurity"?

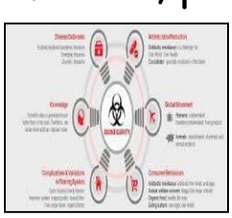
- Protecting your birds from disease
- Preventing or controlling disease transmission by vectors

What is a vector?

 Something that may transmit a disease

 Rodents, birds, insects, shoes, car tires, shared equipment, best friend, pet, feed, water, dust, air….





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138 Biosecurity 2

Sq: Biosecurity

 Preventative strategies to control disease causing organisms and their carriers (vectors).

 Protection of poultry flocks from any type of infectious agent.

✓ Control transmission of disease from flock to flock.

Biosecurity is the cheapest, most effective means of disease control available.

Who should practice good biosecurity?

Everyone!

- Commercial producers
- Small flock/backyard poultry owners
- Hobbyists/breeders
- Youth poultry project owners for livestock shows (4-H and FFA members)

Why should I be concerned about biosecurity?

- **Because of what might happen:**

1. Disease and high mortality in flocks
2. Infected flocks euthanized
3. Loss of income for producers
4. Higher prices of poultry meat and eggs
5. Quarantines and restricted movement/sale of birds
6. Cancellation of poultry shows

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139 Major Components of Biosecurity 1

S.Q:WHAT ARE MAJOR COMPONENT OF BIOSECUIRTY 1?

Major Components OF Biosecurity

- Flock Management
- Isolation
- Traffic Control
- Cleaning/ Sanitation

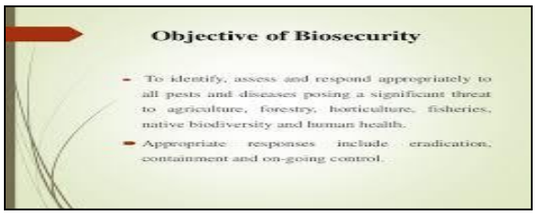


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140 Major Components of Biosecurity 2

S.Q: WHAT ARE POINTS OF BIOSECUIRTY MONITPRING?

Biosecurity Monitoring Points



1. Humans
2. Equipment
3. Environment
4. Animals
5. Contaminated eggs and chicks
6. Contaminated feed

WHAT ARE ROUTES OF DISEASES TRANSMISSION?

Environmental Routes of Disease Transmission ➢

Environmental contaminants:

1. wind
2. water
3. dust
4. feathers
5. manure

❖ Drying and sunlight are very effective at killing many disease causing organisms.

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141 Good Biosecurity Practices 1

S.Q:HOW WE PRACTIVE BIOSECUIRTY?

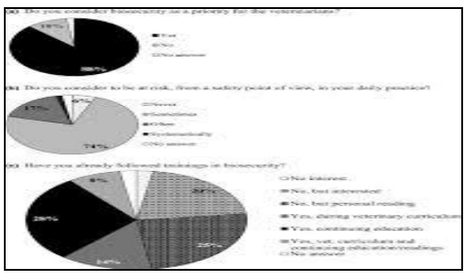
Good Biosecurity Practices

- Keep pets, wild birds and animals and other livestock away from the flock
- No trading or purchase of untested birds – quarantine new birds
- Use dedicated footwear/clothing or disposable coveralls and boot covers when checking your birds
- Hand washing before and after handling birds, manure, coops, eggs, etc.

Isolation Measures

- Confine flocks to controlled environment
- Screen houses to protect flock from wild birds
- Avoid contact with migratory waterfowl and other birds
- Restrict visitors
- Do not share equipment, coops or leftover feed

Reduce exposure Risks



Control vector habitat and attractants

- birds
- rodents
- varmints
- insects

– Make sure pen is animal and bird proof

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142 Good Biosecurity Practices 2

S.Q:HOW WE MAINTAINE FLOCK HEALTH?

Maintenance Flock health

➢ Minimize stressors to prevent increased susceptibility to disease

1. Fresh feed
2. Clean water
3. Clean, dry litter
4. Good ventilation
5. Provide an overall comfortable environment

WRITE STEP OF CLEANING ?

Cleaning and Sanitation

Five steps to cleaning and disinfection (C&D)



1. Dry Clean (remove all organic matter)
2. Soap
3. Rinse
4. Dry
5. Disinfect **Disease/Health Checks**

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Watch for symptoms such as:

✓ Coughing, sneezing, watery eyes, ruffled feathers, loss of feathers, excessive picking, labored breathing and sudden drops in production and feed or water consumption

- Eliminate unhealthy or nonproductive birds

- Check birds periodically for lice, mites, and worms

143 Maintenance of Flock Health

COMMON DISEASE IN POULTRY FARMING

What is disease

Any condition that results in deviation from normal function

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WRITE DIIFERNCE BETWEEN INFECTIOON AGENTS AND NON INFECTION AGENTS?

ETIOLOGY

Infectious Agents	Non-infectious agents
o Bacteria o Viruses o Parasites ➢ Fungi	o Chemical o Physical o Lack or excess of certain vitamins and minerals ➢ Toxins

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144 Common Diseases of Poultry

S.Q:WHAT ARE SYMPTOMS OF DISEASES?

General Signs of Disease

1. Poor appetite
2. Huddling
3. Depression
4. Runting /stunting
5. poor uniformity
6. Ruffled feathers

7. Coughing, sneezing,
8. oculo-nasal discharge,
9. difficult breathing
10. Bloody or wet litter
11. Increased mortality

- VIRAL DISEASES

SQ: Name the viral disease in chickens?

LQ: Write a note on viral disease of chicks?

FOWL POX

- Viral disease of **domestic fowl**
- development of nodular proliferative Skin lesions on the featherless parts of the body.
- fibrino necrotic and proliferative lesions in the mucous membranes



- Family: Pox viridae
- Genus: avipoxvirus
- Double stranded DNA

CLINICAL SIGNS

- Appearance of **nodular lesions**
- combs
- wattles eyelids

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- other unfeathered areas of the body.

PREVENTION AND CONTROL

- Prophylactic vaccination
- M.C.Q:Fowl pox vaccine at 4-6 weeks of age ➢ Second dose M.C.Qat 12-14 weeks of age.

145 Ranikhet Disease

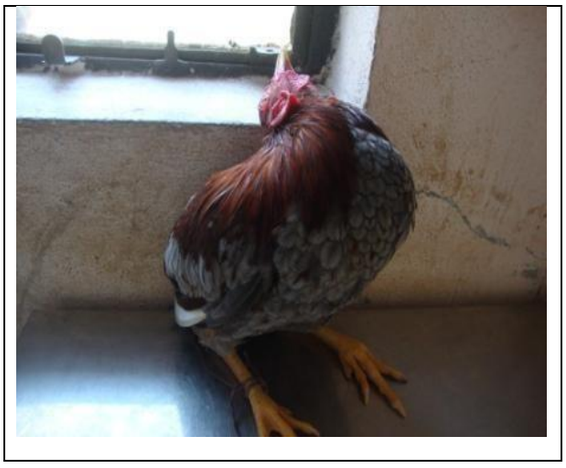
S.Q.:WHAT ARE RANIKHET DISEASE?

RANIKHET DISEASE

- New castle disease
- Viral disease of domestic fowl is characterized by
- respiratory signs
- often associated with nervous and digestive disorders ➢ high mortality.

ETIOLOGY

- Paramyxoviridae
- Paromyxovirus-1



CLINICAL SIGNS

Opisthotonus

1. Listlessness
2. Increased respiration
3. Weakness
4. Edema around the eyes
5. Torticolis
6. Paralysis of legs

Prevention and control

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➢ Prophylactic vaccination

Lentogenic strain (F or B1)

- day old chicks
- intranasal ➢ intraocular route
  ➢ drinking water.

Mesogenic strain (R2B)

- M.C.Q:6-8 weeks of age ➢ intramuscular

 Subcutaneous route.

146 Bacterial Diseases

INFECTIOUS

CORYZA

S.Q:WHAT IS

FOWL CORYZA?

FOWL CORYZA

- Highly contagious
- acute disease of upper respiratory tract of chickens, ➢ turns into a chronic respiratory disease.

ETIOLOGY

- *Haemophilus paragallinarum*
- Small cocoid or gram negative rod
- Non motile
- Exhibits bipolar staining

S.Q:WHAT ARE SYMPTOMS OF DISEASE?

CLINICAL SIGNS

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1. Serous to mucoid nasal discharges with foul smelling
2. Facial edema
3. Conjunctivitis
4. Swollen wattles
5. Diarrhea
6. Reduced feed and water consumption.



S.Q.:HOW WE CONTROL THIS DISEASE?

Treatment and control

1. Gentamicin
2. Penicillin
3. streptomycin
4. delivered in feed or drinking water.
5. Proper Disinfection

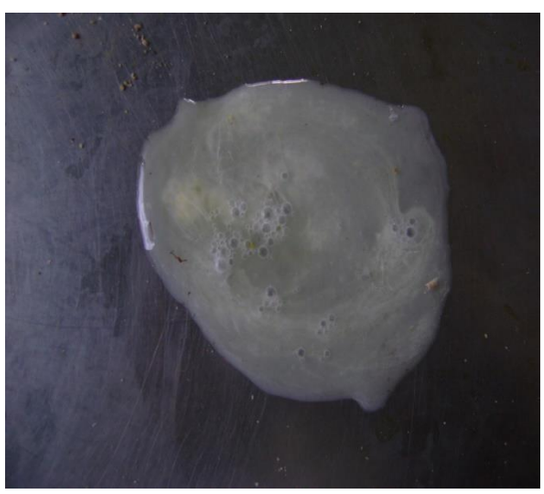
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147 Bacillary White Diarrhea

S.Q:WHAT IS BACILLARY WHITE DIARRHEA ?

PULLORUM DISEASE

- Fatal septicemia of young chicks .
- Salmonella pullorum



S.Q:WHAT ARE SYMPTOMS OF THIS DISASES?

CLINICAL SIGNS

1. Somnolecence
2. Weakness
3. Loss of appetite
4. Chalky white diarrhea

5. Stained greenish brown(sometimes) in and around vent

S.Q.:HOW WE TREAT THIS DISEASE?

TREATMENT

1. Enrofloxacin
2. Parenteral injections
3. Oral liquids
4. Supportive therapy

148 Bumble Foot

S.Q:WHAT IS BUMBLE FOOT?

PODODERMATITIS

Injury to the lower surface of the foot and subsequent infection with*Staphylococcus bacteria*

M.C.Q.Common causes of injury:



- Rough perches
- Splinters
- Wire floors
- Poor litter or bedding
- Quality

CLINICAL SIGNS

- Lameness
- Swelling of the foot pad
- Hard, pus-filled abscess on foot pad

SQ: TREATMENT

- Soak foot in warm water and Epsom salts.
- disinfect with alcohol.
- If skin is open, drain pus from abscess.

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- Flush abscess cavity with hydrogen peroxide to cleanout pus and debris.
- Pack the cavity with antibiotic ointment .
- wrap the foot with gauze and elastic bandage.
- Repeat daily until foot heals.

PREVENTION AND CONTROL

- Provide good quality litter or bedding.
- ➢ Keep bedding clean, dry, and deep.
- M.C.Q.:Keep perches less than 18 inches from the floor to prevent foot damage due to impact from jumping.
- Remove potential sources of injury such as sharp objects and/or surfaces.

149 Deficiency Diseases

S.Q:WHY TOE PARALYSIS OCCOUR?

CURLED TOE PARALYSIS

- Deficiency of Riboflavin
- Poor growth
- Weakness
- Emaciation and diarrhea
- unable to walk as their toes are turned inwards
- Drooping of wings



TREATMENT

- M.C.Q: Riboflavin @3.6 mg/kg of feed in chicks
- M.C.Q.;Riboflavin @ 1.8 mg/kg of feed in growers
- M.C.Q:Riboflavin @ 2.2mg/kg of feed in layers

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150 Cattle Farming Uses

Animal Science and

Livestock Production What is

Animal Science?

- Refers to the total store of knowledge relative to the breeding, feeding, care and management of animals and the marketing and processing of animals and their products as gained through practical experience and research methods.

Animal Use as Food

- Meat – Beef, Pork, Lamb, Goat Poultry
- Milk – Cheese, Ice Cream, Yogurt
- Eggs – Pastries, Mayonnaise, Custards



151 Animal By-Products

S.Q:HOW ANIMAL USED AS A BY-PRODUCTS?

Animal By-Products

- **Bones** – Button, glue, mineral supplement for livestock feed (Ca)
- **Fat** – Chemicals, salves, creams, dressings, lubricants, soaps, food
- **Glands** – Medicines, food additives
- **Collagen** – Glue, Gelatin
- Intestinal & Stomach tissue – lunch meats, surgical sutures, strings for musical and sports instruments

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- Fertilizer

S.QHOW ANIMAL USE AS A WORK?

Animal Use as Work

- Cultivate land
- Transportation
- Control other animals (herding)
- Assist physically & Medically handicapped
    A. Blind
    B. Epileptic & Diabetic

Other Animal Uses

- **Hides** – Leather
- **Hair** – Wool, mohair, fiber
- mcqs: **Lab Animals – Mice, rats, guinea pigs etc.** **Pets**



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152 Live sock Enterprises and Management

- **SQ: NQME THE Numerous livestock enterprises**
    - Traditional
    - Exotic
    - Purebred/Crossbred
    - Recreational

Species

- Cattle, Horses
- Sheep, Goats, Swine

- Poultry, Rabbits

- Different Space Requirements
- Different Nutrition and Management
- Many co-exist well

153 Dairy Farming

S.Q: WRITE Basic Principles of Dairy Farming?

Birth and Puberty

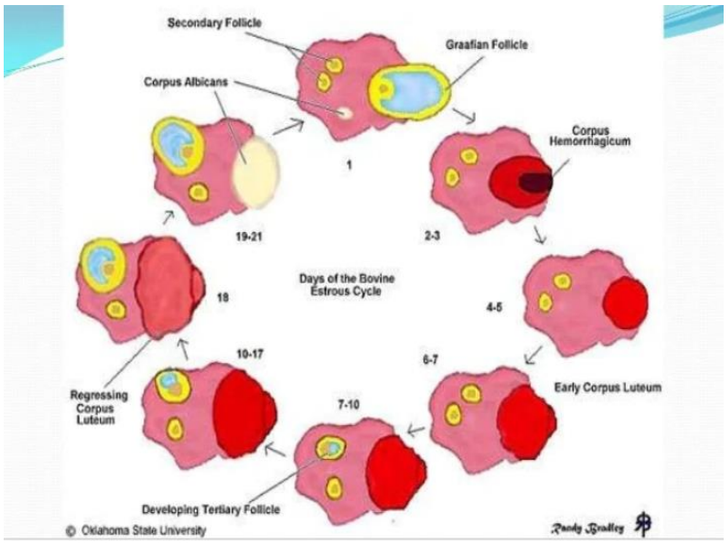
- A female calve (destined for dairying) weigh approx **40kg at birth**.
- If reared correctly, she should reach puberty at **1 year, weighing 250kg.**

Oestrous Cycle

- The length of the oestrous cycle of a cow is 21 days and it lasts an average of 18 hours.
- The gestation period of the dairy cow is 283 days (approx 9.5 months)

The estrous cycle can be divided into four stages: proestrus, estrus, metestrus, and diestrus. During proestrus the CL regresses (progesterone declines) and a preovulatory follicle undergoes its final growth phase (estradiol increases). Ovulation usually occurs during estrus (cows ovulate during metestrus).

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Lactation

- A cow commences producing milk as soon as she has a calf.
- If the cow is used to suckle calves, then she may continue to produce milk for up to 2 years.
- If she becomes pregnant (in-calf) she will go dry two months before calving.
- In commercial milk production, the aim is have the cow calve once a year, around the same time each year.
- This means that she will me milking 10 months of the year or approx 305 days.



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154 Lactataion Yields

S.Q:: WHAT ARE Basic Principles of Dairy Farming

Lactation Yields

- The milk yield of the cow depends on the breed.

- **Holstein**: 5800kg per year
- **Jersey**: 3400kg per year
- **Ayreshire**: 4000kg per year
- **Friesian**: 5000kg per year.
- **Dairy Shorthorn**: 4000kg per year.
- These vary significantly even amongst individual cows – A Friesian may vary from 2000kg a year to 12000kg of milk a year.
- These yields have also increases over the last number of years due to better selection of breeds.

Basic Principles of Dairy Farming

- Lactation Yields also vary with a cows age:

Age	Lactation No	% Of Max Yield
2	1	75
3	2	85
4	3	90
5	4	95
6	5	100

- The yield of the average dairy cow decreases after the 5th Lactation.

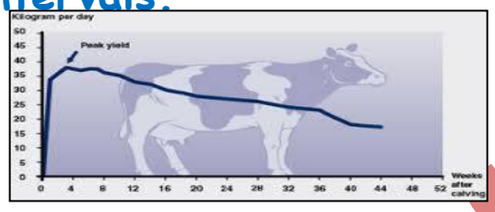
S.Q:WHAT IS THE FREQUENCY OF MILKING?

Frequency of Milking

- Milking empties the udder of the cow, which stimulates **milk-secreting alveoli** to commence producing milk.
- Frequently milked cows produce more milk than those milked irregularly.
- Therefore a cow milked four times a day will produce more milk than a cow milked twice – it will work to demand.

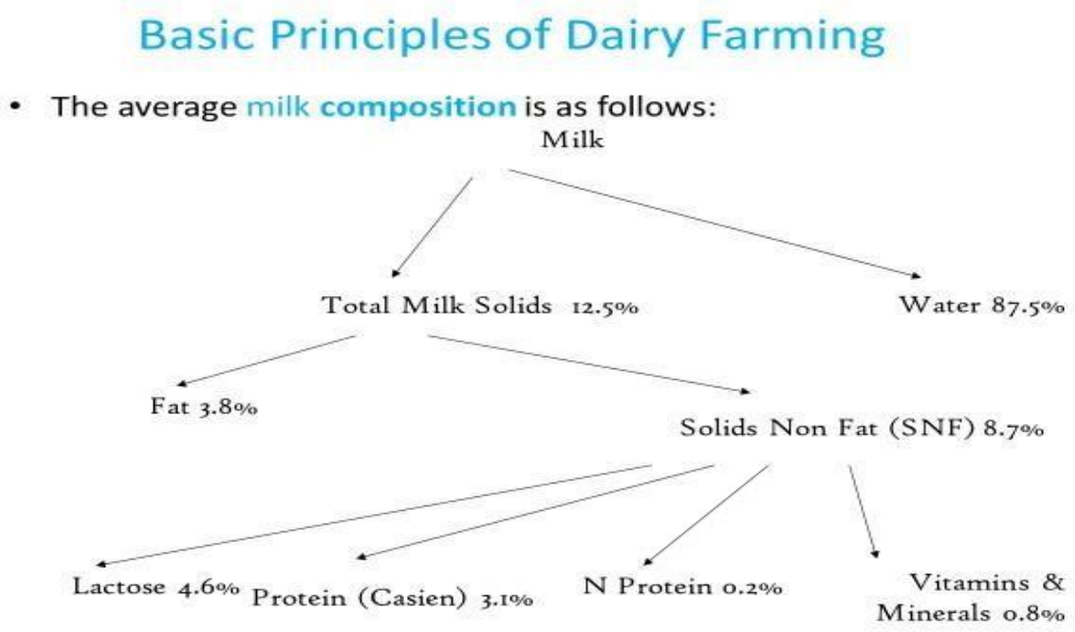
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- **BUT milking** more than twice a day is uneconomical and leaves cows more prone to disease such as **mastitis.**
- With milking twice daily, the **ideal milking interval is 12 hours.** This gives the cow less stress and will get the highest yields.
- But this gives the farmer a very long working day and is uncommon. The most common intervals are **14/10 or 13/11 hour intervals.**



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155 Lactation



Basic Principles of Dairy Farming

- Milk used for bottling or drinking must by law contain 3.6% fat and 8.5% SNF.
- Creameries buy milk in many different ways – sometimes at a flat rate per kg.
- Sometimes, however, it may be by percentage butterfat or protein.

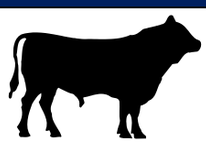
- Milk composition varies also amongst different breeds.
- The milk of the Jersey cow has the highest butterfat and SNF content of any cow.
- Also the composition of milk varies during milking.
- The milk at the start may only contain 1% fat, while the milk at the end ("The Stripping") may contain 10% fat.

Hygiene and Milk Quality

- Dirty milk (containing dirt, bacteria or antibiotics) can cause serious problems when processing.
- Dirty milk is caused by unhygienic milking machines or poor milk filters.
- Antibiotics are found in milk as residues from treatments to cows **for mastitis.**
- When milk is found to be not up to standard, it may be rejected by the creamery or bought at a lower price.

156 Beef Farming

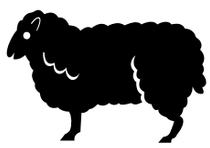
L.Q.:WHAT DO YOU KNOW ABOUR BEEF FARMING?



Beef Farming

- Great for the Part-time
- Labor and facilities can be low cost
- Land is required **1-5 acres per animal**
- Returns can be low and seasonal
- Enterprises
  - Freezer beef, feeders, purebred, contract heifers

Sheep Farming



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- **Low initial investment**

Low labor – expect lambing when

- Little investment
- Meat/Wool
- Not a huge market+

Rabbit Farming

Small acreage requirements

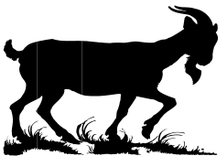


- Minimal cash outlay
- Small market – mostly pets

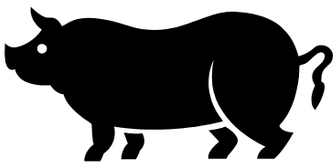
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157 Goat Farming

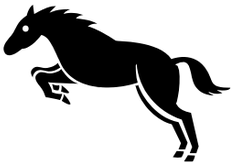
Goat Farming



- Projects, Milk, Meat, Fiber
- Ethnic Populations
- Dairy – High labor
- Should explore markets



Swine Farming



- Low acreage – Higher maintenance
- More management than – sheep, cattle or goats
- More investment in housing, shelter and fencing
- Do not utilize forage

s.q:What are effects of hourse farming?

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- Facility and acreage requirements
- Higher acreage – higher maintenance
- Expense to purchase
- Enterprises – training, boarding, lessons, breeding

158 Livetsock Selection

Livestock Selection

➢ Profitability of any individual animal or of a herd or flock of animals is determined by

- Type or individuality based on the ability to produce high-quality products for a tough market
- Performance or efficiency of production which is the ability to utilize feed efficiently, in producing meat, milk, wool or power.
- **Bases of Selection**

•SQ: **Selection based on**

1. Type and individuality
2. Pedigree
3. Show-ring winnings
4. Production testing

• SQ: **Type or individuality**

➢ Selecting those animals that most closely approach an ideal or standard of perfection and culling out those that fall short.

Pedigree

- Used in most purebred operations
- Based on performance of ancestors

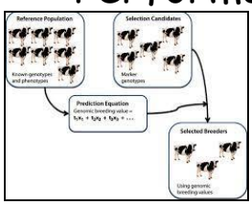
Show ring winnings

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- Implies the animals that have placed well in one or more shows are superior.

Production

- Generally look at economics such as body type and feed utilization
- Performance testing, progeny testing

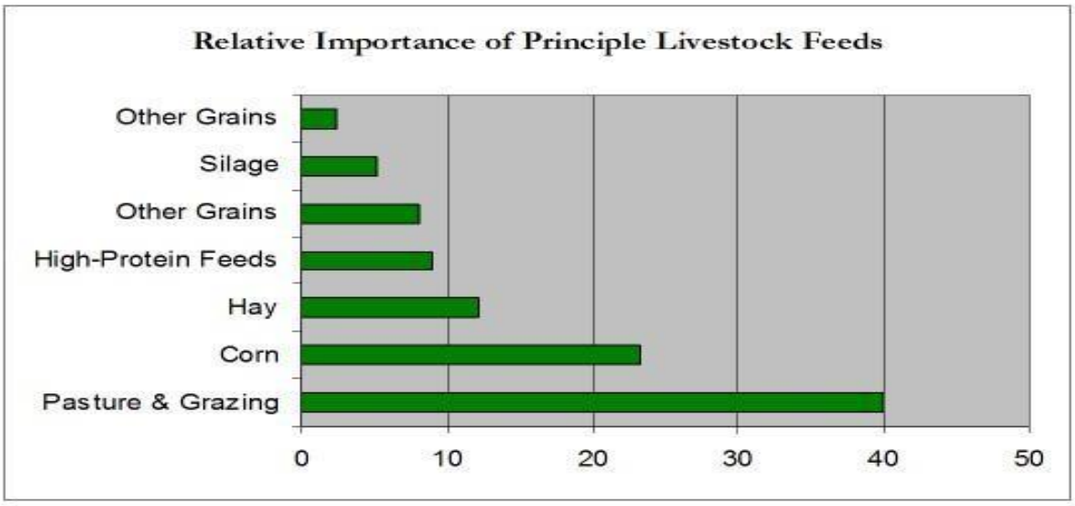


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159 Feeding Livestock

How h

Feeding Livestock



How quality of feed effect?

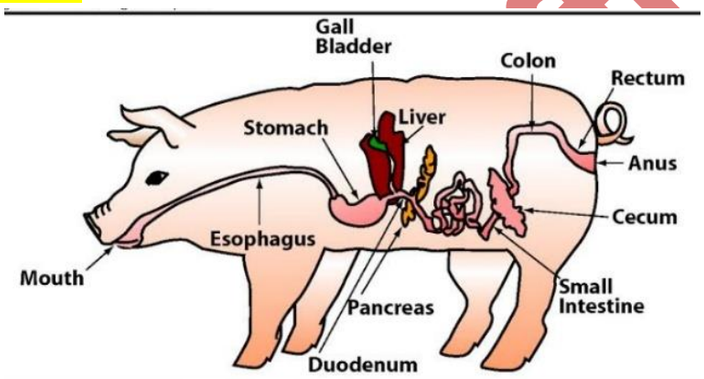
Feed Quality

- Quality of feed affects its value for animal nutrition.
- Quality factors include palatability and nutrient content.
- Palatability refers **to how well the animal will accept feed.**
- Improper harvesting or handling will reduce quality.

- If an animal does not find a feed palatable it will not eat enough, make good gains, grow properly or produce meat, milk or eggs.
- Feeds on produced on well fertilized soils will produce feed, especially roughages, that is of a higher quality.
- Vitamins, amino acid content and minerals will also be affected by soil type.

- **Digestion**

- Monogastric . . . . . e stomach) o

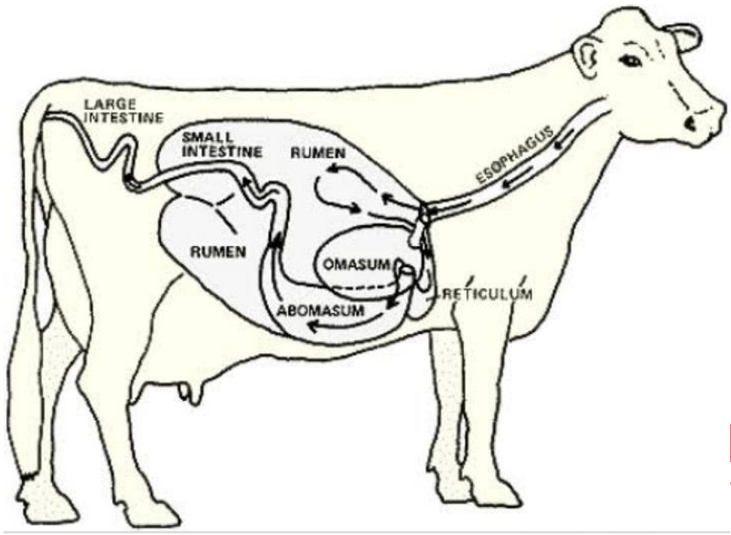


Pig, Dog, Human Digestric

- Avain (poultry) – Gizard
  - Chicken, Turkey, Duck

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Polygastric
(ruminants . o



Cow, Sheep, Goat
o Pseudo-ruminants
(cecum) o Horse,
Rabbits, Hamster

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160 Feeding Goals and proper Animla Nutrition

Feeding Goals

- Maintenance
- Growth
- Finishing
- Reproduction
- Work
- Age

What is Proper Animal Nutrition?

The process by which animals eat and use food.

- **Proper animal Nutrition**

1. Increases feed efficiency
2. Increase Rate of Gain

A Nutrient is:

➢ A substance **that is necessary for an organism to live** and grow

➢ Nutrients make it possible for animals to carry out **life processes**.

➢ Nutrients are provided to animals through?

1. Feed Stuff (Feed & Water)



161 Nutrient Needs of Animals

Nutrient Needs of Animals

s.q:What are essential nutrients of animal/

6 Essential Nutrients

1. Water
2. Carbohydrates
3. Fats
4. Proteins
5. Minerals
6. Vitamins

Nutritive Needs

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- **Energy** o Carbohydrates, Fats
- **Protein**
- **Mineral**

MCQS: o **Macro**

 Salt, Calcium, Phosphorous, Magnesium, Potassium,

Sulfur

MCQS: o **Micro**

 Chromium, Cobalt, Copper, Fluorine, Iodine, Iron, Manganese etc.

 Vitamin

- **Most Important** o WATER
- **Roughages (Forages)**
  - Bulk feeds that are low in weight per unit of volume, >18% curde fiber, low energy
  - Hay
  - Pasture
  - Silage
  - Crop Residues

Hay



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Silage



Rice bran

ad Official

162 Why Roughages and Concentrates

Why are some animals fed roughages and others concentrates?

Types of digestive systems

1. Non- Ruminants

1. **MCQS:** Mono-gastric – has one stomach
2. Avian – has **a crop & gizzard**
3. Pseudo-Ruminants

Ruminants

- **Have 4 compartments or stomachs**

  Ex Sheep, Cattle

Digestion -the mechanical and chemical breakdown of feed into a form which can be absorbed into the blood.

Ration

- The total amount of feed an animal has in a 24 hr period

- **SQ: DEFINE Balanced Ration**

  - Contains all the nutrients that the animal needs in the *correct proportions* .

Water

- Necessary for an animal to live
- Animal can liver longer without food than water
- Water makes up 75% of the weight of an animals body.

Basic functions of water

1. Regulate body functions

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2. Promoting biochemical process



163 Nutrient Composition of Feeds

Carbohydrates

They provide Energy

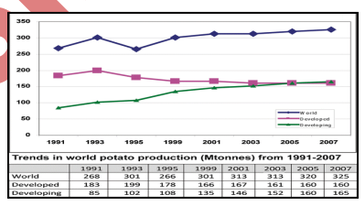
- Should make up 75% of an

animals diet **Types of**

Carbohydrates

1. Sugars
2. Starches
3. Fiber

what are Sources of Carbohydrates?



Trends in world potato production (Mtonnes) from 1991-2007

	1991	1993	1995	1999	2001	2003	2005	2007
World	268	301	266	301	313	313	320	325
Developed	183	199	178	166	167	161	160	160
Developing	85	102	108	135	146	152	160	165

SQ: NAME THE Cereal Grains

1. Corn



2. Wheat
3. Barley
4. Oats
5. Hay

OATS

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6. Rye

7.

Lipids

- Fat is a good source of energy
- Fats have 2.25 times more energy than carbohydrates

Protein

- Needed for

MCQS: Grow new tissue and repair old tissue.

- Highest amount found it muscles

Proteins contain

A. Amino Acids

Building blocks of proteins

B. **MCQS:23 Amino Acids (10 are essential)**

164 Sources of Proteins

s.q:What areSources of Protein ?

6 common sources

1. Soybean Meal
2. Cotton Seed Meal
3. Fish Meal
4. Tankage
5. Skim Milk
6. Alfalfa



Protein is the most common

Nutrient Deficiency

Minerals

- **Macro-minerals**-Calcium, phosphorous

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- **Micro-minerals**
- **Essential for**

1.Skeletal growth

2. Body systems to function properly

Common sources

Alfalfa Hay, Cereal Gains, Bone Meal, Molasses, Salt

Vitamins Functions

1. Help regulate body functions
2. Keep body health
3. Develop resistance to Disease

s.q:What are Types of Vitamins?

1. Fat Soluble

- Vitamin A, D, E, K

2. Water Soluble

- Vitamin C & B

165 Feedstuffs used in Livetsock Diets

S.q:what Feedstuffs Used in Livestock Diets?

- **Feed Classes**

✓ 8 classes grouped by origin and

like characteristics ✓ Dry

Forages & Roughages

- All feeds that are cut and cured
  - All feeds w/ CF>18%
    - Usually low in NE
  - Carbonaceous Roughages
- Generally low in protein

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- Straw
- Stalks
- Weathered grass

Proteinaceous Roughages

A. Legume hays
B. Some grass hays
C. Legume/grass mixtures

- Pasture, Range Plants, & Fresh Fed Forages

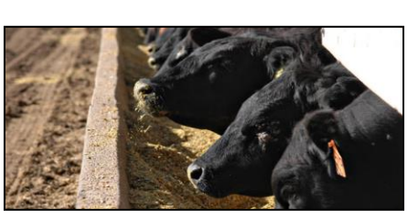
- Pasture grass
- Anything that is not allowed to ferment before feeding

Silages

- **Ensiled forages**
  - **Carbonaceous**
    - Corn silage
    - Grass silage
  - **Proteinaceous**
    - Alfalfa silage
    - Clover silage

Energy Feeds

- *<20% CP, <18% CF*
- May be ensiled
- Carbonaceous Concentrates
  - All cereal grains & sorghums
  - Byproduct feeds
    - Bran
    - Middlings
    - Cobs



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- Molasses

166 Protein and Vitamin Supplements

Feedstuffs Used in Livestock Diets Protein Supplements

- **>20% CP**
- **Vegetable Origin**
  - Soybean Meal
  - Cottonseed Meal
  - Corn Gluten Meal
  - Brewer's Dried Grains

➢ **Animal Origin**

- Animal tissues
  - Meat & Bone Meal
  - Blood Meal
  - Most are banned/restricted from livestock diets – Fish Products
    - » Fish Meal
- Milk Products
  - » Whey protein
- Feather Meal

Mineral Supplements

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- Calcium Carbonate
- Limestone
- Others

- **Vitamin Supplements**
  - ✓ Fish Oil
  - ✓ Others .



- **Additives**
  - Propylene Glycol
  - Titanium Dioxide (coloring agent)

167 International Feed Names

Feedstuffs Used in Livestock Diets

s.q:Write international feed name?

- **International Feed Names**
  - Used to create a "common language" among the feed industry – 6 Facets are included in the naming process
- Original Material
- Parts of Material used as feed (may be affected by processing)
  - Bran
- Processing and Treatments
  - Dry-rendered
  - Hydrolized

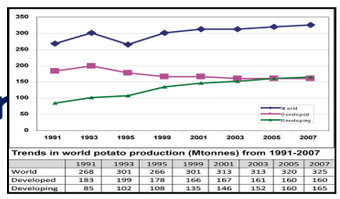
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- Extracted

 Stage of Maturity (plants & animals)

 Cutting

 Grade



Trends in world potato production (Mtonnes) from 1991-2007

	1991	1993	1995	1999	2001	2003	2005	2007
World	268	301	266	301	313	313	320	325
Developed	183	199	178	166	167	161	160	160
Developing	85	102	108	135	146	152	160	165

- **Characteristics of Concentr** .

- Carbonaceous Concentrates

- <20% CP, <18% fiber

- Generally, high energy feeds

168 General Nutritive Characteristics

Feedstuffs Used in Livestock Diets

s.q:What are general nutritive characteristics?

- **General Nutritive Characteristics**

Examples

 High in energy

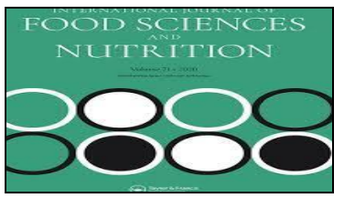
 Low in fiber

 Low in protein

 Low protein quality and high variability ✓ Minerals

» Low Ca

» Med P



- **Corn**

» 80% TDN

» 8-9% CP

» Med P, low Ca

» Recent technologies – high lysine corn, waxy corn, high-oil corn

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» Alternative feeding forms

- **Oats**

» 65-70% TDN

» 12% CP

» Very palatable, more expensive to feed » Where is it most commonly used? -

Dried Beet Pulp

» 65-70% TDN

» 8-10% CP

» Byproduct of sugar beet processing

» ~18% CF

- **Molasses**

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» 55-75% TDN

thick, dark brown juice obtained from raw sugar during the refining process.




» 3-7% CP (mostly NPN)

» Byproduct from same industry as above

» Usually fed in what form?

» What are the advantages to feeding?

- **Animal Fat**

» Byproduct of rendering » Treated w/ antioxidant to prevent rancidity »

Why do we feed it?

» 5% max in ruminant diets, 10% in non-ruminants

- **Dried Bakery Product**

» What might this include?

» Similar to corn in energy, higher in fat, and salt?

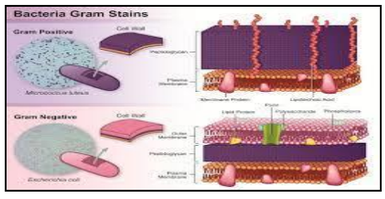
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169 Proteinaceous Characteristics

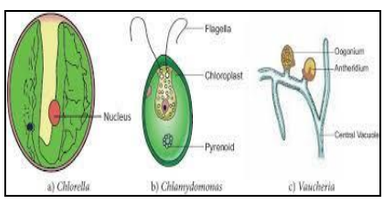
Feedstuffs Used in Livestock Diets

- Proteinaceous Concentrates

- **Quality**
  - Kinds, amounts, ratios of amino acids
    - Essential Amino Acids
      - » Must be supplemented
      - » MCQs: **PVT TIM HALL**
        - » **Phenylalanine, Valine, Threonine, Tryptophan, Isoleucine, Methionine, Histidine, Arginine, Leucine, Lysine**
  - NPN may be used as a protein source (only in ruminants)
- **Examples**



  - **Urea**
    - » 281% CP
    - » Use only in very small amounts
    - » Very effective for feeding rumen bacteria
  - **Soybean Meal**
    - » Most commonly used plant protein supplement
    - » 44% or 48% CP available (depends on how much its diluted w/ soyhulls)
    - » 71-80% TDN



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» Very low in fiber

» Very broad amino acid profile

What other forms are used?

- **Animal/Marine protein supplements**
  - Derived from meat/poultry packing/rendering, or from the marine industry, or surplus milk
  - Used only to improve the CP of basal feeds and improve amino acid profile
  - Balances protein sources (plant vs. animal) – Blood Meal
    - » 80+% CP
    - » Highly unpalatable
    - » High rumen undegradable protein for ruminants
  - **Fish Meal**
    - » 35-70% CP
    - » Excellent protein quality and good source of B vits
  - **Whey protein**
    - » 11% CP, 61% lactose
    - » Used in milk replacers or pig starter diets

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» Highly palatable, excellent source of protein

- **Animal Waste**

» Nutrient content varies

» Used primarily in ruminant diets

» Usually high in NPN **(non protein nitrogen)**

» Has proven to be fairly effective, in certain diets

170 Feed Grain By-Products

Feedstuffs Used in Livestock Diets

- **Feed Grain Byproducts**
  - **Corn byproducts**

- **Corn Gluten Meal**
  - Dried residue remaining after removal of most of the starch, germ, and bran
  - 46-60% CP

- **Corn Gluten Feed**
  - Dried residue remaining after removal of most of the starch, germ, gluten, but contains bran
  - 20-25% CP

- **Distiller's Dried Grains**
  - Byproduct of the alcohol brewing industry
  - **25-27% CP, 9-11% CF**



- **Distiller's Wet Grains**
  - Byproduct of ethanol production

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- Use usually restricted to geographical area close to the distiller
- Will ferment if not used quickly
- **Wheat byproducts**



- **Wheat middlings**
  - Fine particles of bran, germ, shorts, tailings
  - **16-18% CP**
  - More commonly fed in swine diets, unpalatability makes its use limited in ruminants
  - **Soybean Hulls**
    - **12% CP, 78% TDN**
    - Very good for replacing other high fiber feeds, without losing too much fiber
    - Very palatable

171 Characterisitics of Roughage Feedstuffs

Feedstuffs Used in Livestock Diets

What are Characteristics of Roughage Feedstuffs?

- **Characteristics of Roughage Feedstuffs**
  - Generally low in energy
  - Higher in fiber
  - Higher mineral content
  - Extremely palatable to ruminants
  - Nutritive value can be extremely variable
  - Limited inclusion in beef finishing diets, excluded from swine rations

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- Must be present in dairy rations to maintain health rumen and milk fat content
- Proteinaceous Roughages

- **Alfalfa**
  - Hearty, w/ extensive root system
    - May be pastured, baled, mealed, or ensiled
  - **15-25% CP, >50% TDN**
    - High in Ca, fair P
  - ✓ 2-5 cuttings/yr.

- **Carbonaceous Roughages**
  - **<10% CP**
  - Nonlegume forages, low quality roughages Ex. Switchgrass

- **Cool-season Perennial Grass Forages**
  - Particularly productive during spring – early summer, and fall
  - Orchardgrass
    - Grows in clumps
    - **May grow to 2-4'**
    - Grows early in the spring
      - Rapid developer
      - Highly palatable
      - **8-18% CP**
      - Good partner w/ alfalfa for hay or pasture grass

Timothy

- **8-12% CP**

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- Not great for pasturing alone, but good in a mixture
- Cut no later than early bloom for maximum nutritive value

172 Small Grains

Feedstuffs Used in Livestock Diets

s.q:what are small grains include?

- **Small Grains** (wheat, oats, etc.)
  - Generally same seeding rate as for grain, may increase if going to cut for silage
  - Effective in the pasture
  - Harvesting for silage should occur around boot stage
  - 10-22% CP, 62-72% TDN
  - If harvested early, can mimic corn silage
  - Be cautious of low Mg levels, may see grass tetany
  - **Corn Silage**
    - Most popular silage
    - Extremely palatable
    - Moderate to high energy, low in protein
    - May not be most efficient in a finishing diet
    - Many varieties available
    - High grain content is desirable
  - **Corn Stover (Stalklage)**
    - Harvested at or just after grain harvest
    - Ensiled

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- Should be fairly fine-chopped to ensure packing
- Good for wintering cattle, somewhat high in energy

s.q:write **Methods of Utilizing Forages?**

- **Pasture**
- Major feed for dairy, beef, and horses
- For certain situations, the most profitable option
- **Essential Pasture Qualities**
  - Durable both for longevity, and to stand up against the foot traffic
  - Growth start early in spring continue late through fall
  - Plant variety
    - Greater yields
    - Better nutritional value
    - More complete nutritional balance
  - Monitor quality/growth

Use paddocks

- Reduces under or overgrazing
- Increases carrying capacity of the pasture
- Costs more to maintain
  - Manage w/ mowing, fertilization, or herbicides (i necessary)



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- Avoid grazing in wet conditions – Proper stocking rate
  - Varies w/ pasture crops
  - Some 1-2 acres/cow, others 10-20 acres/cow

173 Special Concerns

Feedstuffs Used in Livestock Diets

- **Special concerns**
  - **Poisonous plants**
    - Highest risk w/ poorly maintained pastures or poor cattle management
  - **Bloat**
    - Risk on legume pastures due to amount of lush plant material
    - Can reduce risk by adding other grasses into the alfalfa
  - **Nitrate Poisoning**
    - Accumulation after drought, or cultivated forages
    - Seldom occurs in other types of pastures
    - Causes abortion or death
- **Green Chop**
- What is it? Is it still being used?
- **Advantages**

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- Maximizes yield/ac.
- Less nutrient loss
- Less fencing
- May reduce bloat

- **Disadvantages**
  - No uniform quality
  - Weather
  - Cannot be fed year round

174 Hays and Common Losses in Hay Making

Feedstuffs Used in Livestock Diets

- Hays

- **Should be 15% moisture or less** - Why?
  - What are the risks?
- **Keys to haymaking**
  - Know when you want to mow
    - Stage of maturity directly correlated to nutritional value ✓ Nutritional value decreases quickly ✓ What kind of yield to you want?
    - How does it fit into your feeding program?
  - **Allow for sufficient curing**
    - Conditioning can reduce curing time **by 50%**
    - Maximum moisture **18-22%**
  - **Raking**

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- Should be done prior to the end of curing to minimize leaf shatter losses
- May be necessary to facilitate drying

- Baling

- Squares – should be stored and stacked ASAP out of the weather
- Round Bales – will survive outside storage if stacked in a pyramid shape (sheds water), grass hays last longer than legumes

• Common Losses in Haymaking

- Leaf Shatter

- Leaves have 2-3x more protein than stems
- Higher in most other nutrients, except fiber
- **~20% loss is normal, losses can be as much as 40-75%**

- Heat Damage

- **>25-30%** moisture may lead to mold and excess heating
- During its sweating phase, hay should be not warmer than 84°, above **120° nutrient loss** occurs
- Spontaneous combustion risk is high is hay reaches **160°**

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- Fermentation losses

  - Should account for only 5-7% loss in total DM(dry matter)
  - Converts sugars and starches to carbon dioxide and water
  - Reduced energy content -

Bleaching

  - Loss of color ✓ Some vitamin losses

- Losses under normal conditio



  - 20-30% DM
  - 27-30% CP
  - 25-28% TDN (total digest able nutrient)

175 Additives in Hay Making

Feedstuffs Used in Livestock Diets

- Additives in Haymaking



- Preservatives

  - Propionic acid, acetic acid, etc
  - Allow storage at higher moisture
  - Should be applied at the bale chamber
  - ~20#/t

- Drying agents - Anhydrous ammonia

  - Treatment for low quality hay to improve protein and energy content
  - Must seal and gas for 20-30 min
  - Can increase CP by 3-6%
  - Difficult to handle

- Silages

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- Goes through acid fermentation under anaerobic conditions

- **Types of silos**



  - Upright
    - » Semiairtight
    - » Airtight
  - **Horizontal**
    - Trench/pit
      - Bunker
      - Piles
    - Temporary
      - Bags

- **Silage activity in the silo**
  - Plant cells continue to respire
    - Consume oxygen
    - Produce carbon dioxide
    - Mold inhibited due to lack of oxygen

Temperature increases

» Should increase to 80-100°

» Temperatures >100-120° result in carmelization

Fermentation

Acetic acid – first acid, reduces pH from 6-4.2, in the first 4 days

Lactic Acid – begins on d 3 and lasts 2 wks., most abundant acid, stops around pH 4.0

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 Butyric Acid – forms if an insufficient amount of lactic acid is present, or a result of excess moisture—leads to a putrid smell

 Propionic Acid – very little present

- Lowered pH prevents bacterial growth and further fermentation as long as it is w/out oxygen

 Takes ~21d, but many say 2-3 months is best

 Corn silage can keep for several years, if stored well

176 Crops used for Silage

s.q:which Crops used for Silage?

 Corn

 Legumes

» Haylage is most common, very difficult to make well

Using silages

- **Advantages**

 More feed/ac. Land

 High quality feed

 Harvest is mechanized, range of harvest dates are wider?

- **Disadvantages**

 Once you have it, you got to feed it

 May slow ADG ✓ Considerable cost

✓ Water is a factor?

Making good Silage

- 50-70% moisture range target

 Too wet – sour smell, nutrient loss by seepage

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 Too dry – improper fermentation, difficult to pack
 May add water during ensiling process

- **Proper stage of maturity**
  - When should we chop corn?
  - When should we chop alfalfa?

- **Proper packing**
  - Cut at proper length to ensure packability, but maintain good fiber
  - Distribute evenly in the silo
  - Uprights pack themselves
  - As much weight and time as possible on a bunker

- **Proper drainage** » Slope » Base of the silo



- **Losses from Ensiling**
  - Field losses
  - Gas/fermentation losses – Surface spoilage
    - » Can be as much **as 25% on** a bunker silo
  - Seepage
- **Silage Additives**

Nutrient Additives

» Urea/ammonia – increase

CP(**crude protein**)

concentration

- Preservatives

» Acids, Bacteria, Enzymes, cultures, etc. – all to promote proper fermentation

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177 Ensiling vs Hay Making

General idea is do add/subtract DM, aid in fermentation, or increase nutrient content

Ensiling vs. Hay making

- Advantages

- Wider harvest windows
- More nutrients are accessible/maximized ✓
  Weeds are prevented more effectively ✓
  More yield/ac.

- Disadvantages

- More and more expensive machinery?
- More variability in quality
- Fermentation problems may destroy good feed

Characteristics on Common Nutrient Additive Feedstuffs – Macro-minerals

• What are the major macro-mineral ingredients that will be used?

- Trace Minerals
  - Cu – copper sulfate
  - Zn – zinc oxide
  - Se – sodium selenite
  - Chelates may increase the amount absorbed in the SI

- Vitamins



• Fat soluble .

A

Found in all green plants

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Usually don't need to supplement much

- D
  - Can be synthesized by sunlight on the skin
  - Fed through fish oil



- E
  - Found in germ of grains
  - Green plants

- K
  - » Also found in green plants

May all be generated and fed synthetically

» E is the most supplemented and has the most health implications attached to it (especially in cattle)

Water soluble – Riboflavin

» Milk products, yeast, hay, etc.

- Niacin

» Some present in all feedstuffs

» Much is not available for digestion (especially swine)

- Choline

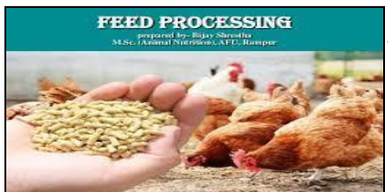
» Present in most feeds of animal origin

- Most all are synthetically made also
- Supplemented at extremely low rates
- Very expensive

178 Methods of Feed Stuff Preparation

s.q:What are Methods of Feedstuff Preparation:

Why process?



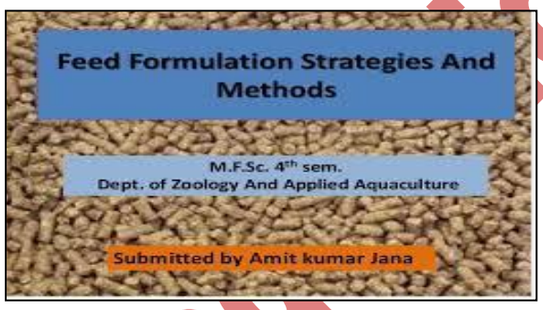
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- Increase ease of handling
- Increase efficiency of utilization
  - Palatability
  - Digestibility

» What does this have to do w/ it?

- Alter the density of the feed/diet
- Improve feed efficiency by **5-15%**
  - May also observe improvements in gain



Processing Methods

- **Grinding – Hammer Mill**
  - Can grind hay, ear corn, grains, etc.
  - Well suited for fine-grinding
  - Not effective for coarse-grinding

– Burr Mill

- 2 plates w/ unmeshed burrs, one moving, one stationary
- Relatively uniform grind
- Not designed for roughage materials
- Higher maintenance cost

– Degrees of fineness

- What are the advantages/disadvantages to a fine grind?

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- Are there health risks?
- Finely ground – pass through 1/8-3/16" screen

<700 microns

- Medium ground – pass through ¼-3/8" screen

8001000 microns

- Coarse ground – pass through ½" or larger screen >1300 microns

Swine

- Medium to fine grind
- Starter diets ~650-750 microns
- Breeding herd maybe 750-900 microns

Beef

- Grinding fineness depends on feeding program
- More forage = finer grind

- Dairy

- Coarsely ground ~1200 microns
- What will happen if it is ground finer?

Dry Rolling/Cracking

- Rolling if the rollers are smooth, crimping if rollers are corrugated
- Can produce similar feed to coarse grinding

179 Pelleting

What is Pelleting?

- Combination of heat, moisture, and pressure forcing feed through a dye
  - ➢ Reduces dust

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- Reduces losses and fine
- Reduces sorting
- Increases digestibility
- Higher cost
- Pellet quality may be variable



Swine diets

➢ Improved performance due to increased consumption

➢ Higher rates of gain and feed efficiencies

Dairy diets

➢ Only recommended for concentrate portions of the diet

➢ Highly effective at improving digestibility of nutrients

Beef diets

- Can improve rates of gain, feed efficiency, etc.
- Not recommended as a large component of a high concentrate diets
- Useful for creep feeding



Horse diets

- Useful for creep feeding
- May pellet the entire diet

Pellet varieties

- Granules
- Crumbles
- Pellets

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➢ Cubes

What are Heat Treatments and Grain Processing?

Steam rolling

- 1-8 min of steam, then roll
- Slight increase in moisture
- Not significant advantage in digestibility

- **Steam Flaking**

- Steam **15-30 min**, then roll
- Thin flakes **w/ 16-20%** moisture
- Gelatinizes starch increasing digestibility
- Common in calf starters, and horse feeds

- **Roasting**

- **Heat to 300°F**
- Partial starch gelatinization
- Caramelized appearance, and sweet smell
- Common for soybeans

180 Extrusion Pelleting

Extrusion Pelleting

- Heat and pressure forcing feed through a small spiral hole
- Produces a flaked feed
- Similar to pelleting

• **High Moisture Grains**

- Harvested at **20-35%** moisture (usually corn)

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- Coarsely ground and stored in a silo
- Does goes through fermentation
- Highly palatable
- Can improve feed efficiency up to 8%

Feed Mixing Methods

- Must have accurate scales



Types of mixers

- Vertical
  - Cylindrical with vertical auger(s)
  - Cannot handle much molasses
- Horizontal
  - Rolling mixer w/ an auger and paddles inside
  - Handles higher molasses feeds better
  - Sequencing of feeds is important
    - Why?



Feed Storage

- Moisture level is key
- What problems can be encountered?
- Must be void of insect problems and rodents

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181 Factors affecting Feed Intake

s.q:What areFactors Affecting Feed Intake?

- Water Availability
- Palatability
- Dietary energy level

- Animals will generally eat to their caloric intake needs
  - Age and Size of animal
  - Temperature
  - Health

Expected Intake guidelines (% of BW)

- Swine 4-5%
- Cattle 2-3%
- Horses 1.5-2.5%
- Dogs 2-3%

182 What is Forage

What is forage?

- Vegetable matter in a fresh, dried, or ensiled state.

What can you do with forage?

- graze it
- machine harvest and store it
- it's animal feed

- Forage allows you to raise an agricultural crop on land where other crops cannot be produced.

Grasses

- Timothy
- Orchard Grass
- Kentucky Blue
- Fescue (endophyte free)





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Legumes

- White Clover
- Alfalfa
- Ladino Clover
- Red Clover





183 Miing Forage Species

Mixing Forage Species

Consider the traits of each species.

Aggressive vs

Passive Maturity

dates: early vs late

Other traits that can be utilized:

- o -N fixation -heat tolerance
- o -fast germination -wet tolerance
- o sod(turf/grass) formation
- - palatability
- o -summer dormancy -
- hay or grazing
- o -life span -fertility needs

Forage Establishment

- Plan at least 1-year ahead.
- Planning includes soil testing.

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 Don't be cheap with fertilizer, limestone, and seed. ✓ Do proper field preparation.

 Select the right forage species and best available varieties of forage. ✓ Need good seed/soil contact.

 Don't bury seed too deep.



s.q:What do you now about harvesting forages?

Harvesting Forages

Making quality forage is an art.

 need to know your forage species keep on top of the weather

 have flexible production options have timely access to equipment

 have equipment in good working order ✓ have healthy forage and few weeds ✓ have some luck!

184 Forage Storage Losses

Forage Storage Losses

 Uncovered bales stored outside

 Hay baled too wet

 Bales stacked in contact with the ground

 Holes in plastic bags and tubes, or leaky silos

 Poorly packed and uncovered trenches

Evaluating Hay Quality



 Leafiness

 Color

 Foreign Material

 Odor and Condition

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- ACTIVITY

185 Why Manage Pastures 1

Why Manage Pastures?

- **Pastures are profitable**
  - grazed forage is good, cheap feed
  - pastures are inexpensive to develop and maintain
  - animals do the harvesting, therefore there is a reduction in the need for machine harvesting and forage handling
  - while on pasture, animals spread manure in the field, reducing hauling
- S.Q:HOW WE **Protects surface and groundwater from nutrient pollution?**
- **Protects surface and groundwater from nutrient pollution**
  - acts as a filler to screen out and traps soil particles which contain nutrients such as N and P
  - the nutrients are then utilized by the pasture plants once these nutrients have moved into the root zone of the soil
- **Reduces soil erosion**
  - the top growth of pasture plants lessons the impact of rain drops on the soil surface and also slows down the surface runoff of water across the field
  - pasture plant root systems bind the soil together, thereby holding it in place
  - most pastures keep the soil covered year around, unlike annual crops
- **S.Q:HOW WE Improves forage yield and quality?**

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- **Improves forage yield and quality**

  - plants that are maintained at the optimum fertility level and are not stressed by pests or by poor grazing management will be more productive
  - healthy, productive plants will provide a quality product healthy plants will have a higher nutritional value

186 Why Manage Pastures 2

Why Manage Pastures?

- **Reduces weeds and improves esthetics**

  - weeds are opportunistic; they will move rapidly into an open area or an area occupied by a weak plant
  - weeds cannot gain a foothold in a field with vigorously growing plants a clean, weed free pasture reflects well on your farm manage and how people passing by view your farm

Maintaining Pastures

1. Rotate
2. Clip
3. Irrigate
4. Drag Manure



Grazing Management

- Protecting pasture plant root reserves and maintaining plants in a vegetative state are keys to a good pasture.
- Overgrazing reduces root reserves which shrinks the root system and leads to fewer leaves being

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produced; plants also take longer to recover from grazing.

- Under grazing reduces quality and yield as over-mature plants become less vigorous and more fibrous.

Forage Re-Growth

- Slow to recover at first
- Rapid growth after recovery
- Slow after rapid growth period

187 How Grazing Affects Root Growth

S.Q:How Grazing Affects Root Growth ?

% Leaf Vol. % Root Removed Growth Stoppage

- 10%.....0%
- 20%.....0%
- 30%.....0%
- 40%.....0%
- 50%.....2-4%
- 60%.....50%
- 70%.....78%
- 80%.....80% ➢ 90%.....100%

All root growth stops **for 12 days with 80% removal & 18 days with 90% removal. When 60 %** is removed, only half of the roots stop growing.

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A Good Grazing Rule of Thumb

Take half ----------Leave half

- In the long run, the animals will have more forage to graze.

S.Q:Why Timely Mowing?

- Mowing prevents plants from becoming over mature.
- Vegetative plants are more palatable and more nutritious.
- Mowing helps to control weeds.
- Flash grazing can work in place of mowing to help reduce excess forage in paddocks.
- Harvesting excess forage for hay is a good way to fully utilize forage resources.

188 Why Control Weeds

Why Control Weeds?

- Some weeds have been declared illegal noxious weeds by the State.
- Weeds look bad, they reflect poorly on your management.

Weeds will:

- Reduce the stand of desirable plants.
- Reduce overall quality and yield.
- Reduce overall animal yield.
- Some are poisonous, or can affect the animal product.
  - Seeds are spread through manure

Weed Management

- Cultural Control
  - mowing
  - grazing
  - over seeding



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- improved fertility

Chemical Weed Control Grazing Restrictions (herbicides)



- Ally……none
- 2,4-D……milk cows, 7 days+
- Crossbow(A **weapon consisting of a bow fixed crosswise on a wooden stock**, with grooves on the stock to direct the projectile.) ……milk cows, 14 days+
- Banvel……milk cows, 7 days+
- Roundup……livestock, 8 weeks
- Spike……none
- Stinger……none

herbicides

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189 Planning A Pasture System

HOW Planning A Pasture System Sources of Info and Help?

- **Resources**
  - soil survey map
  - soil capability assessment
  - aerial farm map
  - walking the farm

- **Assistance**

  - Cooperative Extension ✓
  - N.R.C.S ✓ F.S.A.
  - Farming Supply Companies

WRITE BASIC TYPES OF PASTURE?

Basic Types of Pastures

- Continuous
  - animals are allowed to graze in the pasture for extended periods of time
  - animals often do well in this system since they are allowed to choose the plants they eat
  - plants are often overgrazed and under-grazed in this system

Continuous Grazing



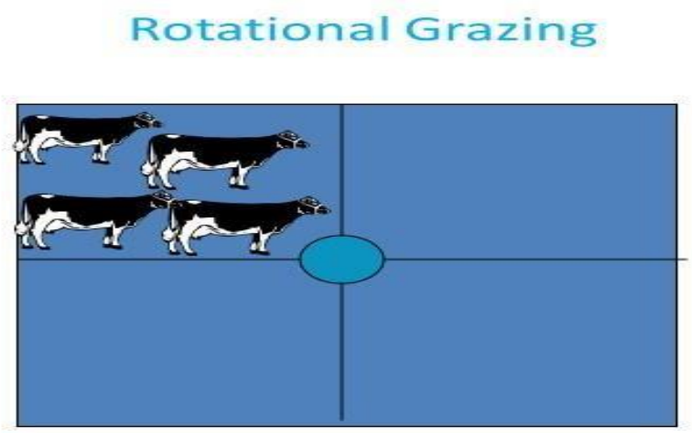
190 Basic Types of Pastures 1

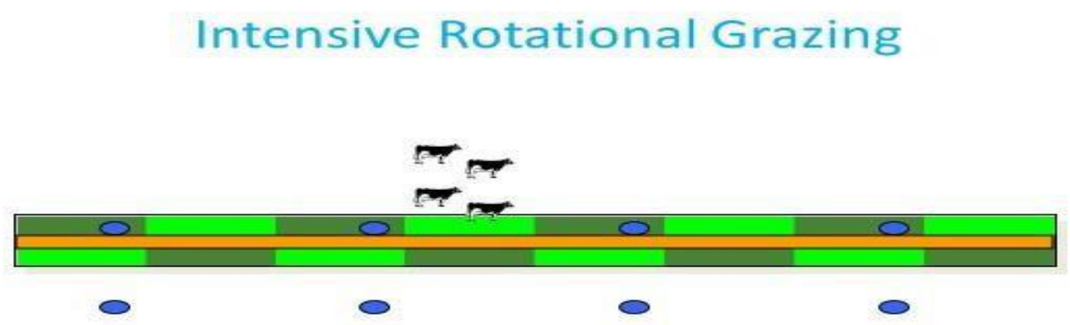
Basic Types of Pastures

- **Rotational**
  - animals are allowed to graze for only a limited period of time and animals are moved when existing forage has been removed
  - intensive rotational grazing systems subdivide pastures into paddocks and use high stocking rates where animals are

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forced to eat all forages ➢ this system is





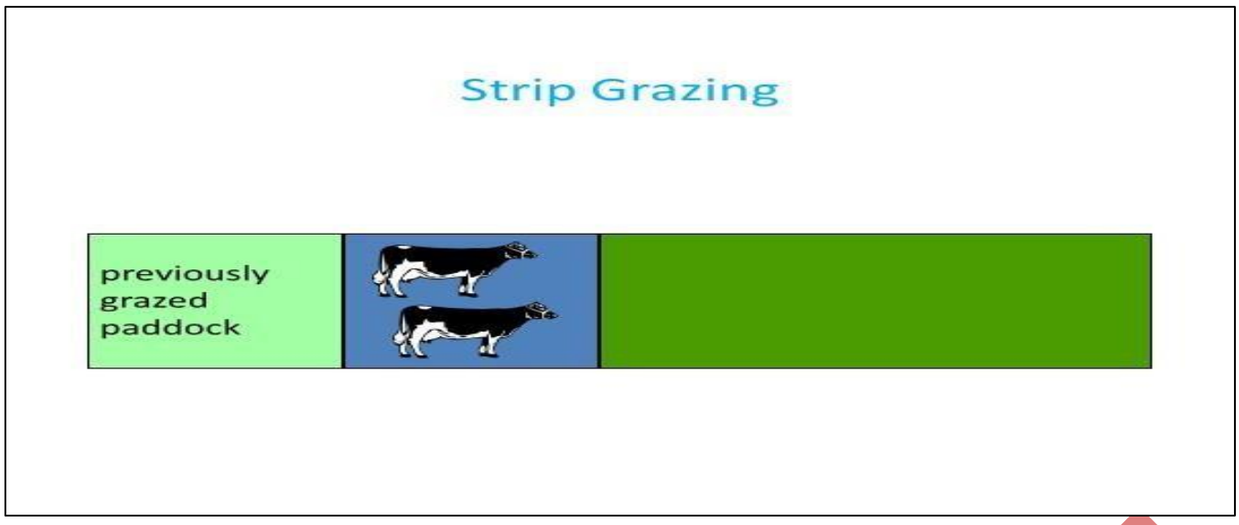
191 Basic Types of Pastures 2

Basic Types of Pastures

most efficient

- **Deferred Grazing**
  - forage is allowed to accumulate in a pasture for grazing at a later date
- stockpiled tall fescue is an example
- **Strip Grazing**
  - high stocking rate of animals are put into a pasture for a limited period
  - usually involves a specially planted crop typically an annual species

i.e. rape, turnips, or summer grasses





Rotating Pastures

- **Benefits**
  - Feeding less grain and hay
  - Reduce pest populations
  - Slow soil erosion
  - Allow daily exercise

BE OBSERVANT and watch your pastures

Setting Up A Pasture System

Recommendations

- Develop a 5 year farm/business plan
- You need to plan ahead
  - plan for when fields need to be renovated
- Use existing resources whenever possible (fences, water, forage crops)
- Establish crops according to your plan

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 Existing pastures can be renovated later if needed now for grazing

192 Pasture System Recommendations

Setting Up A Pasture System Recommendations

- Put your money into good perimeter fence.
  - this will help to keep predators out and your animals in.
- Map out farm, give each field own identity
- Soil test fields individually
  - each has its own personality, so treat it accordingly
- Develop a practical watering system
  - common problem for many
  - there are many factors to consider (costs, environmental,

system)

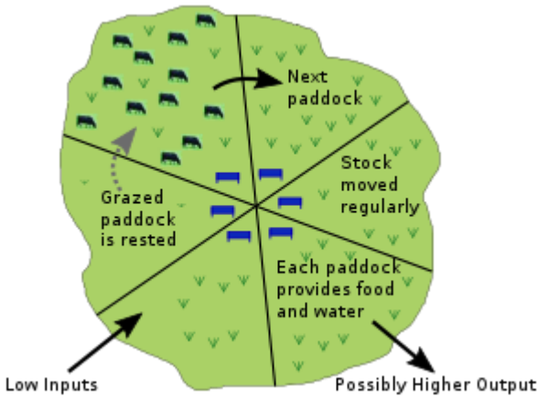
Create a sacrificial area

- this will protect your pastures
- Estimate the carrying capacity of your pastures
  - impacts on the number of animals and paddocks (rotational) (In rotational grazing livestock are moved to portions of the pasture, called paddocks, while the other portions rest. The intent is to allow the pasture plants **and soil time to recover**. Healing native rangeland may require a combination of burning and rotational grazing.)

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- Calculate number of paddocks(enclosures) needed and days/paddock (rotational)



- Temporary fence works well to form paddocks

Sacrifice Area

- This is a part of your pasture system that, just as it sounds, is permitted to become trashed.
- What is important here is that the trashing is confined to one small area where the mess can be controlled.
- Animals are kept in here during periods **(oestrus cycle) (i.e. wet)** when it is not fit to put anim"als in the pasture.

193 Paddocks and Fencing

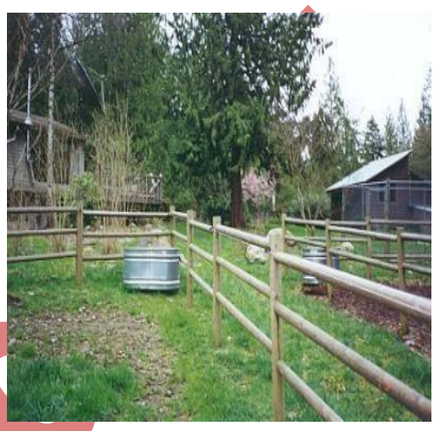
WHAT IS PADDOCKS?

Paddocks

- Sacrifice area
- Turn out lots

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- In a rotational grazing system pastures are divided up into smaller units within the pasture o these smaller units are called paddocks.
- In some smaller operations, permanent fencing is used to divide up the pasture.
- Temporary electric fence is a low cost, effective method of creating paddocks.



Fencing (In agriculture, fences are **used to keep animals in or out of an area**. They can be made from a wide variety of materials, depending on terrain, location and animals to be confined. ... A fencerow is the strip of land by a fence that is left uncultivated.)

- **Considerations**
  a) Safety
  b) Efficiency
  c) Cost
  d) Aesthetics

Fence height should be a minimum of 5 feet.

Fencing Materials

Wood Fencing

a) Different types (3 rail, spilt rail)
b) Low Maintenance

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c) Expensive

d) 20-25 years life expectancy

Other

- PVC
- Plastic grid/mesh
- Electric Tape

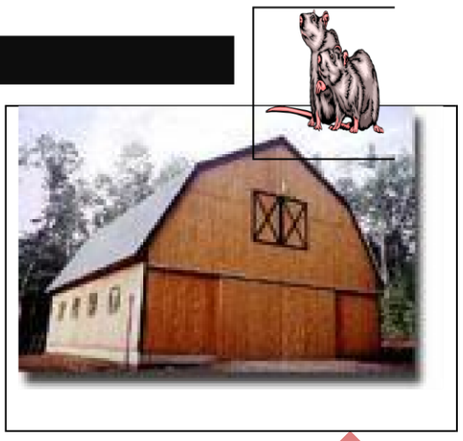
Wire Fencing

 Different Types (board and wire, high tensile, electric, V-mesh) Less expensive

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Maintenance is low to medium
Extended life expectancy

194 Facility Requirements



Facility Requirements

- **Things to think about**
  a) Water
  b) Air/Ventilation
  c) Space Requirements
  d) Shelter

Animal behavior

Animal Behavior

- **Causes**
  - **Genetic**
  - **Simple learning (training or experience)**
    a) Habituation
    b) Conditioning
    c) Reasoning, Insight
    d) Imprinting, Socialization
  - Complex learning (intelligence)

196 Pests and Their Management

WHAT DO YOU KNOW ABOUT Pest Management ?

Pest Control Is An Everlasting Problem

 How does an organism become labeled as a pest species?

 Pest control: winning the battles but losing the war. Designing better mouse traps.

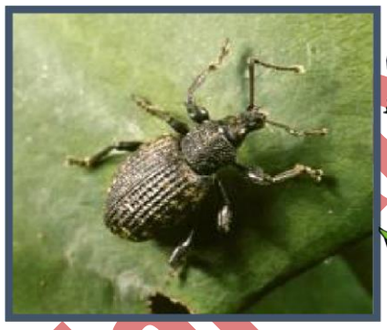
What is a Pest?

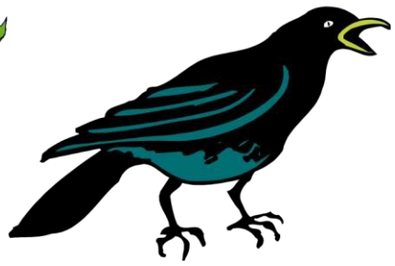
- **Technically,** any organism (bacteria, fungi, plant, animal) that has a negative effect on **human health** or economics (food).

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- Realistically, any organism we don't want around (factors in convenience and esthetics).
- So generally speaking any organism that is detrimental to humans
  - destroys crops & structures
  - poses threats to human health and livestock
  - reduces aesthetic and recreational value
- **Pests** include insects, mites, plant pathogens, weeds, mollusks, fish, birds, and mammals

What is a PEST? **A pest is any organism that is unwanted by humans. Pests may be unwanted due to their destructive nature, the health concerns they pose, or because they're plain annoying.** Insects are not the only kind of pest. Mites, plant pathogens, weeds, mollusks, fish, birds, and even some mammals can be pests. They can be found in waterways, agriculture, landscapes, roadsides, natural areas, stored food, schools, hospitals and even swimming pools.





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197 History Lesson

History Lesson: What is Bubonic Plague ?

Describe the history of Bubonic Plague ?

- **14th Century Europe**: mysterious scourge kills millions
- Centuries later it was found that rat fleas became infected with disease-causing bacteria
- Fleas sought other warm-blooded hosts (humans) when rat numbers declined

- Plague is currently managed monitoring for plague and reducing the number of rodent hosts for fleas

Do you remember anything from your history lessons about Black Plague in Europe?

During the 14th Century there were Plague outbreaks, millions of people died, and at the time it wasn't understood why. Centuries later, it was discovered that Plague was a bacterial disease and it was spread by fleas found in rodent populations. Besides biting rats, infected fleas would bite humans and other mammals. Bubonic plague is still present in some rodent populations today, even here in the United States. But wit h rodent management and plague monitoring, only a few people a year are infected with plague.

History Lesson: Potato Famine

- Late blight, a fungal disease, decimated Ireland's potato crops ✓ Thousands starved; over a million migrated to U.S.
- Today, late blight is still a major problem, but is managed by:
  - resistant cultivars
  - proper sanitation
  - fungicide applications

Plant diseases have been important in our history, too. Late blight of potato is a fungal disease that was responsible for the **Great Potato Famine in Ireland during the 19th century.** Thousands of Irish people starved and more than a million migrated to the United States. The late blight fungus continues to be a major problem of potatoes around the world, but is now

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managed through the use of resistant cultivars, proper sanitation practices, and fungicides.

History Lesson: Pest Control

- **Primitive:** pulling weeds, clubbing rats, plucking insects from foliage
- ➢ Sulfur burning for mites/insects: 2500 B.C.

➢ Lead arsenate in orchards - 1892

- Lime and copper sulfate – Bordeaux mixture
- Early pesticides – plant extracts or inorganics
- World War II: **DDT** and low cost synthetic chemistry

For your information:
DDT was used in the second half of World War II to limit the spread of the insect-born diseases malaria and typhus among civilians and troops. Müller was awarded the Nobel Prize in Physiology or Medicine in 1948 "for his discovery of the high efficiency of DDT as a contact poison against several arthropods".
DDT (dichloro-diphenyl-trichloroethane) was developed as the first of the modern synthetic insecticides in the 1940s.

Our ancestors controlled pests usually using **brute force and manual labor.** Thousands of years ago, chemicals that affected pests were discovered through trial and error. When sulfur was burned, it was found to affect insects and mites. Later several other inorganic materials, like lead and arsenate, were used as pesticides. Toxic plant extracts such as nicotine from tobacco were also used. Modern chemistry was used to develop pesticides during World War II. At the time, DDT was hailed as the **insecticide to solve all insect problems**. Countless other synthetic pesticides were produced. During this time, the modern chemical industry was launched. At relatively low costs,

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pesticides became a successful, primary means of pest control.

198 Concerns with Pesticide Dependence

Concerns with Pesticide Dependence

➢ Pest resistance

- **S.Q:DIFFERENCE BETWEEN Bioaccumulation**:AND Bio-magnification?

Environmental persistence

- **Bioaccumulation**: when a chemical accumulates in animal fat (historical fact)
- **Bio-magnification:** when an organism accumulates residues at higher concentrations than the organisms they consume

However, during the **1970's it became** obvious that relying completely on pesticides for pest management could have some negative impacts.

It was noticed that when certain pesticides were used repeatedly, some pests were no longer controlled by the pesticide. The pests had developed resistance to the chemical. **Since the 1970's, pest resistance** has surfaced in weed species, plant pathogens and insect pests. A prime example is the Colorado potato beetle which has developed resistance to every major group of insecticides, making the older chemical compounds nearly useless as pest management tools for Colorado potato beetle.

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In her book Silent Spring, Rachel Carson discussed environmental concerns about the stability and persistence of DDT and other chlorinated hydrocarbon insecticides.

DDT and its relative compounds were also found to be stored in **fat cells of insects, fish, birds, rodents and other animals.** DDT didn't break down **in fat cells**. This is known as **bioaccumulation**.

Some organisms accumulate chemical residues in higher concentrations than those found in their food source. This process is called **biomagnification**. For example, invertebrates with pesticides in their tissues are eaten by fish, which are then eaten by birds. The birds at the top of the food chain accumulate the highest concentration of pesticide residues. Pesticides that bioaccumulate and biomagnify have been removed from the marketplace.

SQ:LQ: Write a note on Pest Management ?

- Is the pest really causing the problem?
- **1st Step**: Always identify the pest before taking any action! Become familiar with its life cycle and habits
- Use the information to design a pest management plan
- Misidentification results in lack of knowledge = ineffective control of the real pest

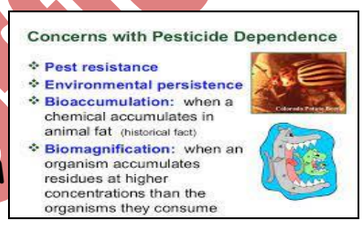
Lets talk about the principles of pest management now. In the old days, pest management may have meant simply using a pesticide to kill a pest. Due to the economics, environmental concerns, and regulations, pest management today involves a decisionmaking process and

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a carefully planned management approach. The first step in any pest management program is making sure you have properly identified the pest. Once the pest is properly identified, the pest manager can study the pest's life cycle, reproductive traits, and its growth habits. Understanding the pest's biology is critical when you are developing a pest management plan. Misidentification of the pest can result in ineffective pest management, wasting time and money.

Philosophies of Pest Control

WHAT IS CHEMICAL TECHNOLOGY?



Chemical technology

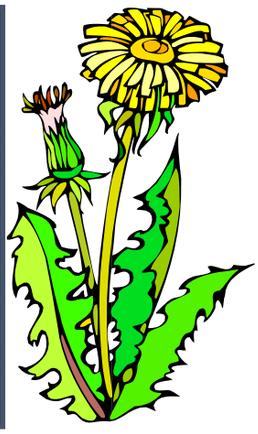
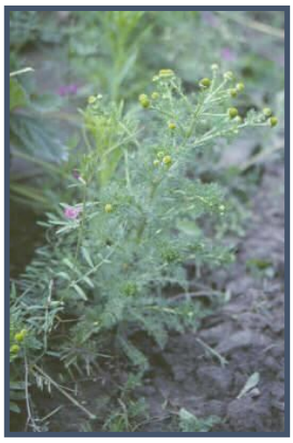
- Use of chemicals to kill large num pest
- Short-term protection
- Environmental and health consequences
- Ecological pest management
- Control based on pest life cycle and ecology
- Control agent may be an organism or chemical
- Specific to pest and/or manipulate a part of the ecosystem
- Emphasizes protection from pest

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199 Four Major Pest Categories

Four Major Pest Categories

1 - Weeds: undesirable plants

We have many types of pests: weeds, insects, fungi, birds, fish, algae, rodents, and plant diseases. For this presentation on pest management, we'll introduce you to the **4 major pest categories.**

Weeds are the most commonly targeted pests for control. A weed may compete with your crop, cause an injury on an athletic field, obstruct a roadway or utility powerline, or be unwanted for many other reasons.

S.Q: WHAT ARE INVERTIBRATES?-

2 **Invertebrates, such as:**

1. Insects
2. Spiders and mites
3. Sowbugs, pillbugs
4. Snails, slugs, and mussels

Invertebrate pests include the wide variety of insects like aphids, caterpillars, grubs, and wasps.

Invertebrates also include the insect-cousins: spiders and mites, sowbugs and pillbugs, and snails, slugs, and mussels.

Pictured here are a cutworm larva which is the immature stage of a moth, a plant-feeding mite as seen from under a

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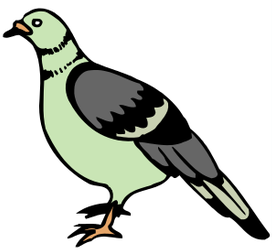
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microscope, and a slug working its way through a backyard lawn.





3 Vertebrates, such as:





4 - Plant Diseases

- **Pathogens – living agents**
    1. Fungi

1. Birds
2. Snakes
3. Fish
4. Rodents and other mammals

Vertebrate pests include all of those animals with a backbone. These include pest birds, snakes, fish, rodents, and other mammals like mice, rats, raccoons and deer.

2. Bacteria
3. Viruses
4. Nematodes
5. Phytoplasmas

- **Non-living agents:** cold, heat, pollutants, dog urine

Plant diseases are the fourth major category of pests. A plant disease can be caused by disease organisms called pathogens, or they can be caused by non-living agents.

Disease pathogens or living agents include fungi, bacteria, viruses, nematodes, and phytoplasmas. These organisms infect plants and often depend on specific

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host species to survive. Rose mosaic virus symptoms are pictured at the top. The bottom image shows fruiting bodies of the fungus that causes **Nectria twig blight** in some fruit and ornamental tree species.

It's important to remember that non-living agents including temperature extremes, air pollution, and a lack of or excess levels of nutrients, can also cause damage to desirable plants resulting in 'plant disease'.

200 Pest Identification is Critical

S.Q:HOW YOU IDENTIFY THE PESTS?

Pest Identification is Critical

- Understand that all stages of a pest do not look the same Know the host of the pest
- ➢ Use books, extension bulletins, field guides, Web, etc. Have pests examined by specialists
- Handle samples carefully

As mentioned earlier, it's important to correctly identify the pest in order to develop an effective management plan. Determining the cause of the problem or identifying the pest may be difficult. Immature stages of weeds and insects may not look at all like the mature stages.

Once you have properly identified the pest, learn about its biology. Resources that can help you identify and learn more about pests and host plants **include books,**

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extension bulletins, and field guides that contain pictures and biological information. Search the Internet. Illustrations of immature pest stages are helpful. Remember, weed seedlings and rosettes often don't resemble mature plants, and caterpillars don't resemble adult moths. If you can't identify the pest, take it to a specialist. When collecting specimens for identification, make sure you collect several specimens and store them in suitable containers. Don't forget to record important specimen information like the host plant and location where the specimen was collected.

WHAT ARE CHARACTERISTICS SIGN?

Look for Characteristic Signs

1. **Birds and rodents:** unique nests
2. **Insects:** feeding damage
3. Fecal materials are distinctive – insect frass or bat guano
4. **Weeds:** particular flowers, seeds, or unusual growth habits
5. **Pathogens:** unique patterns or growths on plant tissue

Pests often leave signs of their presence or damage patterns that provide clues to their identity. Birds and rodents build nests that are often unique to each species. Insects have characteristic feeding damage such as notching or skeletonizing of leaves or leaf mining. Rodents and moles dig distinctive burrows in the ground and may leave gnaw marks on tree trunks or other objects. Fecal materials, like the frass from insects, droppings from rodents, or guano from bats, are distinctive. Weeds may have easy-to-identify

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flowers, seeds, or growth habits. Fungi and other pathogens often cause specific types of plant discoloration, damage, or deformation.




201 Pest Control Methods

Natural Pest Controls

- Wind

SQ: What is natural pest controle?

Temperature

- Humidity, rain
- Rivers, lakes, mountains
- Pathogens, predators
- Food supply of the pest
- Cultural control
- Control by natural enemies
- Genetic control
- Natural chemical control

Natural controls are out of man's control – Mother nature dishes these out. They are without a doubt the leading factors that manage pest populations. Natural controls include wind, temperature extremes, humidity levels, and rainfall. A cold snap in the spring or a wet spring can largely reduce insect pest numbers for that season.

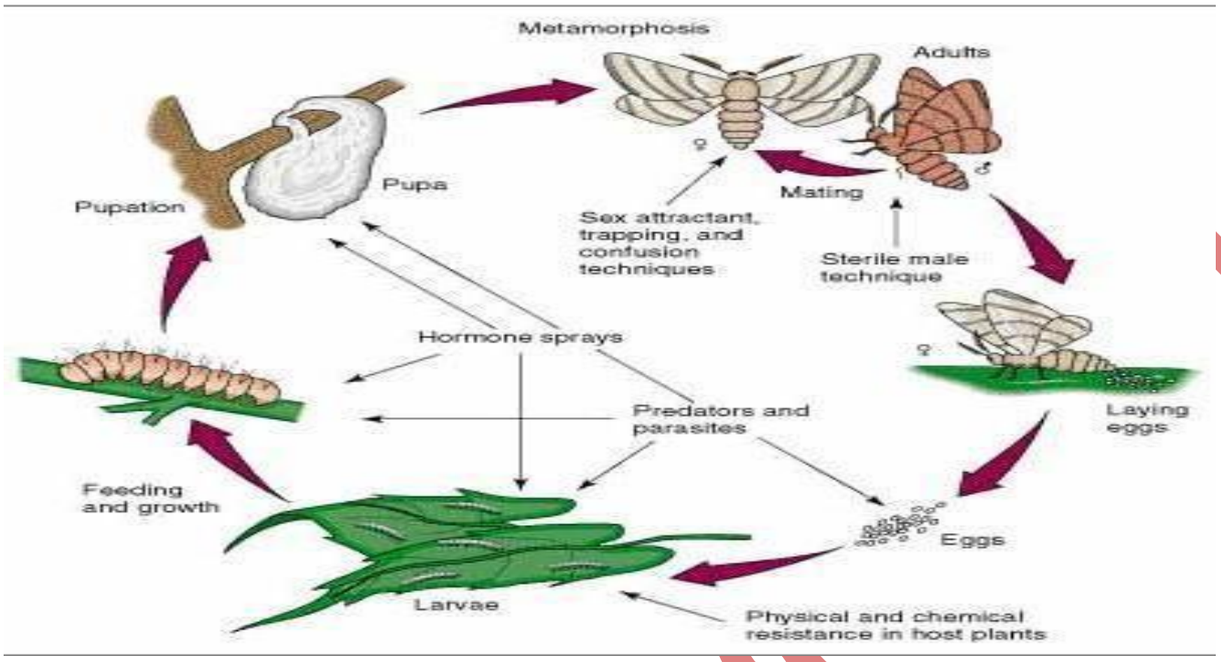
Rivers, lakes, and mountains provide significant barriers to pest movement and expansion. Naturally-occurring predators and pathogens hold down pest populations.

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The natural food supply of the pest also affects population density.

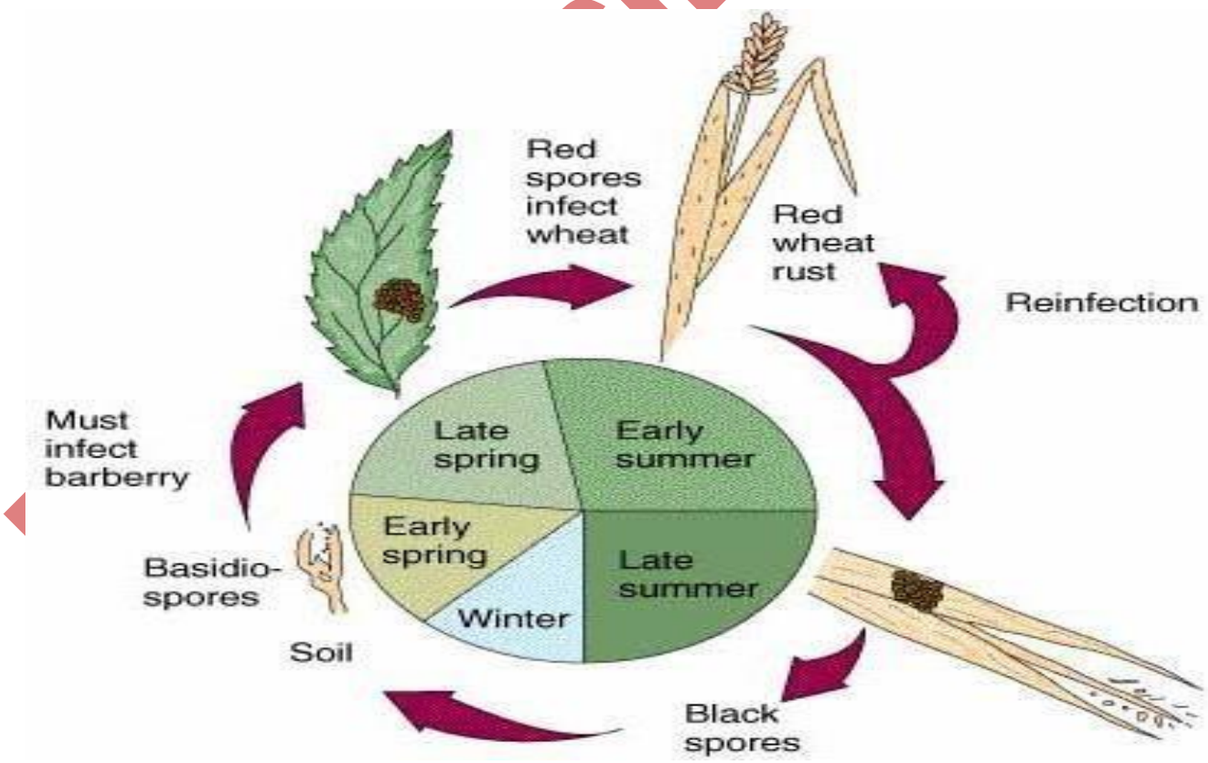
S.Q:WRITE LIFE CYCLE OF INSECTS?

Insect Life Cycle



Cultural Control

Get rid of the alternative host!

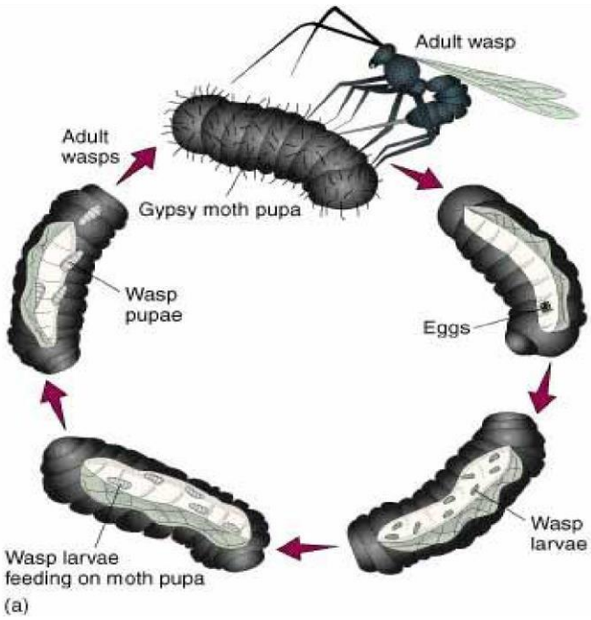


202 Control by Natural Enemies

SQ: How Control by Natural Enemies

Genetic Control

- Plants or animals are bred to be resistant to the attack of pests.



  - Chemical barriers. (Chemical barriers against infection include **enzymes in tears, saliva and mucus** that break down the surface of bacteria. The acid in sweat and in the stomach kills cellular pathogens and there are anti-bacterial proteins in semen (the fluid that contains male sperm).
  - Physical barriers. (Physical barriers are described as **structural obstructions in both humanmade and natural environments** that block effective communication so that messages cannot be sent from the sender to the receiver. Some of the physical barriers are noise, technological problems, and organizational environment)

- Introduction of genes into crops from other species: transgenic crops
- Sterile males are released into pest population.

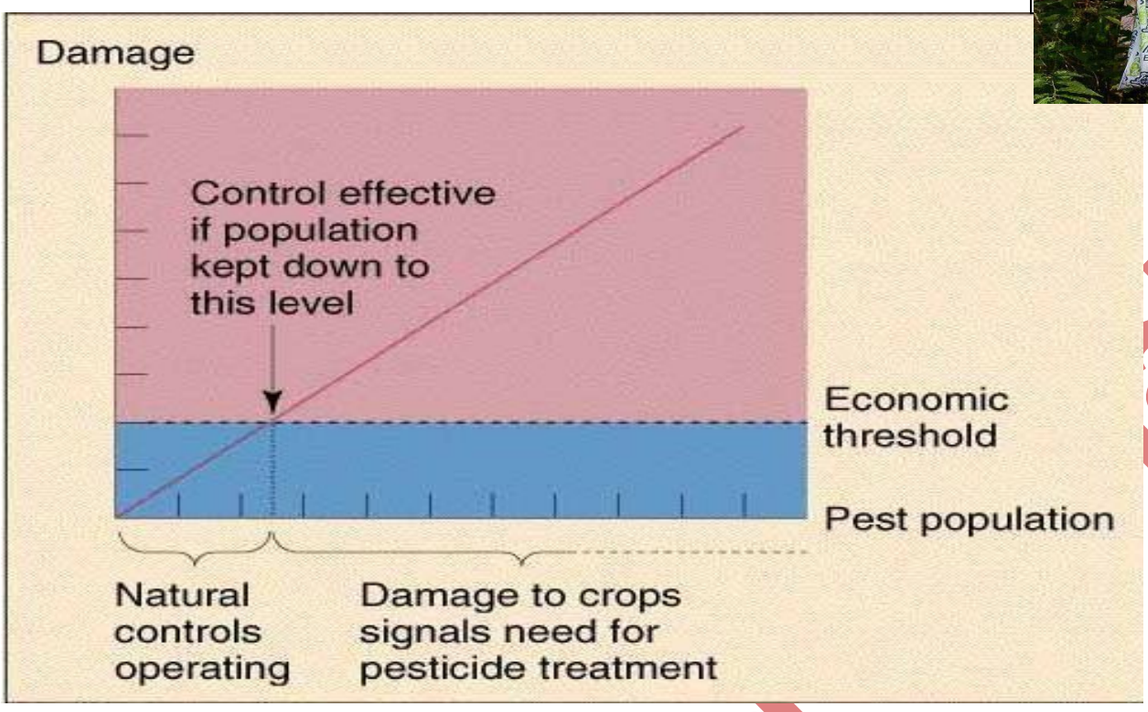
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Natural Chemical Control

- Manipulation of pests' hormones or pheromones to disrupt the life cycle.
- Japanese beetle trap.



Economic Threshold of Pest Control

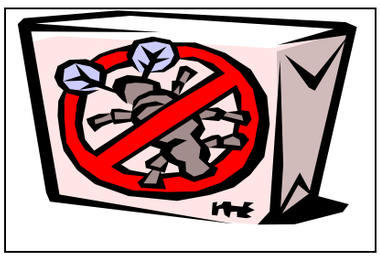


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203 Human Applied Controls

S.Q:HOW Human-applied Controls?

Human-applied Controls



1. Biological
2. Mechanical
3. Cultural
4. Physical
5. Genetic
6. Chemical
7. Regulatory

Sometimes natural controls don't provide sufficient suppression of pest numbers. Then WE get involved and implement control tactics to reduce the pest populations. Applied controls include biological, mechanical, cultural, physical, genetic, chemical, and regulatory methods. We'll look at each of these applied controls now.

What is Biological Control?





- Usually, pests are not native to area
- Locate pest's native homeland and find natural enemies
- Before releasing natural enemy, evaluate if suitable
  - ➢ Rear, release, redistribute

Biological control relies on the pest's natural enemies, such as birds, insects, fungi and viruses.
Applied biological control is often directed against pests that are not **native to a geographical area.** When a non-native pest invades, it often doesn't come with its complement of predators, parasites and pathogens. So scientists visit the native pest range, usually in Eurasia, and collect the natural enemies. They evaluate the suitability of releasing the natural enemies into areas where the new pest has become established. As the newly released natural enemy population grows, scientists redistribute these beneficial organisms to other sites where the pest has become established.

Biological Control Results

✓ Release natural enemies may become established and reduce infestation levels ✓ May not require any additional releases

BEWARE:

The cane toad was introduced in Australia in 1935 to control two pests of sugar cane, but later emerged as an invasive species itself!

In undisturbed natural areas, forests, rangeland, and other sites, biological control has a good chance of

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becoming self-sustaining; in other more disturbed sites like **landscapes and greenhouses**, the pest manager may need to release bio control agents regularly. Introduction of bio control agents is strictly controlled because of the concern for the bio control agent to run amuck. There are several instances where introduced bio control agents have become pests themselves, like the cane toad in Australia.

SQ: Using Biological Control

- Periodic mass release from cultures o Natural areas, greenhouses, orchards
- Recognize naturally-occurring organisms
- Manage to conserve native beneficials
- Avoid broad-spectrum insecticides eating a lygus bug.

➢ **Use non-chemical strategies**

Again, depending on the establishment level of biocontrol agents, periodic releases may be required in natural areas, greenhouses, landscapes, and agriculture. Any person considering a pesticide application should assess the population of beneficial organisms to determine if an application is necessary. Your management plan should be designed to conserve the natural enemies. Environmental modifications, such as planting cover crops and groundcovers, may enhance natural enemy populations.

If you have predatory or parasitic insects present, be very careful and selective in the type of insecticide you use. Broad spectrum insecticides often kill natural enemies. Some selective insecticides won't harm beneficials.

Consider non-chemical control options to ensure conservation of the natural enemies.

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204: Applied Control: Mechanical

S.Q:WHAT IS APPLIED CONTROL?

1. Applied Control: Mechanical

Use of devices, machines, and other

physical methods to reduce pest

populations or to alter the

environment

Mechanical control is everyone's favorite – stomp or squash that bug! Mechanical control involves using devices, machines, or other physical methods to reduce pest populations. You may also use mechanical means to alter the environment so that it's unsuitable for pests to survive.

DESCRIBR MECHANICAL CULTIVATION?

2.Mechanical: Cultivation



Disrupt soil conditions for weeds and insects

1. Hoes
2. Plows
3. Disks

Control growth or destroy plants

- Mowers

Cultivation is one of the more common forms of mechanical control. Hoes, plows, disks, mowers, cultivators, and bed conditioners are all mechanical devices that physically disrupt soil conditions. If you disturb the soil you can manage certain types of weeds,

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germinating weed seeds, and soil insects. Mowing can suppress certain types of weeds and destroy others

DESCRIBE MECHANICAL EXCLUSION?.

SQ: 3.Mechanical: Exclusion




• Prevent pests from entering or traveling

- Nets, screens, air curtains
- Caulking, steel wool
- Metal tree collars
- Sticky materials
- Sharp objects

Exclusion is the use of a device that blocks the entry of pests into a specific area. This may be a screen on a window to keep out mosquitoes, or caulking to keep rodents and ants out of structures. Sealing cracks and crevices keeps insects, rodents, bats, and birds out of buildings. Metal tree collars can keep squirrels from climbing trees and sticky material painted onto trees, posts, and wires provides a barrier that many insect pests won't cross. Using sharp objects on roosting areas prevents birds from congregating.

SQ: WHAT IS MECHANICAL TRAPPING?

4.Mechanical: Trapping



- Use of mechanical or sticky device
- Captures pests in a holding device
  - Restrains the pest
  - Kills the pest

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Certain **rodent and insect pests** may be successfully managed with trapping. The trap may consist of a holding device like a snare or leg-hold trap. Some traps capture live animals for **relocation or euthanasia**. Traps that have a sticky surface don't allow the insects or rodents that come into contact with them to escape.

205 Applied Control Cultural

WHAT DO YOU KNOW ABOUT CULTURAL?

Applied Control: Cultural

Alter conditions or pest behaviors



1. Mowing ➢ Tolerant crop varieties
2. Irrigation ➢ Planting timing
3. Aeration ➢ Crop rotation
4. Fertilization ➢ Trap crops
5. Mulching

Cultural controls alter the environment in which the pest lives. The environment may be changed so it is unfavorable for the pest or the environment may be altered so the host plant grows more healthfully and is less susceptible to attack by the pest. For example, when turf is managed properly through mowing, irrigation, aeration, and fertilization, it's less likely to have weed, insect or disease problems. Mulching in a landscape bed prevents light penetration and greatly reduces weed seed germination. Selecting crop cultivars

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or landscape plants that are not susceptible to pest pressures is a key factor in preventive pest control. Simply by changing planting dates you may significantly reduce pest problems. The pest may emerge prior to the host plant being present, it dies off and then the crop germinates. Or you many plant early to get the crop a head start so it can outcompete the pest. In agriculture, use of crop rotation and trap crops can reduce the pest pressure on your crops.

S.Q:WHAT IS SANITATION?

2. Applied Control: Cultural

- **Sanitation**: eliminate food, water, and shelter
    - destroy infected crop residues or infected ornamental plant materials
    - weed to reduce pest harborage
    - manage manure
    - seal garbage cans
    - remove soil near siding

A pest cannot survive if it doesn't have food, water and shelter. Practicing sanitation reduces the source of the pest or it eliminates food and water sources. If you have infected plant residues, they must be removed from the site and destroyed. Weedy areas near buildings, crofields, or landscapes provide areas for pests to hide. If rodents and some insects have access to food, like manure or garbage cans, they'll be present and reproduce. Having soil near siding provides habitat for certain structural pests. Learn about the habits of your key pests. Evaluate your area for places where sanitation can be effective in reducing pest numbers.

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Physical control:

Physical control is the altering of the physical environment – changing humidity, temperature, air flow, water and light.

Managing a pest may be as straightforward as pruning landscape plants for increased air flow and light penetration. More air movement and light in a plant's canopy reduces the amount of free moisture present and makes it unfavorable for pathogens. Lowering humidity and temperatures in stored grains prevents mold and insect infestations. Increasing air circulation in structures suppresses wood rot caused by fungal growth. Bright lights in attics are used to deter bats.

S.Q:WHAT DO YOU KNOW ABOUT Genetics or Host Resistance?

4.Applied Control: Genetics or Host Resistance

- Add or modify genetic material in crops and ornamental plants
- Breed or select plants for resistance

Plant breeding and selecting resistant crop or landscape cultivars can prevent pest problems. This is true for livestock as well.

Traditional breeding programs select for characteristics that prevent attack by diseases and parasites. Genetic manipulation is becoming more common to acquire plants that tolerate pest presence or crops and turf that can tolerate herbicide treatments.

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Resistance to Pesticides

- Chemical pesticides **lose effectiveness**
- Resistant pest populations produce next generations

207 Genetics of Pest Resistence

S.Q:WHAT IS Genetics of Pest Resistance?

Genetics of Pest Resistance

- There are two copies of each gene. Each gene may have several variations (called alleles).Some alleles are more dominant than others.
- Consider a gene for an insect's reaction to a pesticide. See how the alleles are distributed among offspring if susceptible bugs (RR) mates with a resistant bug (RR)

RR x rr Susceptible x Resistant	R Susceptible allele	R
r resistant allele	Rr Mildly resistant offspring	Rr
r	Rr	Rr

Populations will have different proportions of dominant and recessive traits depending on selective pressures in the environment.

Consider what happens if mildly resistant bugs (Rr) mate with each other; and then offspring are exposed to a dose of pesticide.

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Rr x Rr Heterozygous mildly resistant	R	r
R	RR dies	Rr May survive
r	Rr May survive	rr **SURVIVES!**

How many exposures does it take to make your population resistant?

Expose a population to a pesticide several times with mating in between exposures and see how many exposures it takes for resistant bugs to become the majority.

Sq: • Rules of the Game:

1. Toss your population (pennies) on the ground, and consider all with painted sides showing as exposed to pesticide
2. All exposed yellow (RR = susceptible) die so remove these pennies.
3. All exposed red (rr = resistant) survive and double; add a red penny for each exposed red penny.
4. For every three exposed blue (mild resistance) add one of each color. (Keep track of any remaining blues (<3) to ad to those on the next throw)
5. Keep track of the exposure count and repeat until most pennies are red.

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WHAT IS Applied Control: Chemical?

5. Applied Control: Chemical

- **Sq: mcq: Pesticide: any material that is applied to kill, attract, repel, or regulate pests**
  - Disinfectants, fungicides, herbicides, insecticides, repellents, defoliants, piscicides, etc.

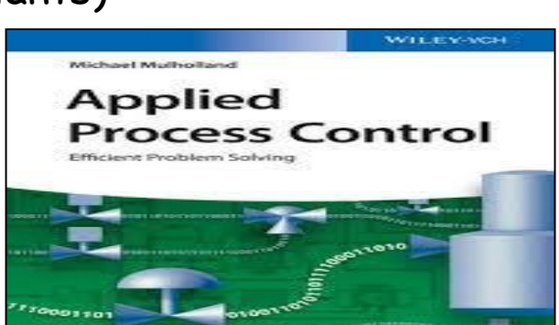
- **Advantages**: effective, fast, easy

Chemical control can be important to any IPM program when used as a preventive or curative method. A **pesticide** is defined as any material that's applied to plants, the soil, water, harvested crops, structures, clothing and furnishings, or animals to kill, attract, repel, or regulate or interrupt the growth and mating of pests, or to regulate plant growth. The broad category of chemicals termed pesticides includes disinfectants, fungicides, herbicides, insecticides, nematicides, repellants, defoliants, and desiccants. Unfortunately, people sometimes use the term pesticide and insecticide as synonyms when in fact an insecticide is only one type of pesticide. Pesticides can be quite effective, quick-acting, and relatively easy to use.

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Pesticides (Biocides)

1. Insecticides (insects)
2. Herbicides (not just the weedy plants)
3. Rodenticides (mammalicides)
4. Fungicides (mildews and rusts)
5. Acaricides (ticks and mites)



6. Bacteriocide (e.g. antibiotic)

The Early Years of Chemical Pest Control

- **First-generation pesticides (inorganic)**
  - First attempt at chemical technology
  - Included heavy metals **such as arsenic, copper and lead.**
  - Toxic to humans and agricultural plants.
  - Pests developed resistance.

209 Pesticide Improvements

SQ: Pesticide "Improvements"
(?)

- **Second-generation pesticides**
  - Organic chemical (organochlorines).
  - Used after WWII (presently in developing countries)
  - Synthesis begins with petroleum ("oil") ✓
    Mechanism of actions often unknown.
  - Bioaccumulation & Biomagnification.
  - Toxic to animals (humans) and agricultural plants. Pests developed resistance.

WHAT IS Smarter Pesticides (?)

- **Third-generation Pesticides**
  - Organophosphates and carbamates
  - Less persistent in environment (good deal)
  - Acutely potent nerve toxins
  - More lethal in low dose than organochlorines
  - Fourth-generation Pesticides

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- Endocrine disruptors (hormonal chaos) ✓
  Target a critical life cycle stage of insects.
- Not direct killers per say.
- Reduce reproduction (fertility) of population.

210 Chemical Technology Problems

WHAT PROBLEMS ARE OCCOUR IN CHEMICAL TECHNOLOGY

Chemical Technology Problems

- Development of resistance by pests
- Resurgences (pest comes back stronger)
- Secondary pest outbreaks (different pest)

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- Adverse human health effects
- Adverse environmental health effects

S.Q:HOW IT EFFECT HUMAN HEALTH?

Human Health Effects

- **Acute:** high dose, short-term response, rapid onset (headache, nausea, vomiting, respiratory failure, death). Agricultural workers suffer acute poisoning during pesticide application.
- **Chronic:** low-dose, long-term exposure, outcome takes many years before noticed (cancer, dermatitis, neurological disorder, birth defects, sterility, endocrine system disruption, immune system depression). Neighborhoods downwind of agricultural use; farm families; the innocent.

HOW IT EFFECTS ENVIRONMENT?

Environmental Effects

- **Bio-concentration:**
  - Movement against a concentration gradient; typically fat soluble.
- **Bio-magnification:**
  - Movement through the food chain to higher trophic levels; typically persistent.
- **Bioaccumulation:**
  - Combined effect of both; chemicals are typically fat soluble and persistent.

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- **Resurgences**: after "eliminating" a pest, its population rebounds in even higher numbers than previous levels. Why?

- **Secondary outbreaks**: outbreaks of species' populations that were not previously at pest levels. Why?

Think about mechanisms of environmental resistance on any one population.

211 Pesticides Vary By

Sq: Mode of action: how they work to control the pest

- Systemic pesticides are absorbed through tissues and transported elsewhere where the pest encounters it through feeding
- Used on plants or livestock
- **Contact** pesticides must come in direct contact with the target pest

Pesticides vary in how they work to actually control the pest. We call this the mode of action – how they kill or control the pest.

Some pesticides are systemic and actually move through animal or plant tissues and the pest encounters it when feeding on the animal or plant.

Other pesticides work by contact action like battery acid on your pants. You must make sure that you apply the material so that it comes into direct contact with the pest.

Selectivity: **what range of pests they affect?**

- **Non-selective** – kills all related pests – for example some herbicides kill all green plant that gets a sufficient dose

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- **Selective** – kills only certain weeds, insects, plant pathogens – for example other herbicides only kill broadleaf weeds not grasses

Pesticides also vary in their selectivity or how many different pest species they target.

Some affect all related organisms equally and are called non-selective. A good example is glyphosate, the active ingredient in Roundup Original. Glyphosate is non-selective when applied to green plants. ANY green plant that gets a sufficient dose will die whether it's a broadleaf or a grass plant.

In contrast, a selective pesticide only kills a small group of related organisms and does not harm others. The herbicide **2,4-D is a good** example. It only kills broadleaf plants and does not harm grass plants when applied according to label directions.

2,4-Dichlorophenoxyacetic acid

A 2,4-D stands for **2,4-Dichlorophenoxyacetic acid**. It is a plant growth regulator. It also acts as herbicide.

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WHAT IS PERSISTENCE?

Persistence: how long they remain active in the environment

- **Residual pesticides** – remain active for weeks, months, years – for example herbicides used around road guard rails

- **Non-residual** – inactivated immediately or within a few days – for example – some herbicides do not remain active in the soil once applied

Persistence is the term used for how long a pesticide remains present in the environment. DDT, which is no longer used, was a very persistent insecticide that's still found in the environment today. Many of the new products break down in a matter of hours or a few weeks. There are some persistent materials on the market today; we call them residual insecticides and herbicides and use them where we want long-term control.

Applications of residual herbicides are made to rights-of-ways so repeated applications are not required throughout the year.

Non-residual materials are also available, they break down very quickly.

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212 Regulatory Pest Control

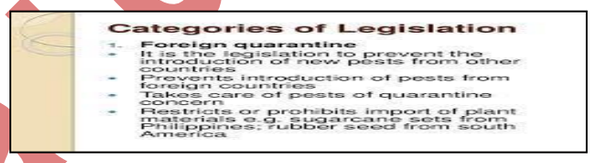
WHAT IS Regulatory Pest Control?

- **Quarantine prevents** pests from entry to an area or movement from infested areas.
  - Monitor airports, ocean ports, borders
  - Nursery stocks and other plant materials
- **Eradication programs** eliminate a pest from a defined area
- Mosquito Abatement used for public health

The government uses several regulatory control programs to manage pests. Quarantines are used at

airports and ocean ports to prevent entry of foreign pests. State borders and areas within states may be under a quarantine status to prevent the movement of pests from an infected area. Nursery stock and other plant material may be regulated to ensure no pests are transported on plant material. Federal and state governmental agencies may be involved in small to very large scale eradication programs with the goal being the total elimination of a pest from a defined area. Regulatory mosquito abatement districts and programs are established to manage for large-scale public health concerns.

SQ: LQ: what is Integrated Pest Management IPM: a balanced, tactical approach



- Anticipates and prevents damage
- Uses several tactics in combination
- Improves effectiveness, reduces side effects
- Relies on identification, measurement, assessment, and knowledge

Integrated pest management or IPM is a decision-based management style that implements a balanced, tactical approach. The decision-making is based on planning to anticipate damage to prevent it from occurring and if damage thresholds are met, using a variety of control tactics to suppress the pest population. When using a decision-based system, you increase effectiveness and reduce the negative side effects. IPM is dependent on proper identification and understanding of the pest

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biology along with proper measurement of population densities and assessment of control tactics.

213 Why Practice IPM

Sq: Why Practice IPM?

- Maintains balanced ecosystems
- Pesticides alone may be ineffective
- Promotes a healthy environment
- Saves money
- Maintains a good public image

The shift from using only a single control tactic to IPM provides for a more balanced ecosystem, since factors other than damage alone are part of the decision-making. We know that continued use of pesticides, without thought to resistance management and effects on the environment, can result in the product becoming less effective. Using IPM promotes a healthy environment…and it saves you money over time.

WRITE ,METHODS OF CONSIDERATION?

Considerations for Choosing Control Methods

- Determine damage level you can withstand
- Determine desired control outcomes
  - Prevention of pest outbreaks
  - Suppression to acceptable level
  - Eradication of all pest organisms
- Manage for pesticide resistance
- Estimate costs
  - Monetary

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- Environmental impacts

Management is decision-based. Once the pest is identified and its biology reviewed, assess the amount of damage you can tolerate. In some cases, this may be the presence of only one pest organism, in others you may be able to tolerate **100s of individuals**. Next, set your goal for control. Are there strategies you can use to prevent the pest from becoming a problem? Can you tolerate to just knock the pest population back a little – suppress it, or do you actually need total elimination or eradication of the pest organism? This may depend on agricultural markets or a customer's demand for a pest-free environment. When selecting management tactics, most specifically when selecting pesticides, consider pesticide resistance. Is it wise to go back and treat with the same chemical that you have used the last few treatments – or should you change product types?

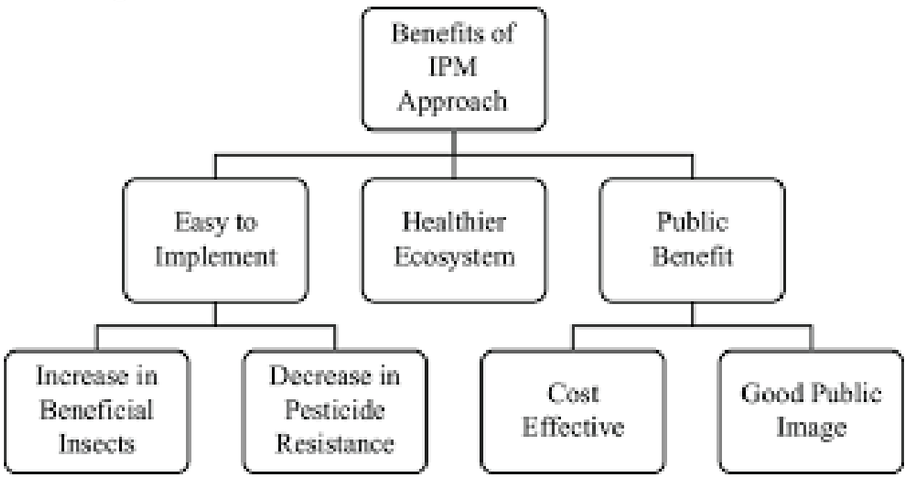
Ultimately you'll want to consider the cost to your pocketbook and the impacts on the environment.

Integrated Pest Management is Driven by Decisions

1. Identify the pest and know its biology
2. Monitor and survey for pests
3. Set IPM goal: prevent, suppress, eradicate
4. Implement
    1) Select control strategies

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2) Timing



3) Economics

4) Environmental impacts

5) Regulatory restrictions

5. Evaluate

Here are the major factors that drive the decision-making within an IPM program.

First and always foremost is pest identification and understanding.

Once you know what, when and where about the pest – you can develop a monitoring system using several survey techniques.

You need to set your management goal and decide what level of control is acceptable for your situation – a golf green will have a different goal than a fairway.

Once you set your goals and understand the pest, you can look at the factors that impact your selection of control tactics. These factors may include timing, economics, environmental considerations and regulatory constraints.

Don't forget to evaluate the entire process and record your findings. You can then tweak the plan in future years once you understand your successes and failures.

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Lets look at some of these considerations in a little more detail.

214 Components of IPM

WHAT ARE Components of IPM?

1. Identify and Understand

- Is it a pest, beneficial, or just there?
- Study pest biology
  - Pest classification
  - Life cycle
  - Over-wintering stage
  - Damage impacts
  - Environmental needs
  - Vulnerable control stages/timing

Pest identification is critical. It's the most important step. You must know what pest is causing the problem, or what pest may potentially cause a problem. Often an insect or plant may be present, but it poses no threat whatsoever and requires no action. Some may even be beneficial organisms.

Once you have identified the pest, take some time investigating its classification and life cycle. They are key to mechanical, cultural, and chemical management options. Knowing the stage of the pest that overwinters may provide a key as to the timing for control measures to be taken.

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Investigation is critical to understand what pest population levels trigger concern and what levels trigger action.

Having some knowledge of the pest and its environmental needs may shed light on what population levels may be anticipated due to the environmental conditions. Did the cold winter knock the population back? Did the early warm spring allow for another generation to be produced?

Knowledge of the life cycle may provide the key to timing your control activities to the pest's most vulnerable stage.

WHAT IS KEY PESTS?

- **Key pests**
  - Prior knowledge of which common pests may pose a problem
  - Recognition of damage symptoms
  - Recognition of diseases
  - Recognition of beneficial insects
  - Frequent monitoring

Using the concept of key hosts and key pests is a good way to start your IPM planning. Know what crop you are growing or what landscape plants you have growing or plan to grow. Do some reading and go to some seminars to learn what pests are the most common and most damaging for the plants you grow. Know what to look for and when to monitor.

What symptoms does the plant express when the pest feeds on it? Can you recognize them? What symptoms does the plant express when it's infected by a disease?

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What do the common and not so common beneficial predators and parasites look like that are associated with the pest.

DIFFERENCE BETWEEN OCCASIONAL PESTS AND SECONDARY PESTS?

Occasional pests may become troublesome from time to time

Secondary pests become problems when key pests are controlled or

eliminated – such as spider mites

Other than the key pests, have a working knowledge of the occasional pests that appear every so often, such as an ant infestation. Their appearance may depend on the weather or even your use of chemicals.

Be aware that when you suppress the key pest, another low-level pest may become very problematic. We call these secondary pests. For example if you manage for codling moth in apples with a broad-spectrum insecticide, the insecticide will kill beneficial predatory mites as well. This sets you up for an explosion of the secondary, plant-feeding spider mites.

HOW WE MONITOR THE PESTS?

2. Monitor the Pest

- Use scouting, trapping, weather data, models

- Economics or aesthetics trigger need for action

How many **pests need to be present before action**

 Pest population

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Beneficial population

 Geographic location Plant variety is taken?

 Plant type & stage of growth

 Cost of control measure(s)

 Value of plant or crop

If you have a working knowledge of the pest, you can learn about scouting and trapping techniques. Some scouting is dependent on weather data since growth of pest populations depends on temperature. Several factors will play into the decision when evaluating the trigger levels for control action. These may be based on the economics of lost yield and harvest costs, or they may simply be based on aesthetics. You have to know the answer to the question – how many pest numbers need be present before control action is warranted?

- **SQ: Define Action threshold:** unacceptable pest level – do something
- Sometimes the action threshold may be zero!
- Action thresholds vary by pest, site, and season

6 aphids per wheat plant = no problem - no action

15 aphids per wheat plant = hits the pocketbook take action

The action threshold is that pest number that triggers you to initiate corrective measures. Those corrective measures may be mechanical, cultural, or chemical. In agriculture, the action threshold is usually well-defined based on economics. Depending on the pest species, the site, or even the season, the trigger level may differ. At six aphids on a wheat plant, the plants can tolerate the feeding and the loss of yield does not approach the cost

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to control the aphids. However, once the aphid population **builds to 15 aphids** per plant, it makes good economic sense to take action to hold down your losses.

215 Treatments of Action Threshold

S.Q:WHAT IS Treatment or Action Threshold ?

- **Economic Threshold**
  - pest population density when *control is necessary* to prevent economic injury
- **Economic Injury Level**
  - when the cost of losses equals the cost of control measures
  - Apply control measure prior to reaching economic injury level

In agriculture we base these action triggers on economics. The 'economic threshold' is when the pest population reaches a specific density that causes significant damage and if not held in check, 'economic injury' will be realized. **'Economic injury' is a threshold of pest number just above the 'economic threshold'.** At this point, the cost of control and the cost of the damage are equal. If you did not start control measures at the lower, economic threshold, you will end up with damage that exceeds the costs of control.

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WRITE Components of IPM?

2. Monitor the Pest

Action Threshold is also based on aesthetics or public health issues At what point does the cost of control ward off future expenses

For turf, ornamentals, and even public health, we simply call the threshold an action threshold. Ask yourself, at what point does the cost of control ward off future expenses or unacceptable damage levels? If you manage turf on a golf course green your tolerance for weeds is very different than for weed levels in the fairways and your action threshold is different.

Components of IPM

3. Develop the IPM Goal

- **Prevention**: weed-free seed, resistant plants, sanitation, exclusion, pesticide treatments
- **Suppression=reduction** cultivation, biological control, pesticides
- **Eradication=elimination** small, confined areas, or government programs

You need to set a goal. Prevention is an important goal. During your planning period for crop production or prior to seeding or establishing an ornamental bed or turfgrass, consider what actions you can employ to prevent pests from becoming established. Consider planting certified weed-free seed or use resistant plant cultivars. Build exclusionary devices to keep pests out. If you had a previous infestation, consider sanitation measures to remove infected source material. You may even want to consider some preventive pesticide

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treatments like using treated seed or using a preemergence herbicide application.

Again, prior to the growing season, set your pest management goal. Do you require total elimination of the pest, or would just suppressing the population offer sufficient protection? There are many tactics you can consider to hold a population in check.

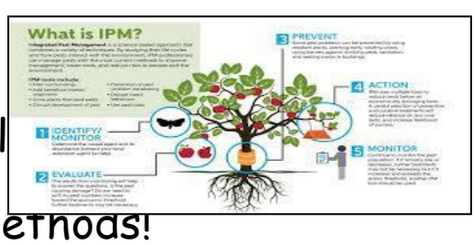
If your goal is eradication, you'll have to develop a very intense, methodical plan to ensure the total elimination of the species and all of its life stages. **Eradication is typically a goal when you have a very small, contained pest population or when the government is managing the program and they can employ wide-scale control activities on properties of private citizens.**

HOW WE. Implement the IPM Program?

Components of IPM

4. Implement the IPM Program

- Make sure you have taken initial steps
  - Identification and monitoring
  - Set action thresholds
  - Know what control strategies will



- Select effective and least harmful methods!
- Observe all local, state, federal regulations!

Always follow these steps before implementing an IPM program:

 Identify the pest and set up a monitoring program.

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- Know the economic or action threshold, as well as the economic injury level.
- Consider the available management options.
- Evaluate the benefits and risks of each method.
- And make sure your control strategy falls in step with any regulations.

WHAT DO YOU KNOW ABOUT .
Record and Evaluate Results?

Components of IPM

5. Record and Evaluate Results

- Know what worked and what did not
- Some aspects may be slow to yield results
- Might be ineffective or damaging to the target crop, beneficial insects, etc.
- Use gained knowledge in future planning efforts

A very important step in any IPM program is recordkeeping and evaluation of your efforts.

Keep track of climatic conditions. Record when and where actions were taken. Track the pest populations. Near the end of the season, review your records. Some control strategies may not be quick-acting, but may prove effective given sufficient time. Other management strategies may have caused negative impacts that need to be considered when making future decisions.

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216 Considerations for Pesticide Use

S.Q:WHAT DO YOU KNOW ABOUT Considerations for Pesticide Use?

- Identify the pest and select the appropriate product o old or new infestation
- Avoid developing resistant pest populations
- If using pesticides, use the correct application rate (dose) and timing

When you look to chemicals as a management option, there are several considerations. First make sure you have identified the pest. Make certain that the product you select is effective against the pest and is labeled for the site. If you have had a continual problem with a pest, your decision on a chemical may be slightly different than if you are targeting a new, invasive pest.

If you have used chemicals in the past to treat a pest population, consider using an alternative type of product to reduce the possibility of pesticide resistance. If you keep using the same product type repeatedly, year in and year out, resistance will likely develop.

Make sure you read the label carefully to understand the timing for the application and make sure you use the correct application rate. Using a dose that is over the label rate is illegal and can cause significant impacts. For example, using too much of a selective herbicide may result in damage to your crop or turf. Making an application with too low of a dose or at the wrong time can be a waste of money.

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Be A Professional IPM Practitioner

- Careful observation
- Knowledge of the pest, control options
- Professional attitude

With an IPM plan, you must keep your eyes open. Pay attention to all factors that contribute to pest populations. Be knowledgeable about the pest and the various management options you can employ when necessary. Be a professional – Be smart – Be careful -- and Be accurate.

Sq: Pesticide Resistance:the ability of a pest to tolerate a pesticide that once controlled it

- Intensive pesticide use kills susceptible pests in a population, leaving some resistant ones to reproduce
  - Use of similar modes of action
  - Frequency of applications
  - Persistence of the chemical
  - Pest rate of reproduction & offspring numbers

We've talked a little bit about pesticide resistance. Let's look at this phenomena that is a concern whether applying insecticides, herbicides, fungicides, or any type of pesticide. Any time you apply a pesticide, not all of the individuals in that population will be killed and removed. One in a million may resist the treatment. That one in a million is not obvious to the pest manager the first few years. But after repeated applications and the reproduction of the resistant individuals, a population of resistant pests becomes apparent. The chemical that historically controlled that population is now ineffective.

If you use products that have similar modes of action, those resistant individuals will not be controlled. So

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alternate between chemicals having different modes of action to manage for resistance. If pesticides are frequently applied during a growing season, resistance can show up more quickly.

Chemicals that are persistent tend to have resistance show up more readily than products that **don't remain active over time**.

The pest biology is also a **key factor**. If the pest has a high rate of reproduction and reproduces often, those resistant individuals show up more quickly as the population of those individuals increases. Also pests that have many offspring per generation tend to have a greater chance of resistance developing.

The pest manager must consider resistance management techniques when using pesticides in an IPM program.

WHAT IS RESISTANCE MANAGEMENT?

Resistance Management

- Do not use products repeatedly that have similar modes of action
- Allow some pests to survive o Limit treatment areas o Consider using lower dosages
- Use caution: new compounds having very specific actions - may develop resistance more quickly
- Use non-chemical means to control resistant

pest populations Again, to manage for pesticide resistance. Alternate your chemicals and don't use

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pesticides with similar modes of action repeatedly. Break the cycle.

Instead of treating an entire block, consider leaving an untreated section to allow those susceptible individuals to reproduce to keep the susceptible gene pool present. Using lower doses may increase survivorship.

If you use pesticides that are highly selective with specific modes of action, there's an increased concern for resistance management – again use alternate modes of action to break the cycle.

You can also use other methods of control to break the cycle and kill off the resistant individuals.

217 Arthropode Vector Biology

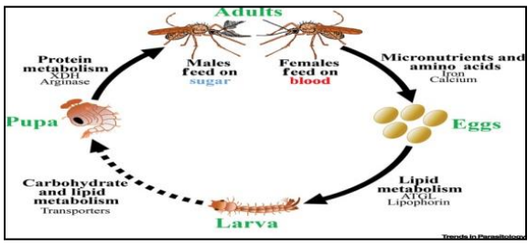
S.Q:WHAT IS MEDICAL ENTOMOLOGY?

- the study of diseases caused by arthropods

- Public health entomology - arthropods and human health
- Veterinary entomology - arthropods and pets, livestock and wildlife
- These fields of study are linked by the ecology of most arthropod transmitted pathogens and parasites.

Arthropods affect the health and well-being of humans and animals in several ways:

- Direct Causes of Disease or Distress
- Vectors or Hosts of Pathogenic Organisms
- Natural Enemies of other medically harmful insects



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WHAT ARE CAUSES OF DISEASES OR DISTRESS?

Direct Causes of Disease or Distress:

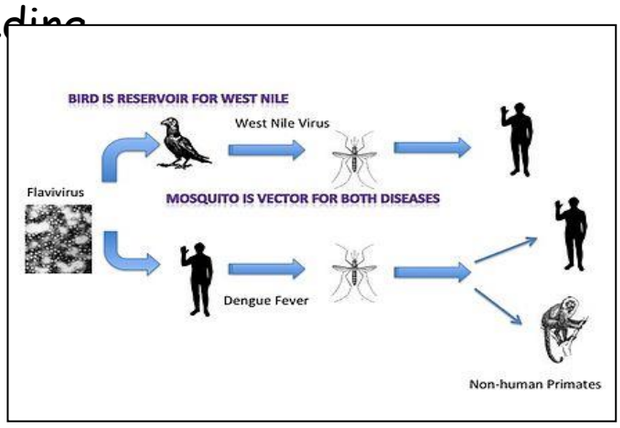
1) Ectoparasites - ticks, fleas, mites
2) Endoparasites - chigoe flea, myiasis
3) Envenomization - wasps, bees, spiders
4) Allergic Reactions - dust mites
5) (**Action**)Annoyance - mosquitoes, black flies
6) Delusory parasitosis (DP) - psychosis

Envenomation is **the process by which venom is injected into some animal by the bite of a venomous animal**.
Delusional parasitosis is **a psychiatric condition where people have the mistaken belief that they are parasitized by bugs, worms, or other creatures**.

218 Vectors or Hosts of Pathogenic Organisms

Vectors or Hosts of Pathogenic Organisms:

- Arthropod serves as intermediate host and vector of pathogenic microorganisms
- Vectors and hosts – blood feeding
- Hosts only - no blood feeding



Natural Enemies of other medically harmful insects:

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- Mites parasitic on mosquitoes
- Fire ants consume tick eggs

219 History of Medical Entomology

S.Q:WRITE History of Medical Entomolog?:

- References to associations between humans and arthropods – historical

(Homer and Aristotle, among others, wrote about the nuisance caused by flies, mosquitoes, lice and/or bedbugs.)

- **Important discoveries:**
  - Microscope - Leeuwenhoek 1700's
  - Infectious Disease - Koch et al. 1800's

History of Medical Entomology – 2:

- Mosquitoes (*Culex pipiens*) and filarial worms (*Wuchereria bancrofti*) - Manson, 1877
- Tick (*Boophilus annulatus*) and Texas cattle fever (piroplasmosis) transmission - Smith & Kilborne, 1891



ticks

- Mosquito (*Aedes aegypti*) and yellow fever virus - Finlay, Reed, Carroll, Agramonte and Lazear, 1900
- Trypanosomes in cattle blood - Bruce, 1895
- Tsetse fly (*Glossina sp.*) transmission of trypanosomes - Bruce, 1896

Official



tsetse fly

- Tsetse fly transmission of trypanosomes to humans (African Sleeping Sickness) - Bruce, 1903

History of Medical Entomology - 3:

- Malaria parasites in human blood - Laveran, 1894
- *Anopheles* mosquitoes with malaria parasites - Ross, 1897
- Transmission of bird malaria by *Culex* mosquitoes - Ross, 1898
- Complete development of human malaria parasite in mosquitoes - Grassi, 1898
- Transmission of human malarial parasite by mosquitoes - Sambon and Low, 1899
- Only *Anopheles* mosquitoes transmit human malarial parasites - Watson and Christophers, 1899

History of Medical Entomology - 4:

- Mosquito transmission of dengue virus - Graham, 1902
- Fleas and plague - Liston, Verjbitski et al., 1895 - 1910
- Triatomine bugs and trypanosomes (Chagas disease) - Chagas, 1908

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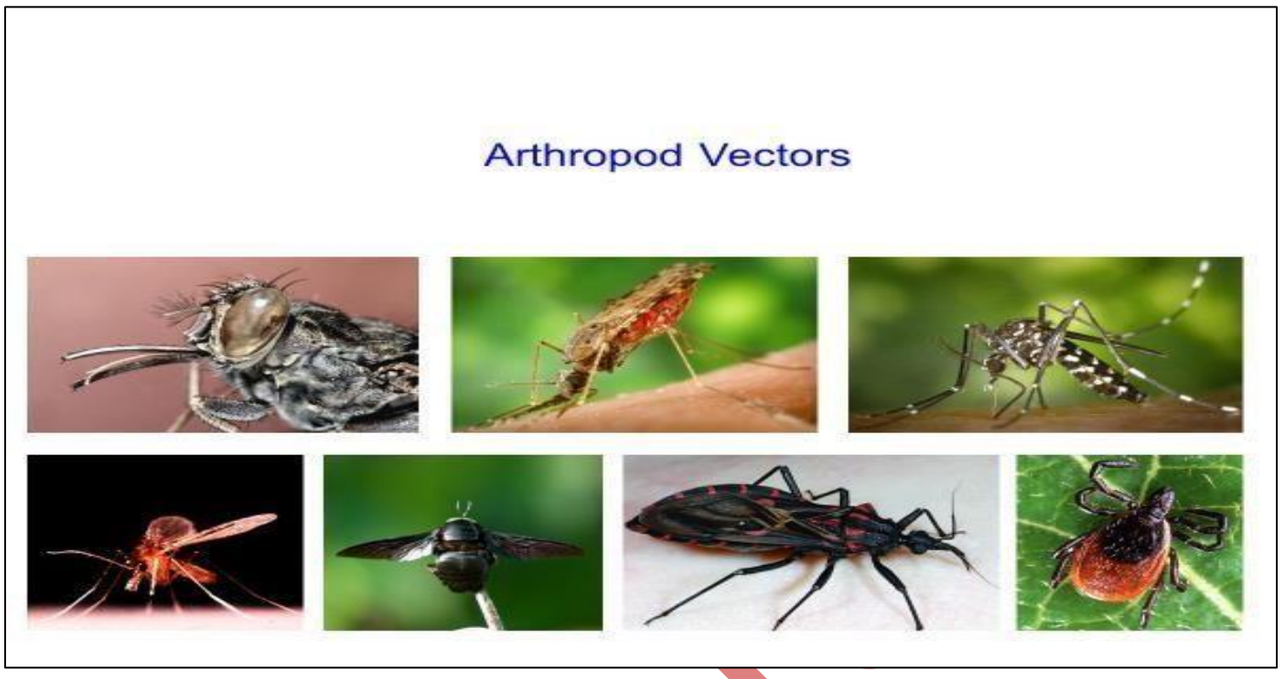
Triatomine bugs

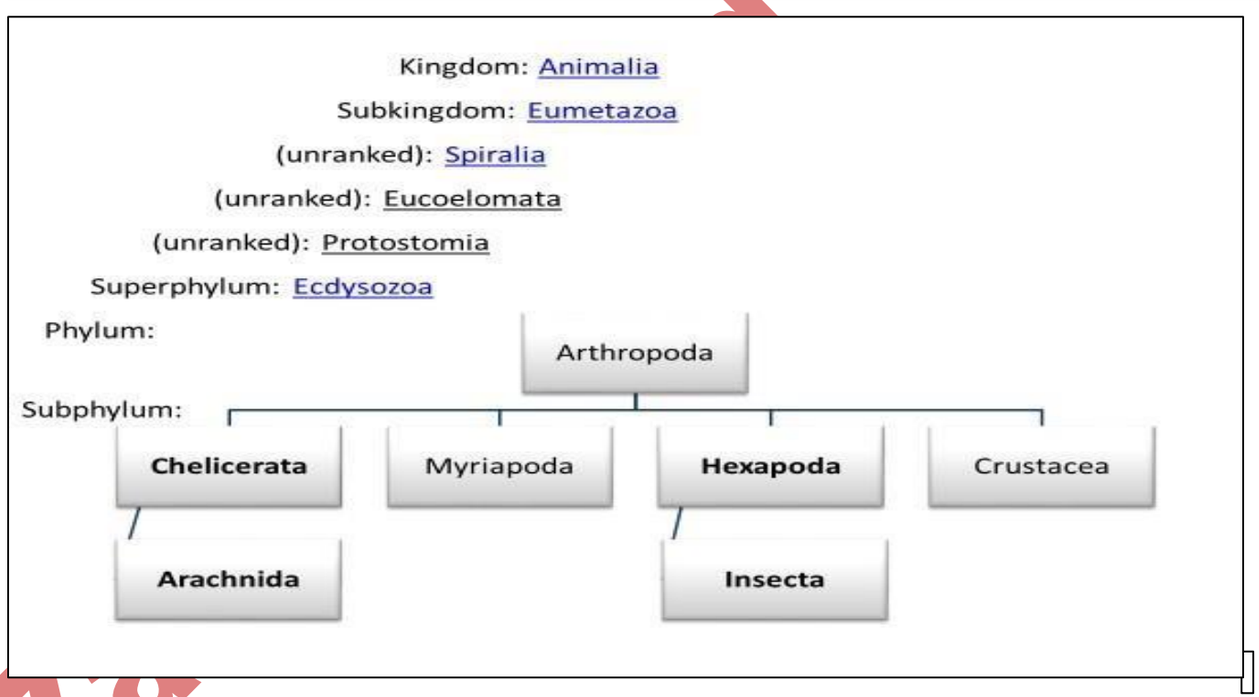
- Black flies and onchocerciasis (river blindness) - Blalock, 1926
- Mosquitoes and viral encephalitides - Hammon and Reeves, early 1940's
- Ticks and Lyme disease - Spielman, early 1960's

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220 Arthropods and Insects Taonomy and Characteristics

Arthropods and Insects Taxonomy and Characteristics





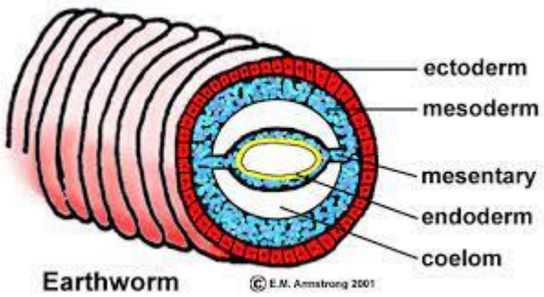
Write general characteristics of arthropods?

Arthropods

General Characteristics:

1) highly successful (⅔ of all known organisms are Arthropods)

1) segmented Eucoelomata with exoskeleton (Arthropod Metamerism)



2) Tagmata/Tagmatization (segments grouped into body regions)
3) Molting, Ecdysis (Ecdysozoa) **(the manner in which an animal routinely casts off a part of its body (often, but not always, an outer layer or covering))**
4) Hemimetabolous (gradual) / holometabolous (complete)
5) Cephalization (highly developed sensory organs)
6) Open circulatory system (hemolymph / hemocoel)
7) Respiratory system (trachea / book lungs)
8) Malpighian tubules / Coxal glands
9) Involved in virtually every kind of parasitic relationship ✓ definitive and intermediate hosts for protozoans, flatworms, nematodes,arthropods ✓ function as vectors (blood-feeding / need for protein for development)

For your information:
hemimetabolous **Describing insect development in which there is incomplete or partial metamorphosis**, **Holometabolism**, also called complete metamorphosis, is a form of insect development which includes four life stages: **egg, larva, pupa and imago or adult**

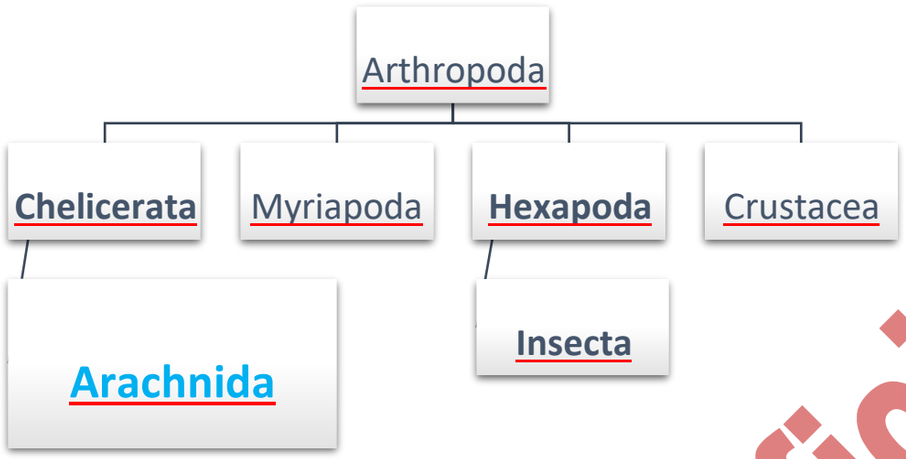
Malpighian tubules are **small slender tubes located posteriorly in the alimentary canals of arthropods**, such as insects, myriapods, arachnids, and tardigrades. The tubules are extensions of the alimentary canal to the exterior.

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221 Taxonomy and Characteristics

Class Arachnida:

- Prosoma (Cephalothroax) – Opisthosoma (Abdomen)

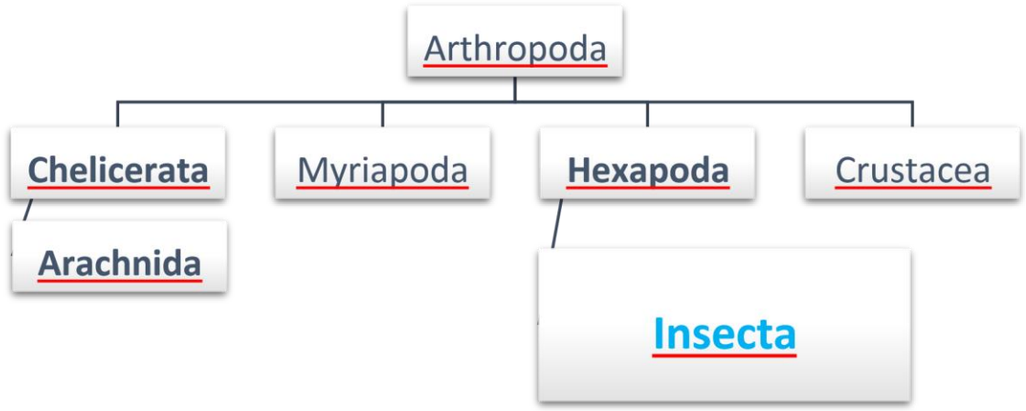


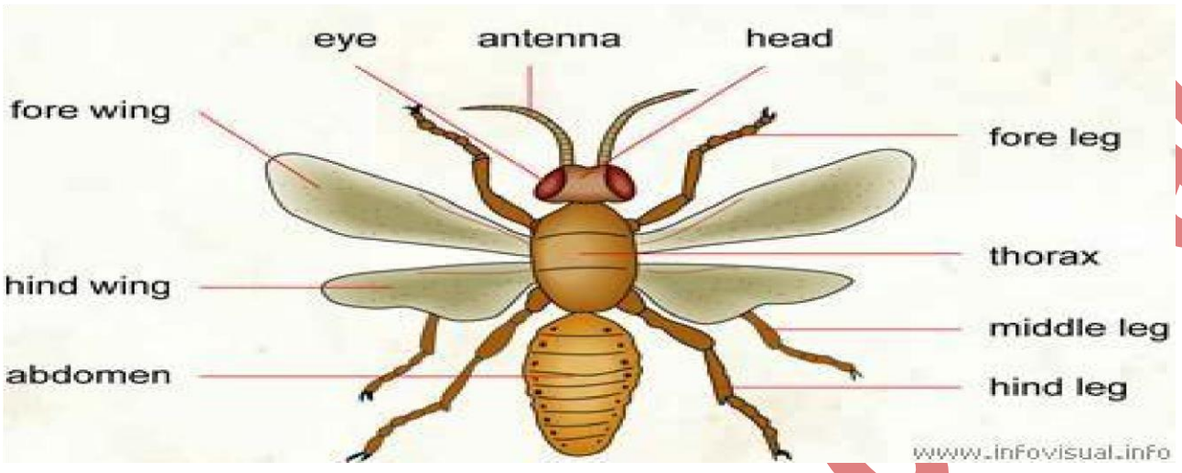
- Subclass
-
-
-
-

Acari: Capitulum (head) – Idiosoma; transmit a large number of disease agents (viruses, rickettsia, apicomplexa, nemotodes)

- Ticks are pool feeders (once attached to the host, the tick will inject saliva into the wound, which liquidizes the tissues and allows them to suck up the fluid)
- Ticks sense CO_2, warmth and movement from animals
- Ticks are divided into two families: Ixodidae (hard ticks) and Argasidae (soft ticks).

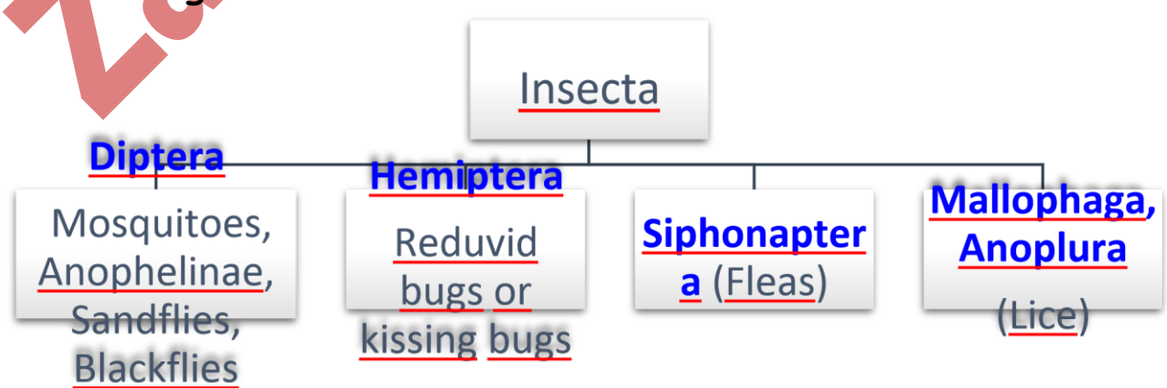
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- outnumber all other known animals put together (almost 1million species described)
- Head – Thorax – Abdomen
- eyes, antennae, sucking , chewing mouthparts, 2 pairs of wings, 3 pairs of legs
- can vector many diseases (mechanical / biological)
- Blood-feeding (need for protein), Flexibility in development, Response to host presence

  +/- low host specificity, can live near or in human dwellings…

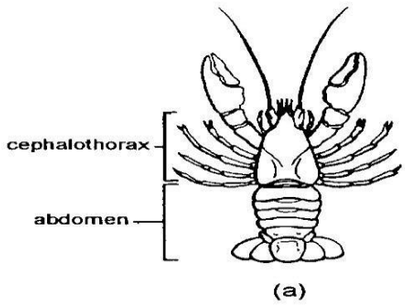


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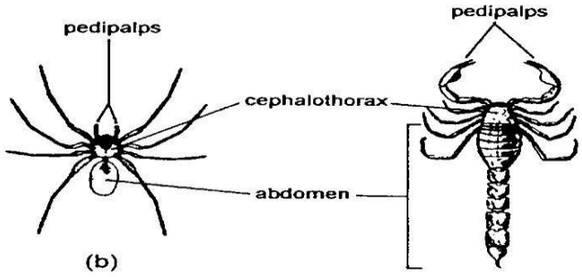
222 Arthropods Classes and Representatives

Phylum Arthropoda (classes and their Representative in Brief)

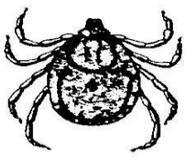
1) Class Crustacea - lobsters, crabs, etc.
2) Class Chelicerata - spiders, mites, ticks, scorpions, etc.
3) Class Diplopoda - millipedes
4) Class Chilopoda - centipedes
5) Class Insecta - beetles, flies, moths, etc.

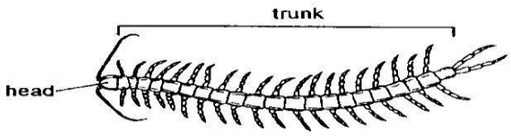


(a)

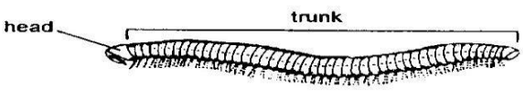


(b) (c)

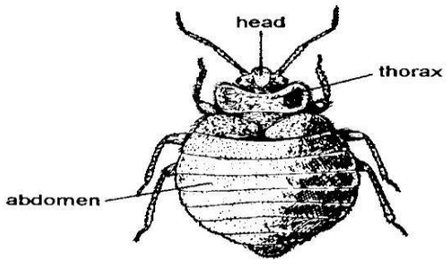

(d)



(e)



(f)



Sq: Phylum Arthropoda: Salient Features

- The phylum is probably monophyletic, but with 4 distinct groups.
- It includes lobsters, crabs, shrimp, centipedes, millipedes, daddy longlegs, insects, ticks, and mites, and spiders.
- There are over 1 million species of arthropods, making up the largest phylum in the animal kingdom.

Phylum Arthropoda - 2:

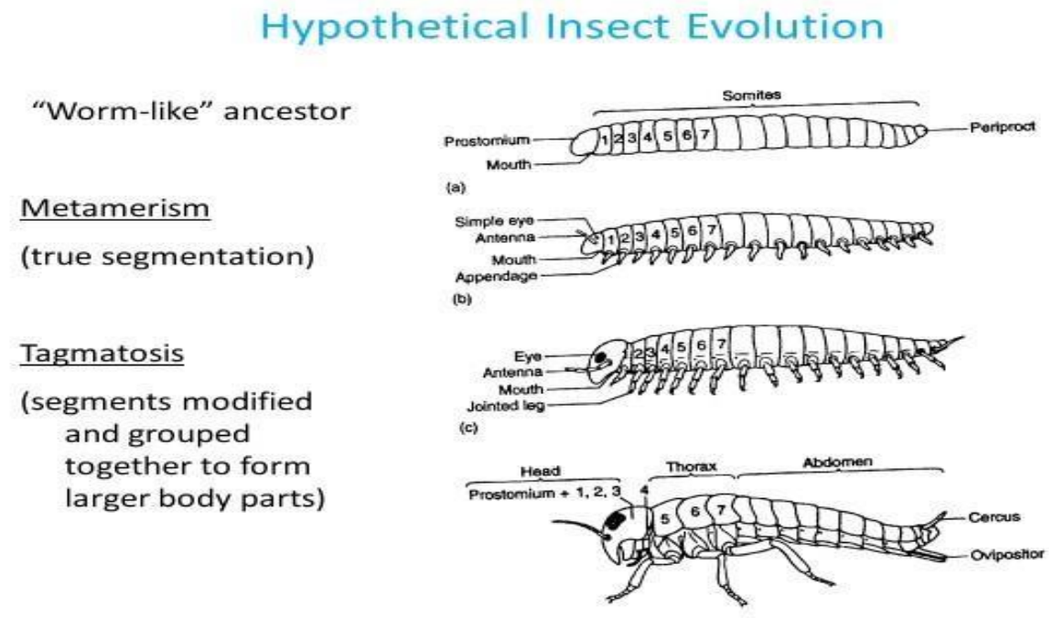
- Metameric - (body exhibits true segmentation - replication of muscles and nerves)
- Tagmatosis - (segments of the body are modified and grouped together to form mouthparts and body regions such as the thorax)
- Chitinous exoskeleton – nitrogenous polysaccharide

223 Phylum Arthropoda

What is Phylum Arthropoda?

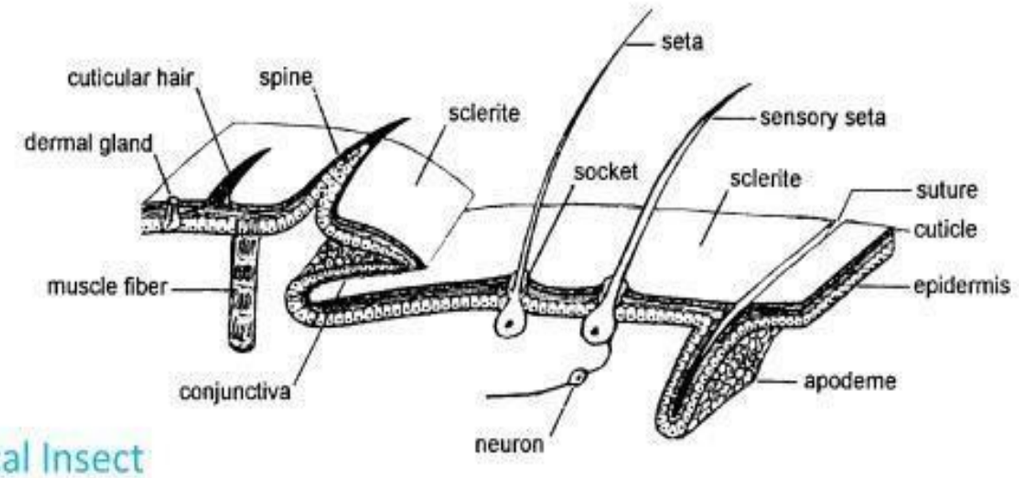
1) Bilaterally symmetrical
2) Jointed legs
3) Dorsal heart – open circulatory system
4) CNS (organized central nervous system)
5) Striated muscle

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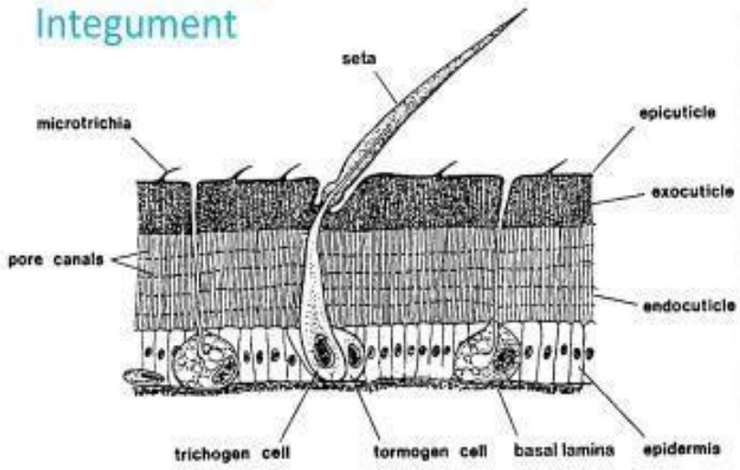


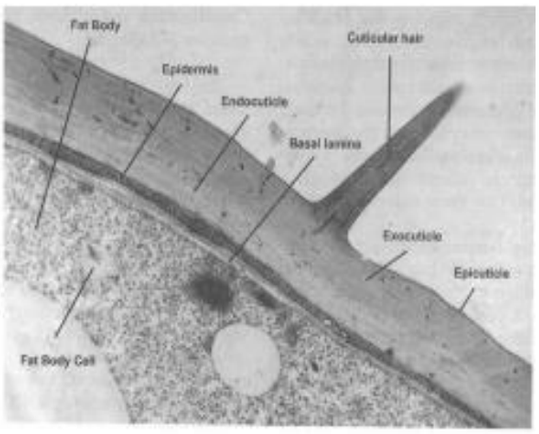
Insect Characteristics Insect Anatomical Characteristics THREE distinct body regions:

1) Head (feeding, sensory, CNS)
2) Thorax (locomotion, respiration)
3) Abdomen (feeding, reproduction)



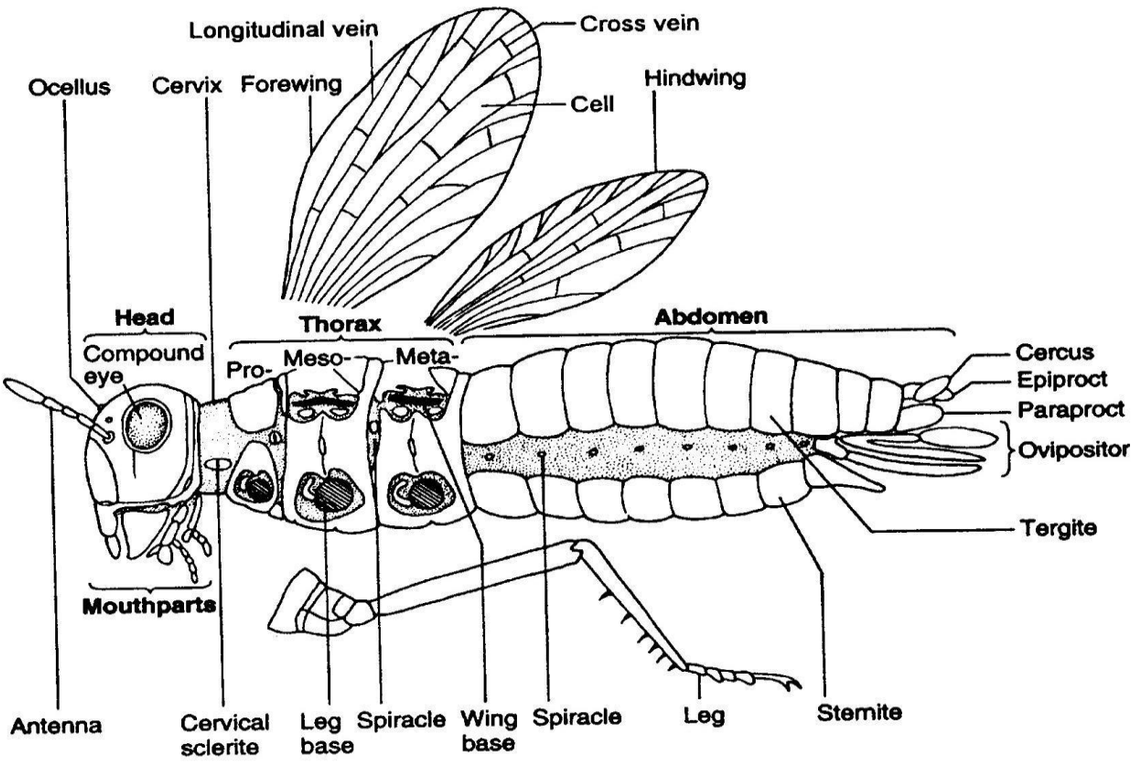
Typical Insect Integument



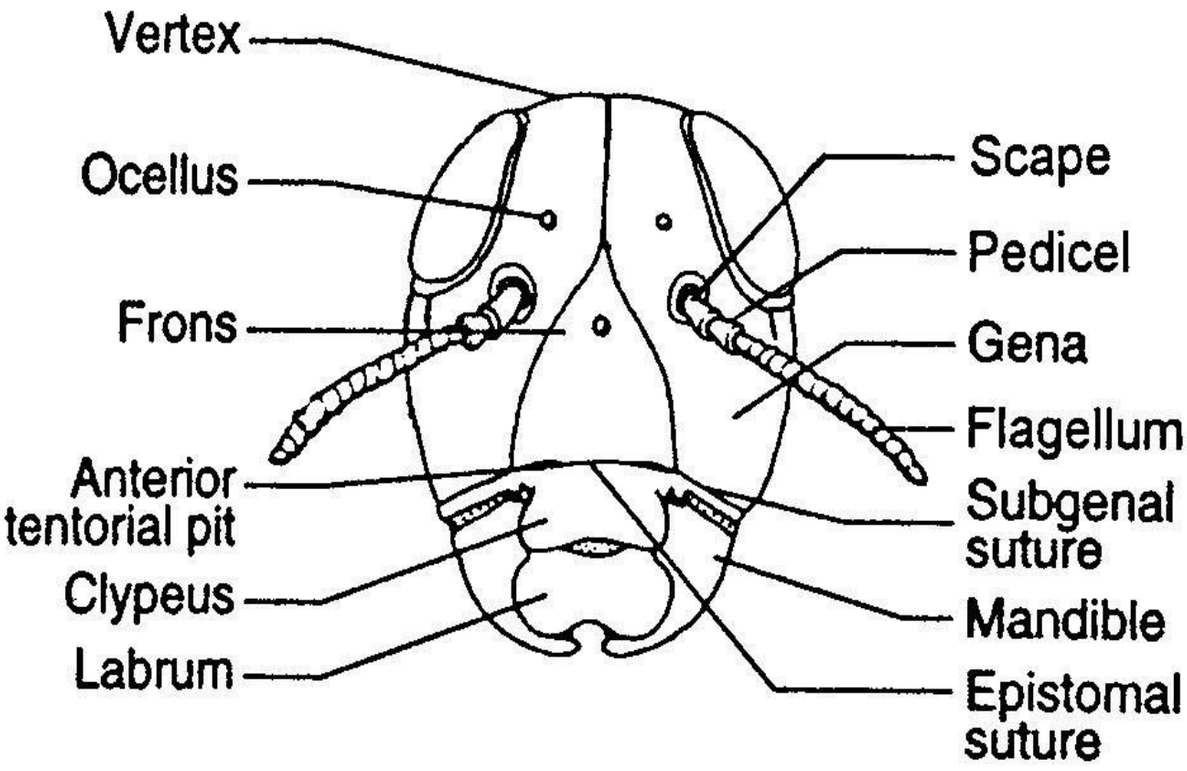


224 General Insect Structure

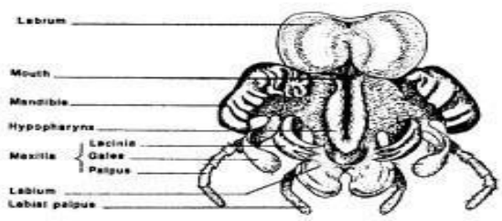
1.General Insect structure



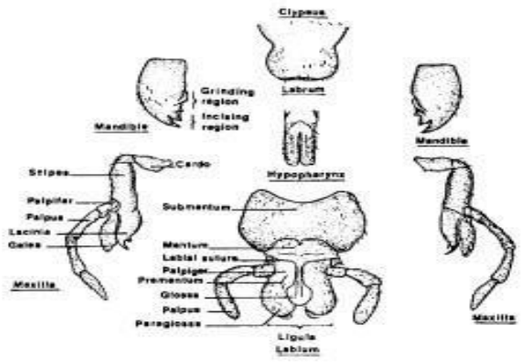
2.General Insect Head



Grasshopper Mouthparts



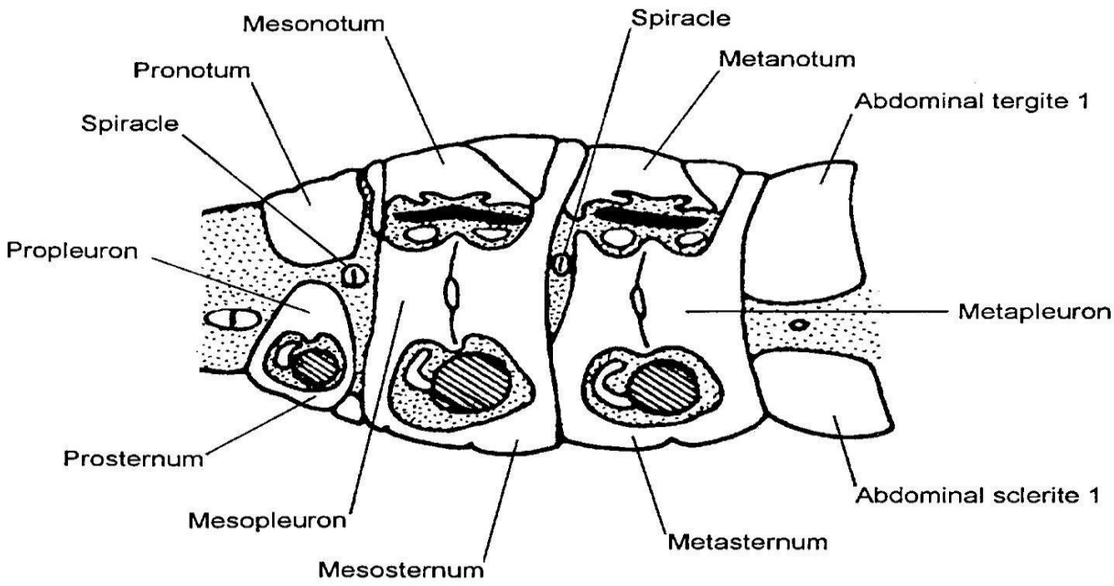
Cricket Mouthparts Dissected



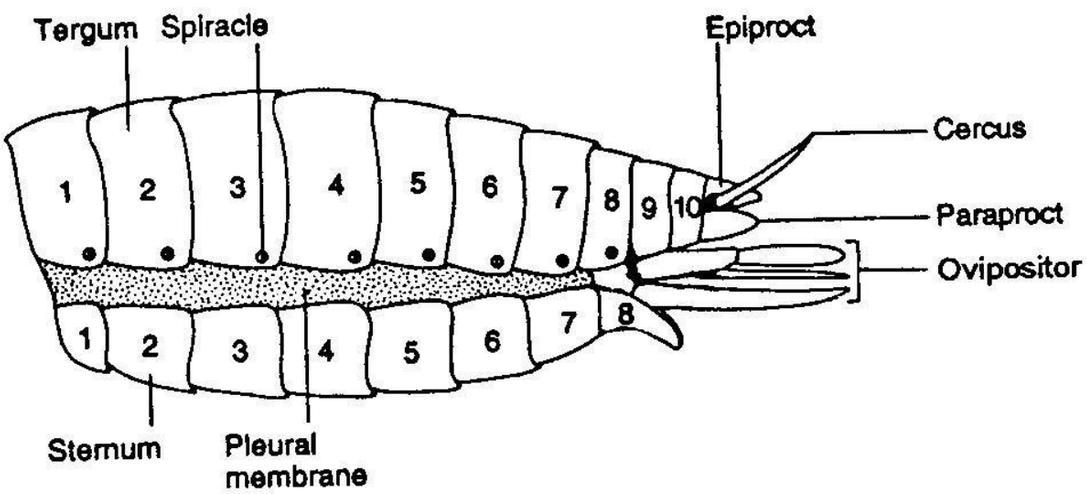
Zain

225 Insect Thorax

1. Insect Thorax

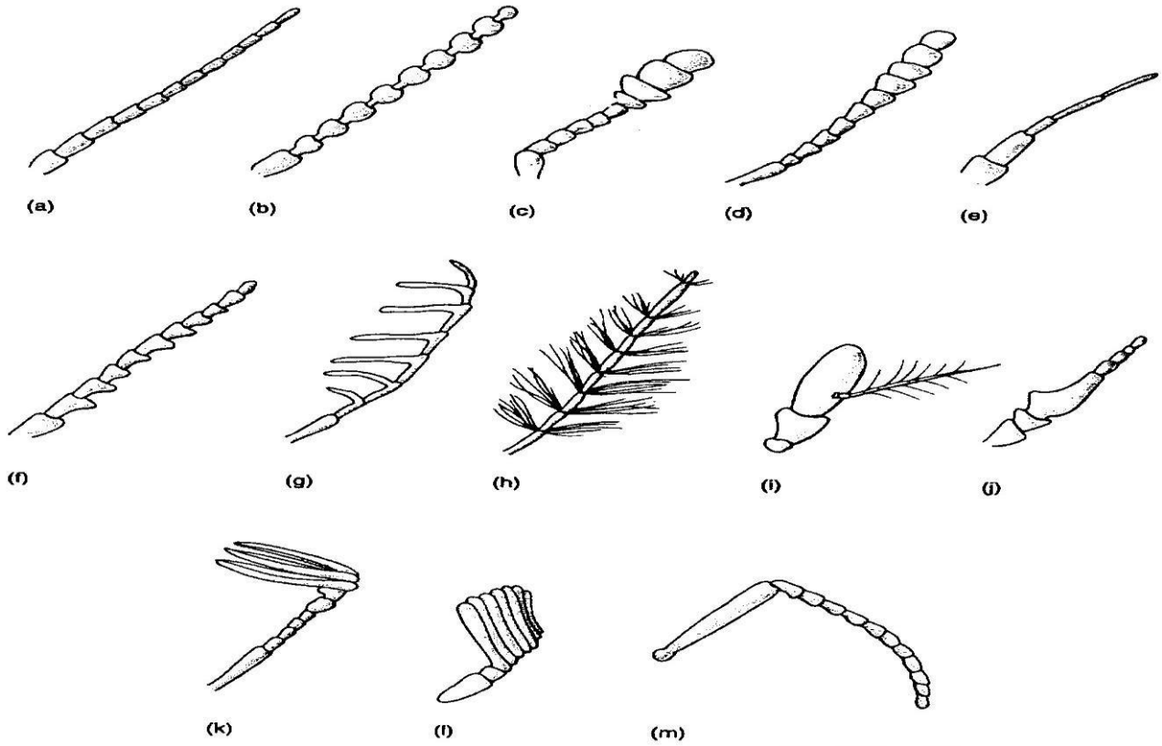


2. Insect Abdomen

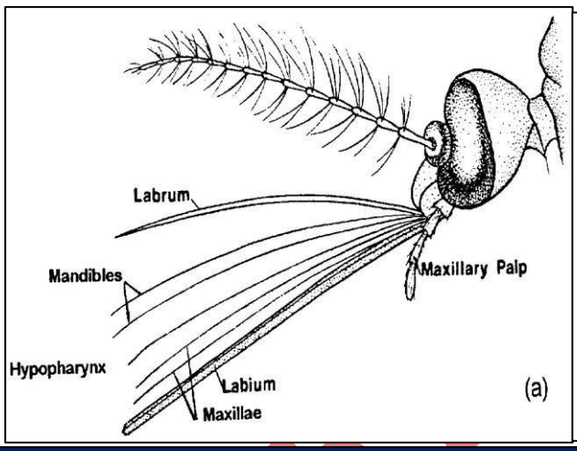


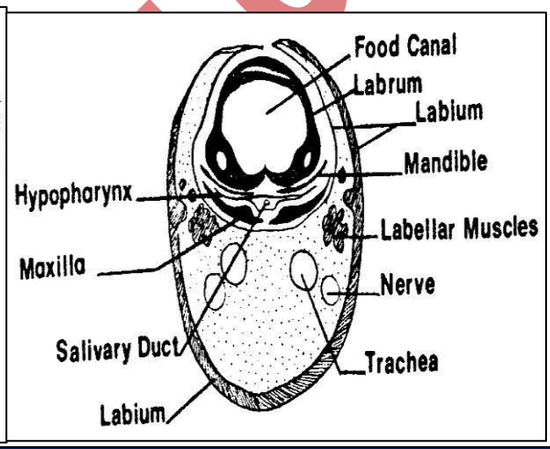
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3. Insect Antennae



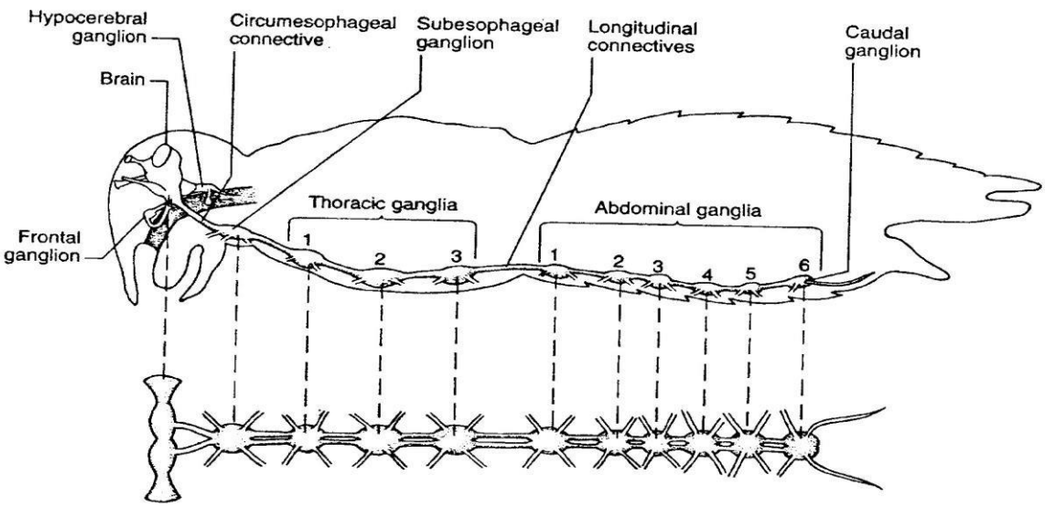
Piercing/sucking Mouthparts





226 Insect Nervous and Circulatory System

1. Insect Nervous System



2 Insect Endocrine Regulated Process

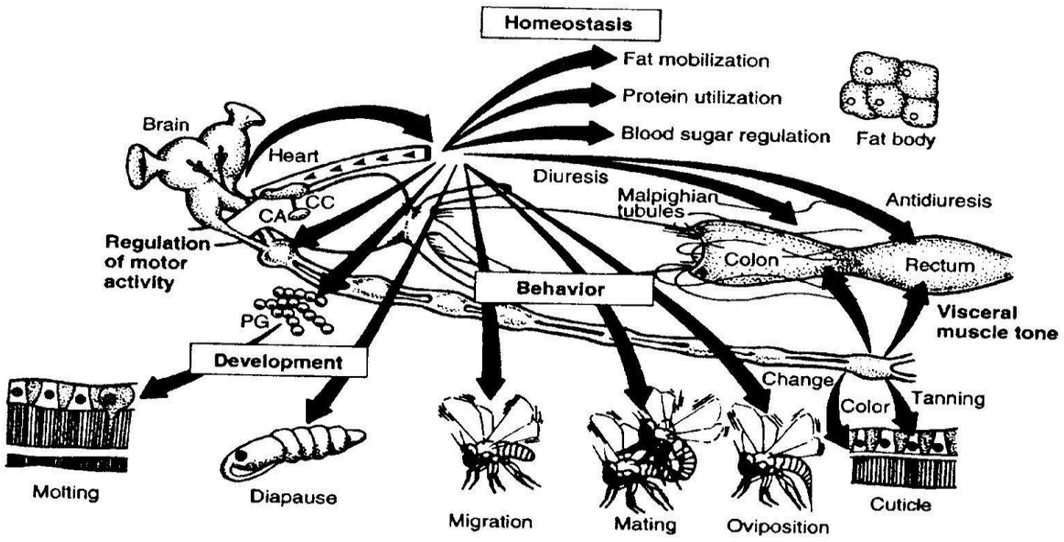
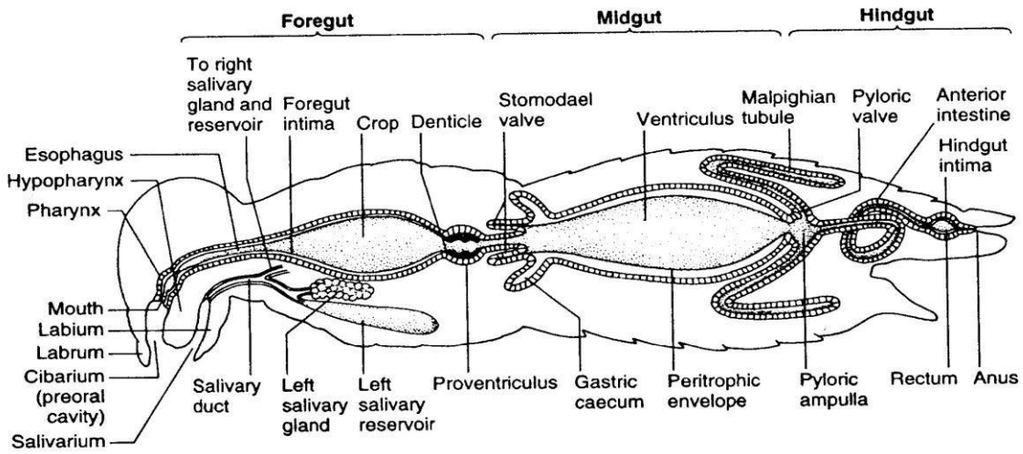
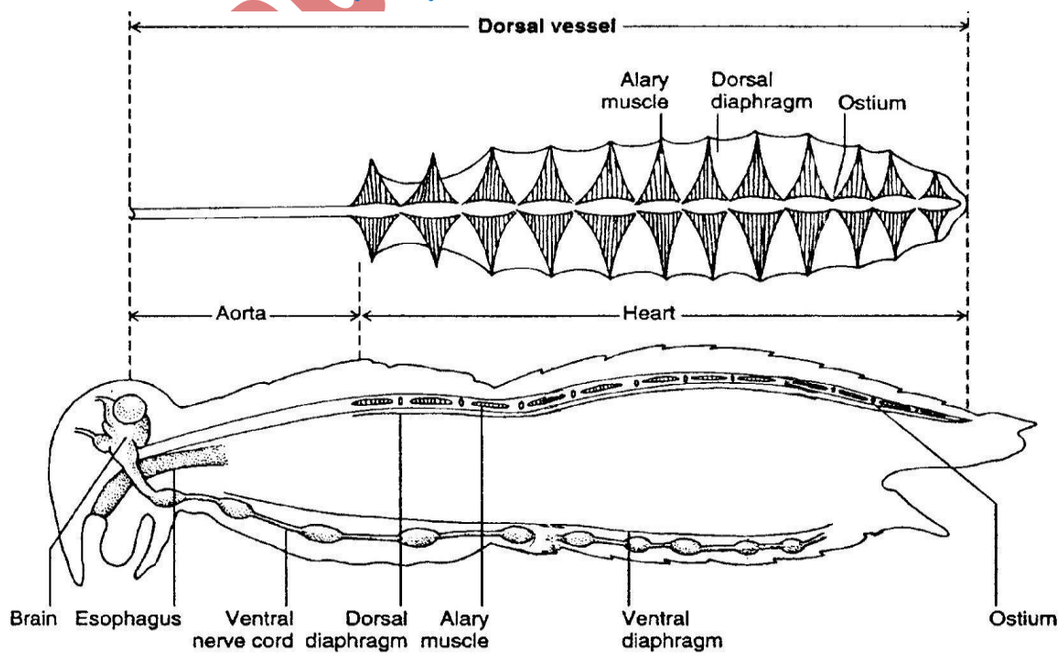


Figure 2.15. Endocrine regulated processes in insects (from Cook and Holman 1985, with permission from Pergamon Press Ltd., Headington Hill, Oxford, UK).

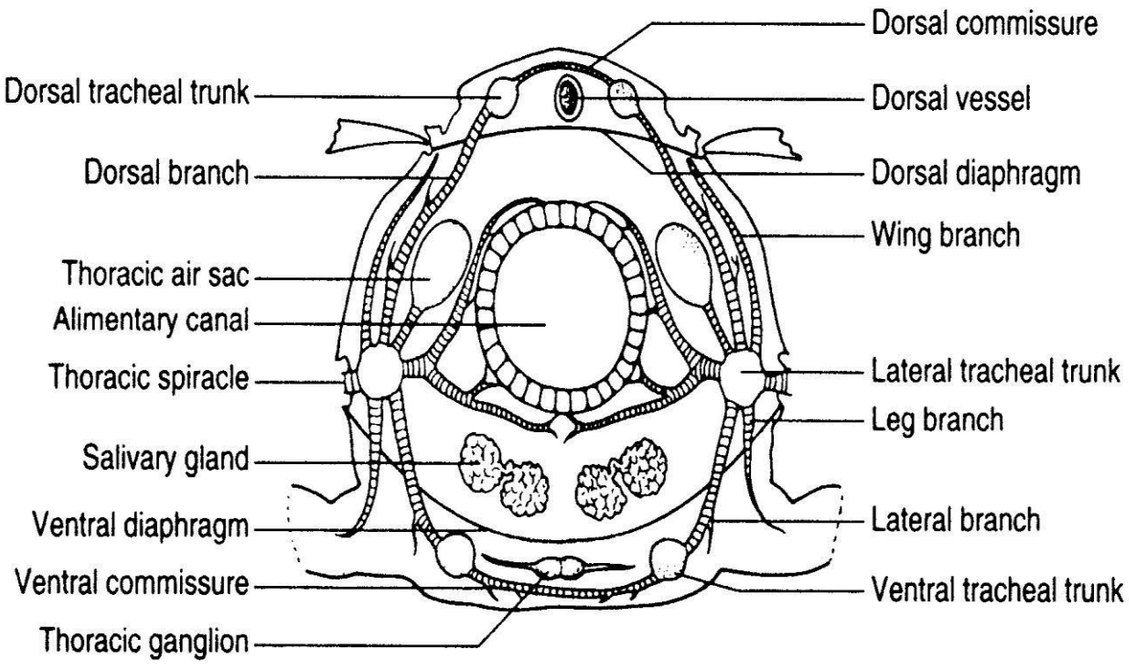
3 Insect Alimentary Canal



4 Insect Circulatory System

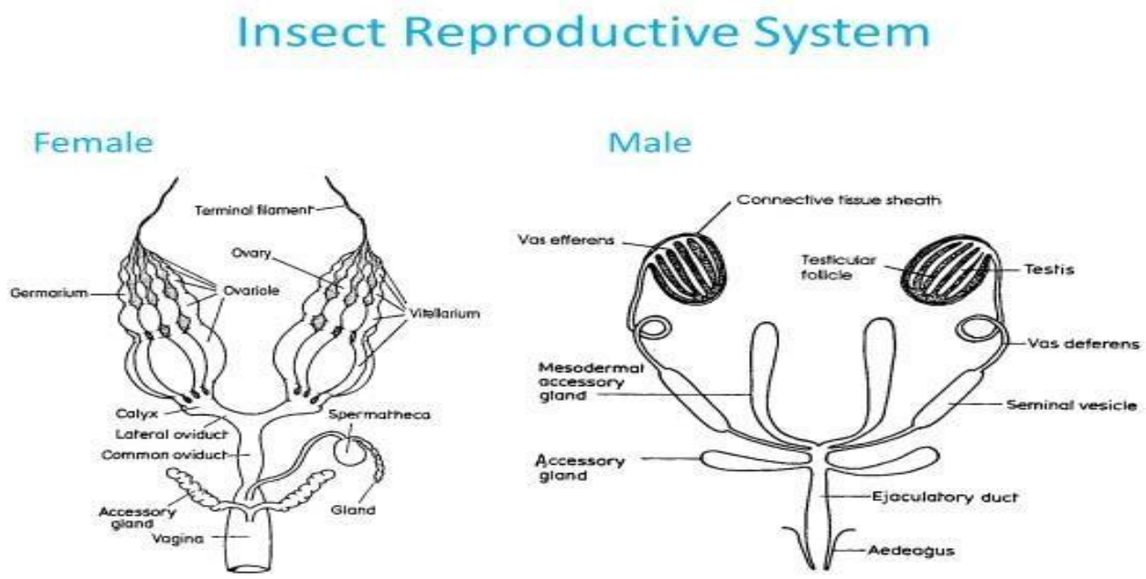


5 Insect thorax showing tracheal branches

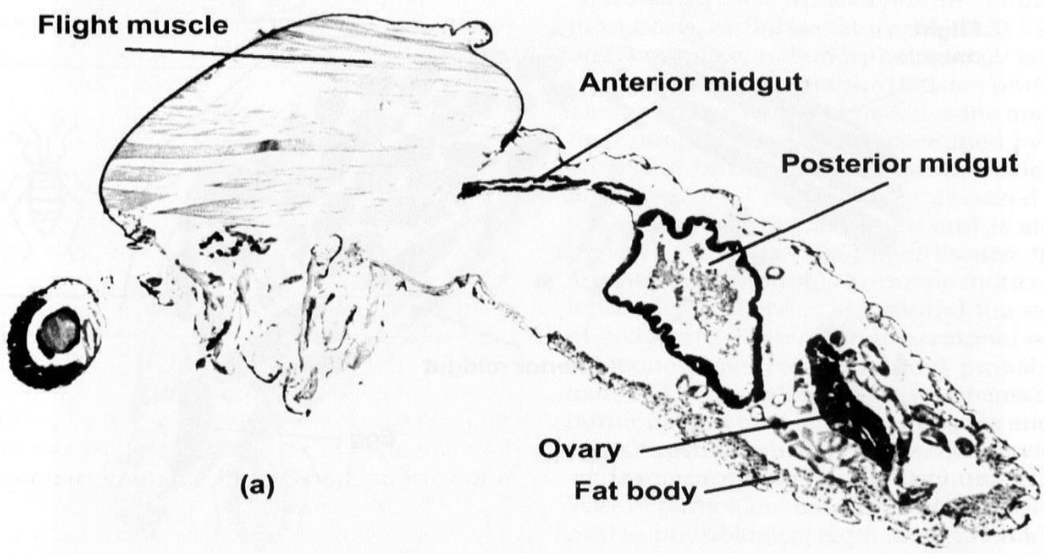


227 Insect Reproductive System

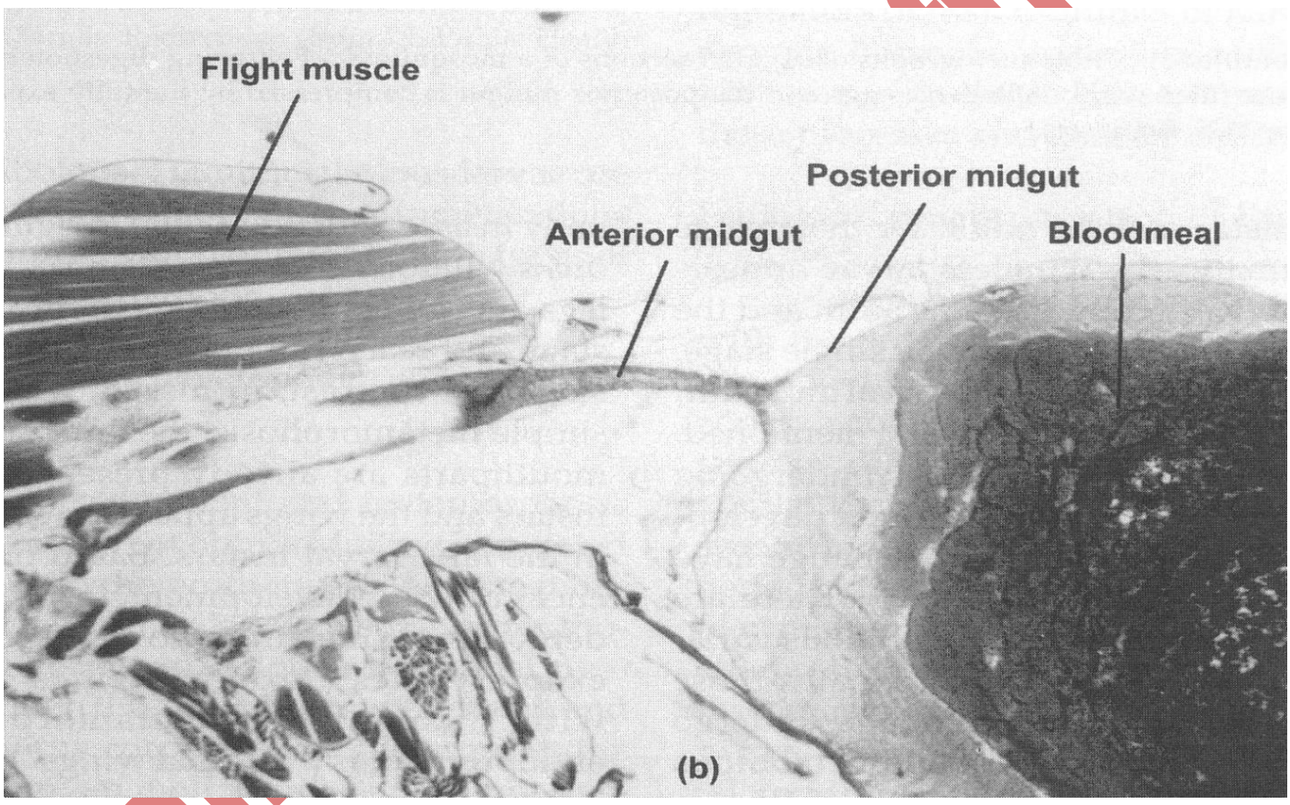
Insect Reproductive System



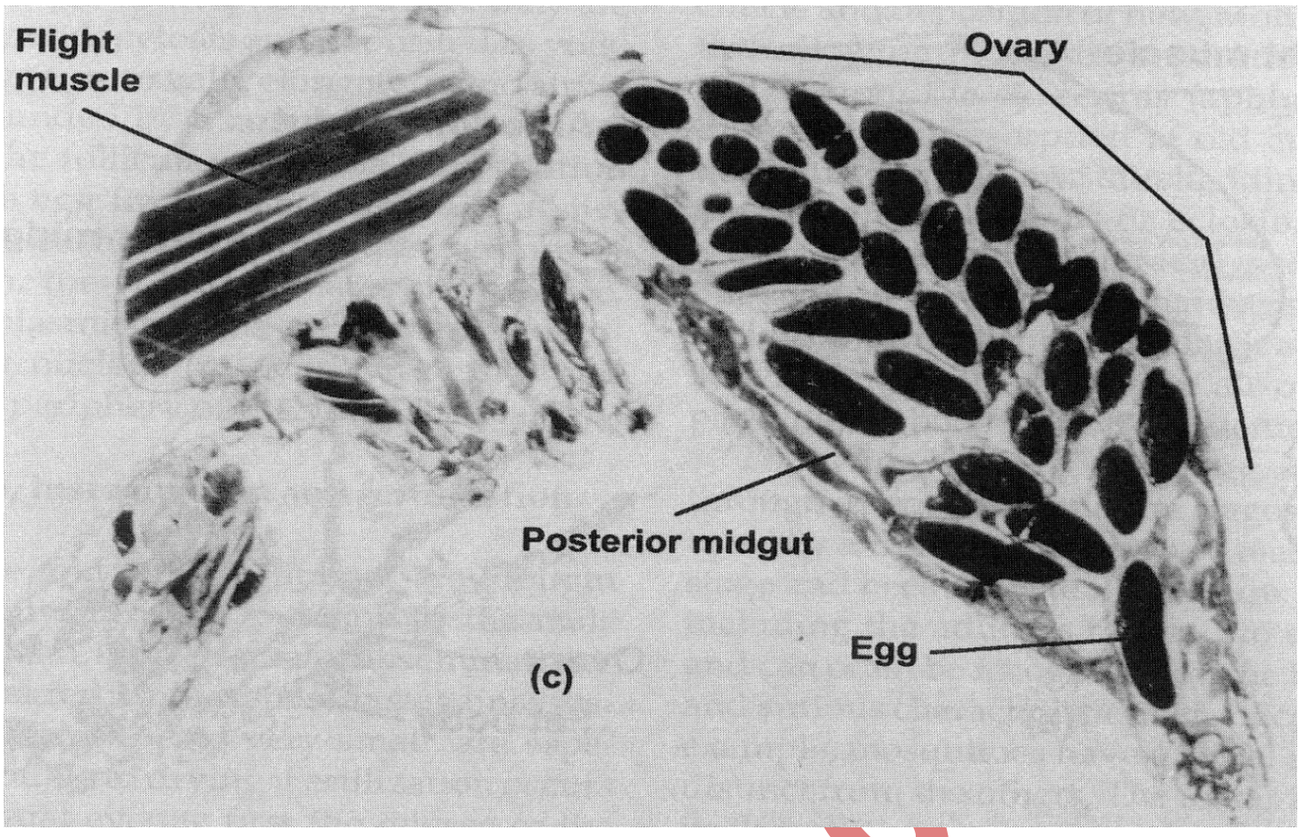
Cross Section of Unfed Mosquito



Blood fed Mosquito



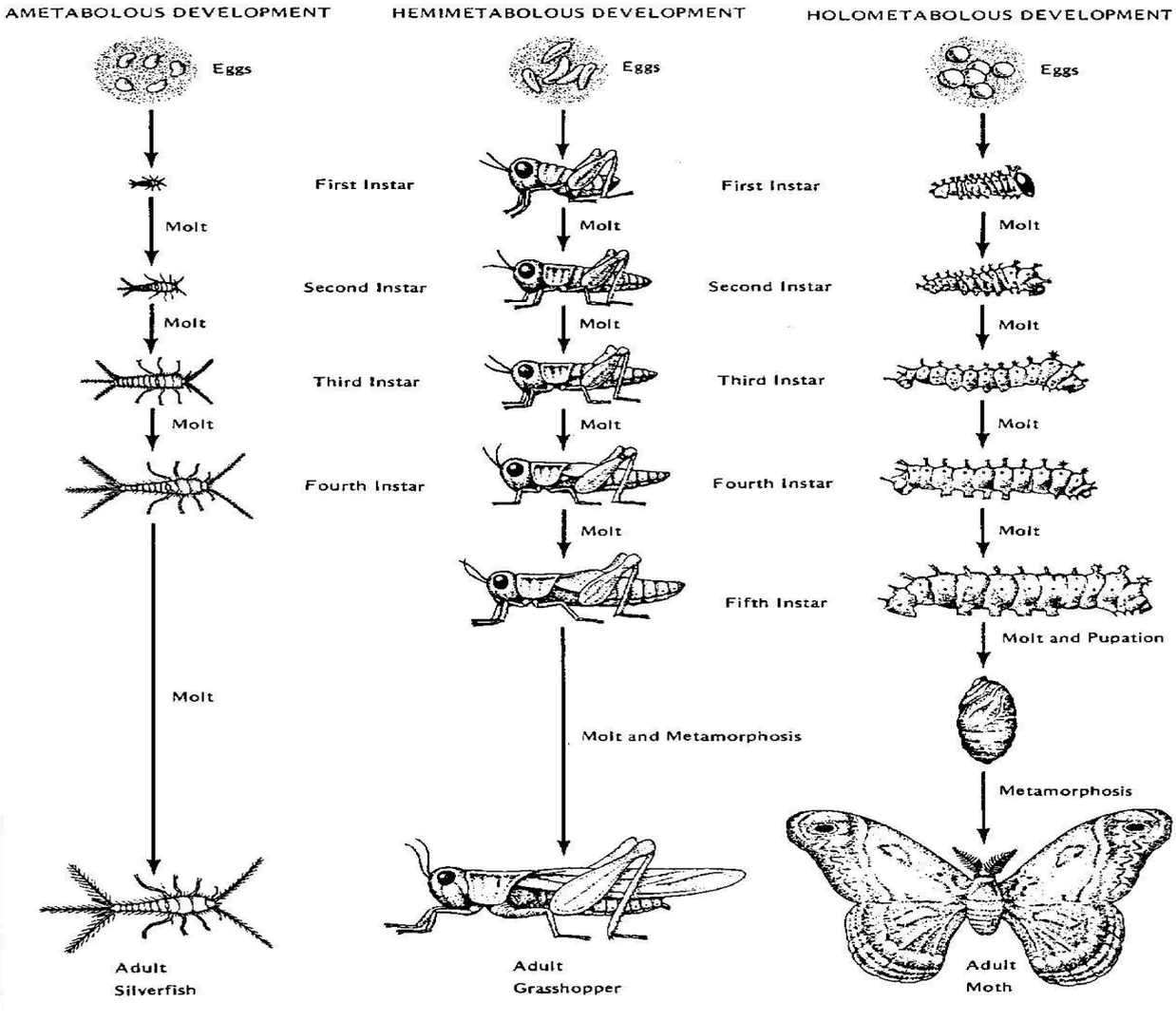
Gravid Mosquito



228 Types of Insect Development

Types of Insect Development

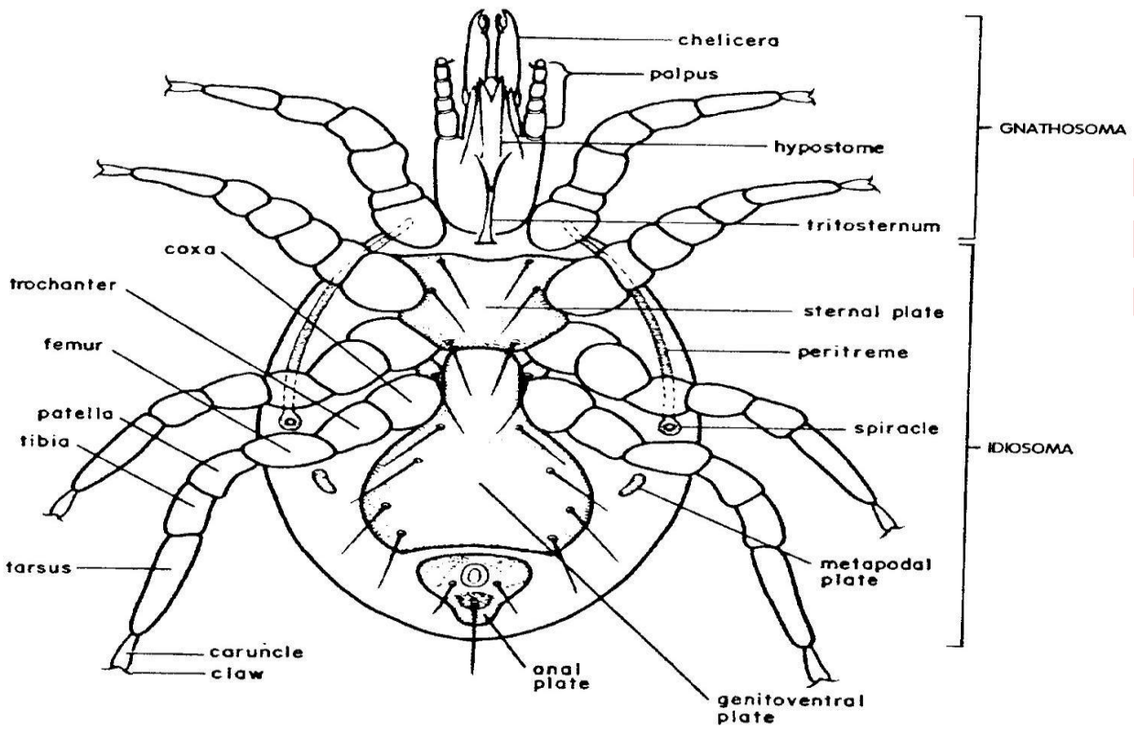
"A" – (lacking) "Hemi" -(incomplete) "Holo" -(complete)



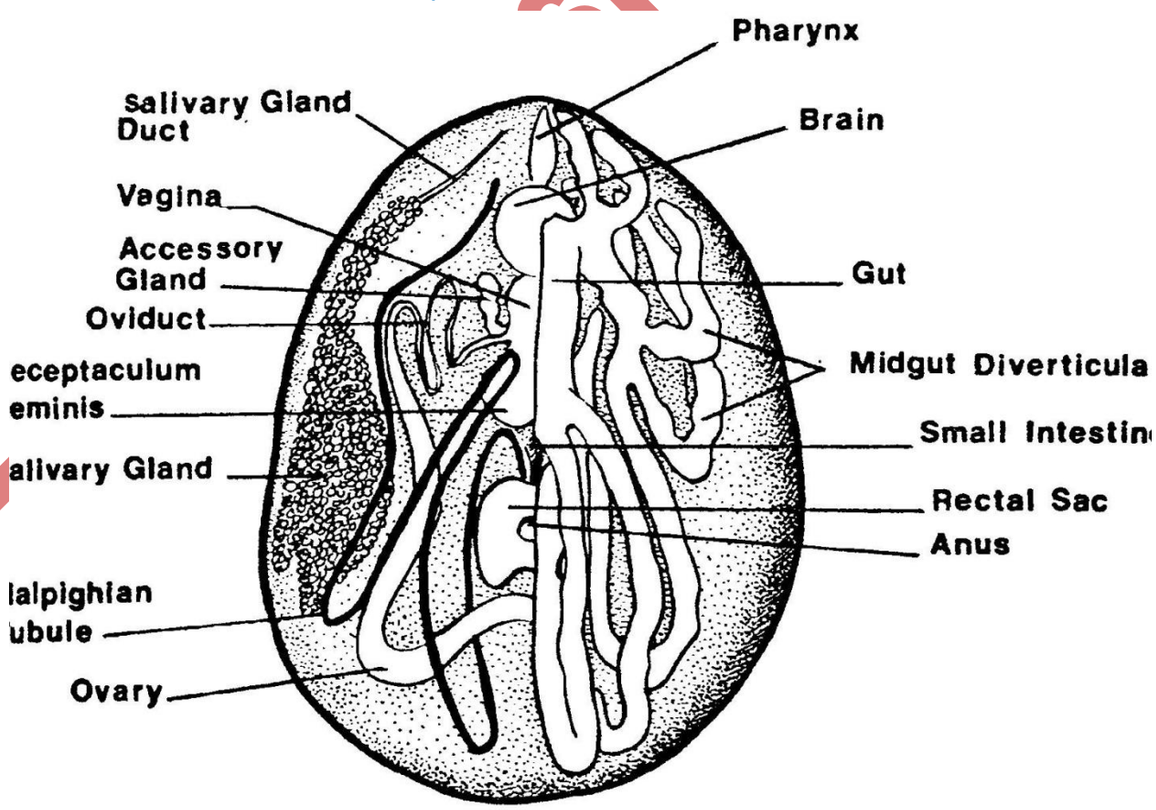
Other Medically Important

Arthropods (non-insects) Mite

External Anatomy

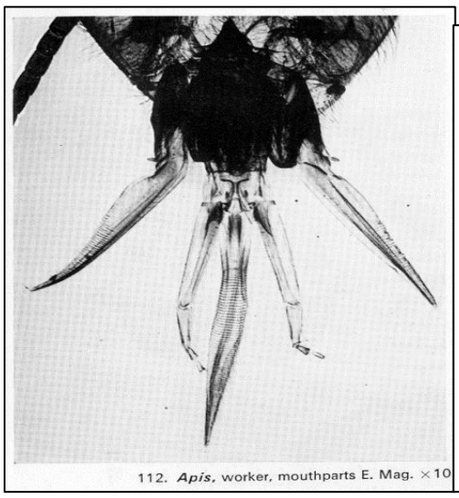


Tick Internal Anatomy

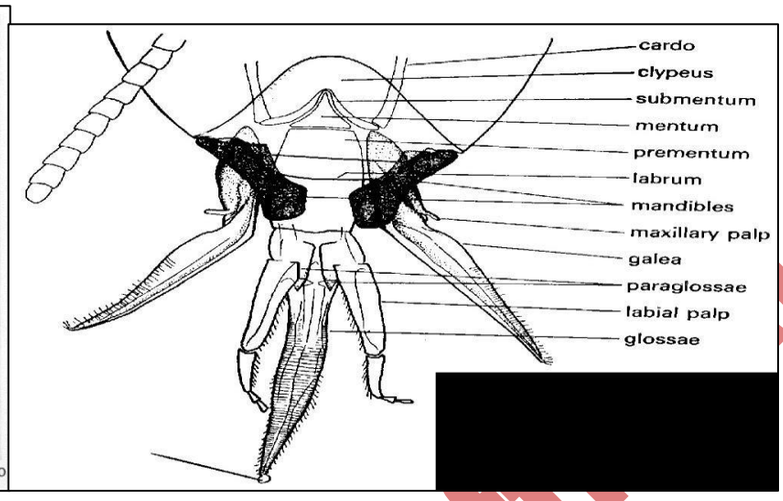


229 Insect Mouthparts and Their Importance

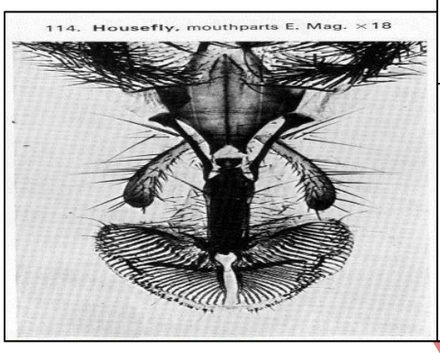
Insect Mouthparts and their Importance *Apis* (honeybee) mouthparts



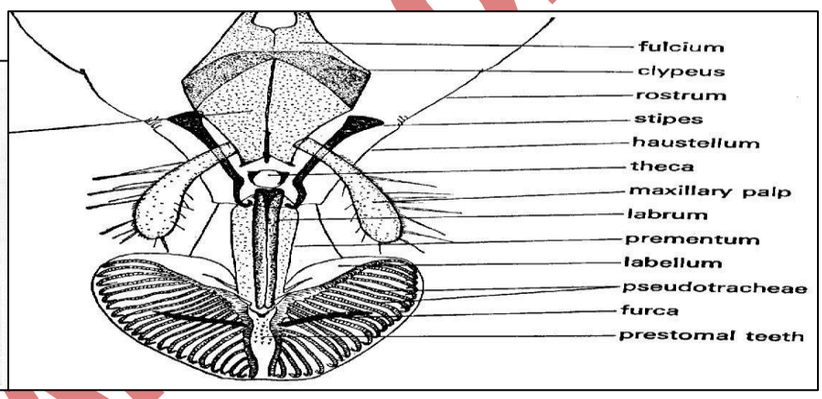
112. *Apis*, worker, mouthparts E. Mag. ×10



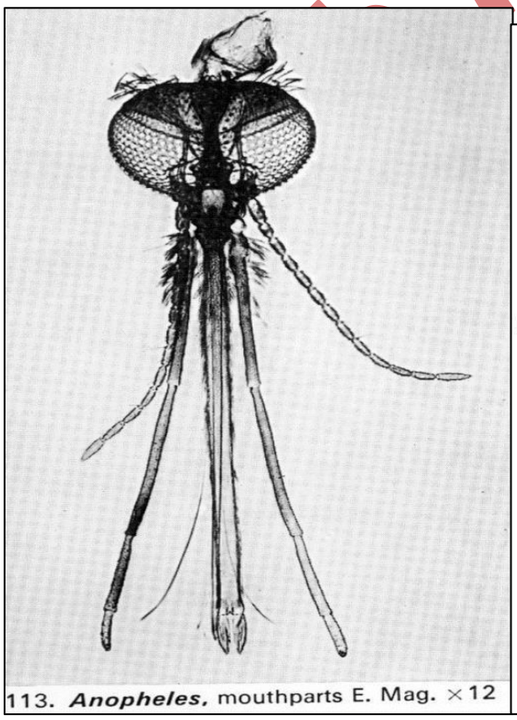
Housefly mouthparts



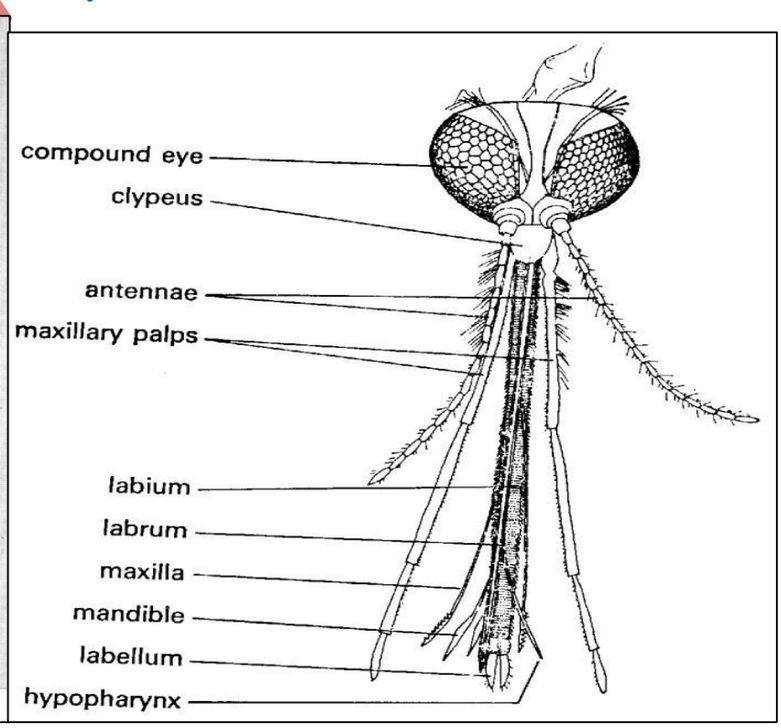
114. Housefly, mouthparts E. Mag. ×18



Anopheles (mosquito) mouthparts



113. *Anopheles*, mouthparts E. Mag. ×12



Composed by Zainab Arshad

230 Concepts in Vector Born Diseases 1

s.q:Write Concepts in Vector-Borne Diseases? Transmission efficiency:

- Geographic or host distribution of the parasite
- Incidence of any given parasite and associated host
- Parasite enhancement of transmission

Transmission frequency:

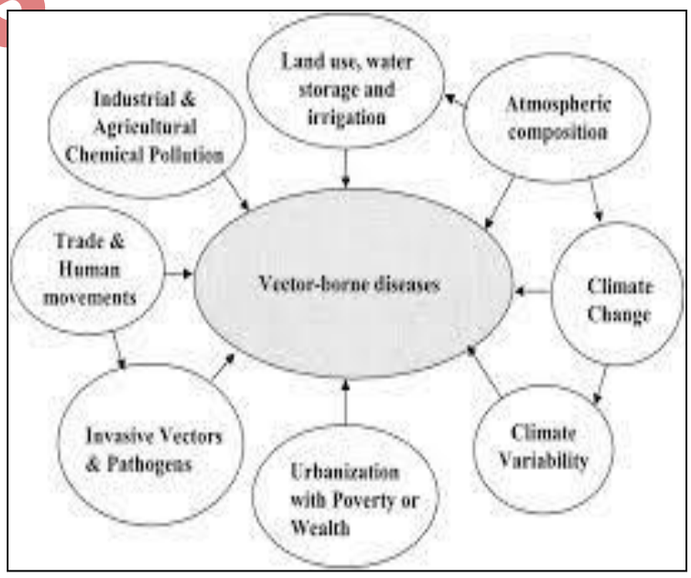
- Shorter life cycle of parasite = more frequent and more efficient transfer to be successful
- Both transmission efficiency and frequency related to blood feeding frequency and efficiency of the vector. These are important factors in vector capacity.

Host

- reservoir host
- disseminating host
- dead-end
- (aberrant) host

Vector:

- primary vector
- secondary vector
- maintenance vector



231 Concepts in Vector Born Diseases 2

Vector Biting Activity:

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- nocturnal
- diurnal
- crepuscular

Host specificity (blood meal source):

- anthropophilic
- anthropophagous
- ornithophilic
- ornithophagous
- zoophilic

Feeding location:

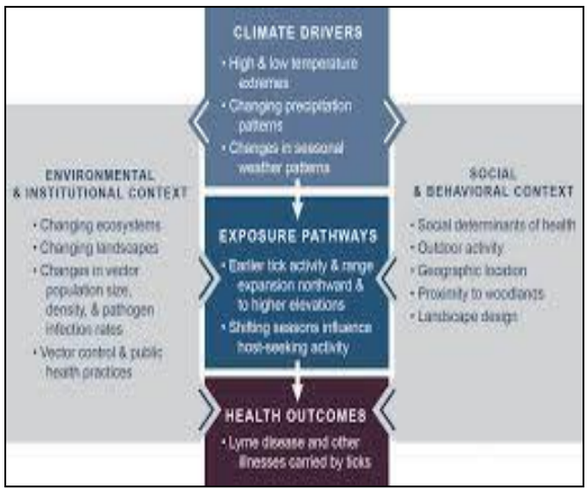
1. exophilic
2. endophilic

Incubation periods:

- extrinsic incubation period (in arthropod vector)
- intrinsic incubation period (in vertebrate host)

sq: Autogeny vs. Anautogeny Number of blood meals:

- ovarian scar/blood meal
- parity status



Determines:

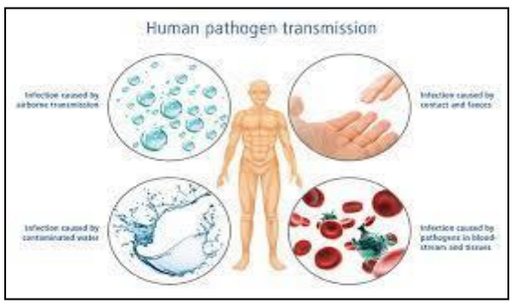
- age of vector
- blood feeding aggressiveness o vector importance

232 Types of Pathogen Transmission

Write Types of pathogen transmission?

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1. mechanical
2. biological



Biological:

- propagative
- cyclopropagative
- cyclodevelopmental

Pathways of biological pathogen transmission:

Vertical transmission:

- transovarial transmission

Horizontal transmission:

- venereal transmission
- transstadial transmission

233 Intrinsic Barriers in Transmission

Intrinsic barriers to transmission in the vector - (genetically and environmentally controlled)

- midgut infection, midgut escape
- salivary gland infection, salivary gland

escape ➢ insect immune response, parasite

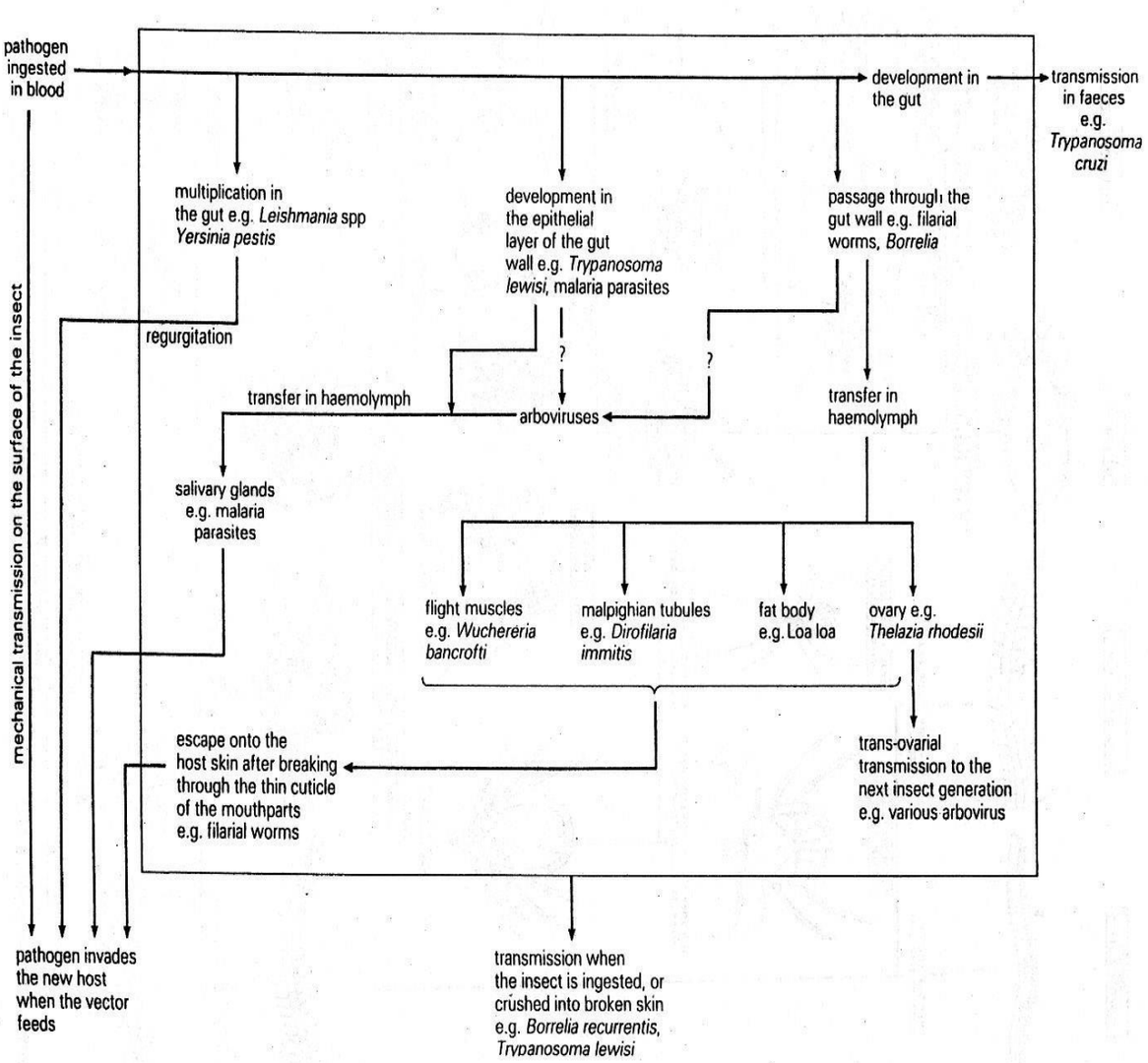
encapsulation

Vector competence vs. vector capacity

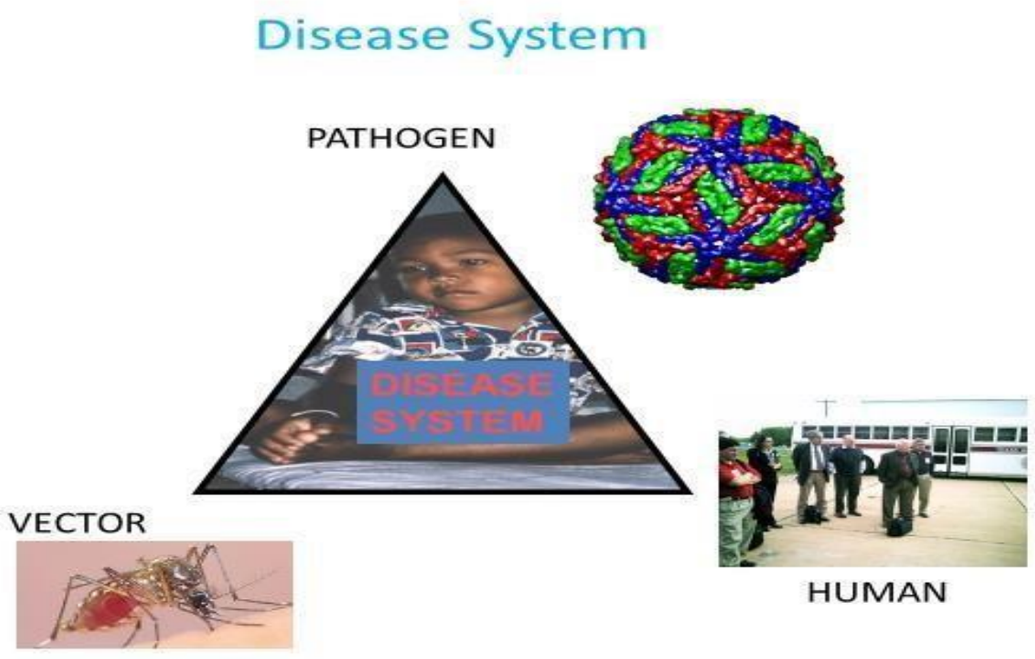
Capacity can be measured in the field using components of number of vectors per human, number of human blood meals per day per vector, daily survival rate, and the extrinsic incubation rate of pathogen; vector efficiency is expressed in terms of low - high capacity

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Competence can be expressed in the laboratory, but a competent lab vector is not necessarily important in disease transmission in the field.

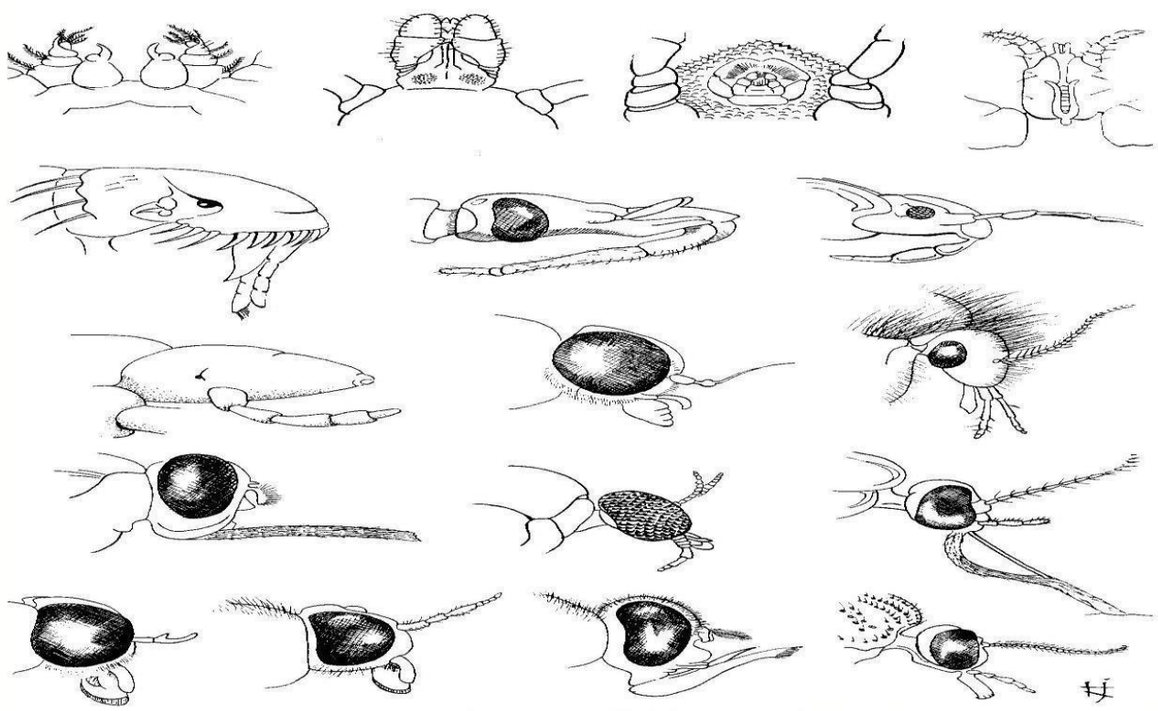


234 Disease System



COMMON NAMES	SCIENTIFIC CLASSIFICATION	
	Order	Family
Arachnids		
Mite	Prostigmata and Mesostigmata	many
Chigger	Prostigmata	Trombiculidae
Tick	Ixodida	Ixodidae and Argasidae
Insects		
Bed bug	Hemiptera	Cimicidae
Kissing bug, vinchuca	Hemiptera	Reduviidae (subfamily Triatominae)
Fly	Diptera	many
Mosquito	Diptera	Culicidae
Black fly	Diptera	Simuliidae
Sand fly	Diptera	Psychodidae (subfamily Phlebotominae)
Biting midge, sand flea, sand fly, flying teeth, black gnat	Diptera	Ceratopogonidae
Snipe fly	Diptera	Rhagionidae
Horse fly, deer fly, cleg	Diptera	Tabanidae
Stable fly, dog fly	Diptera	Muscidae (species *Stomoxys calcitrans*)
Tsetse fly	Diptera	Glossinidae
Flea	Siphonaptera	many

Not Official



235 risk from Insect Vectors

Risk from Insect Vectors

- **Background**
  - Overall death rate

- Influenza in U.S. commonly 8.5% *per week*
- Tuberculosis cases in U.S. 4.6 cases/100K
- Traffic fatalities in U.S. 14.7/100K

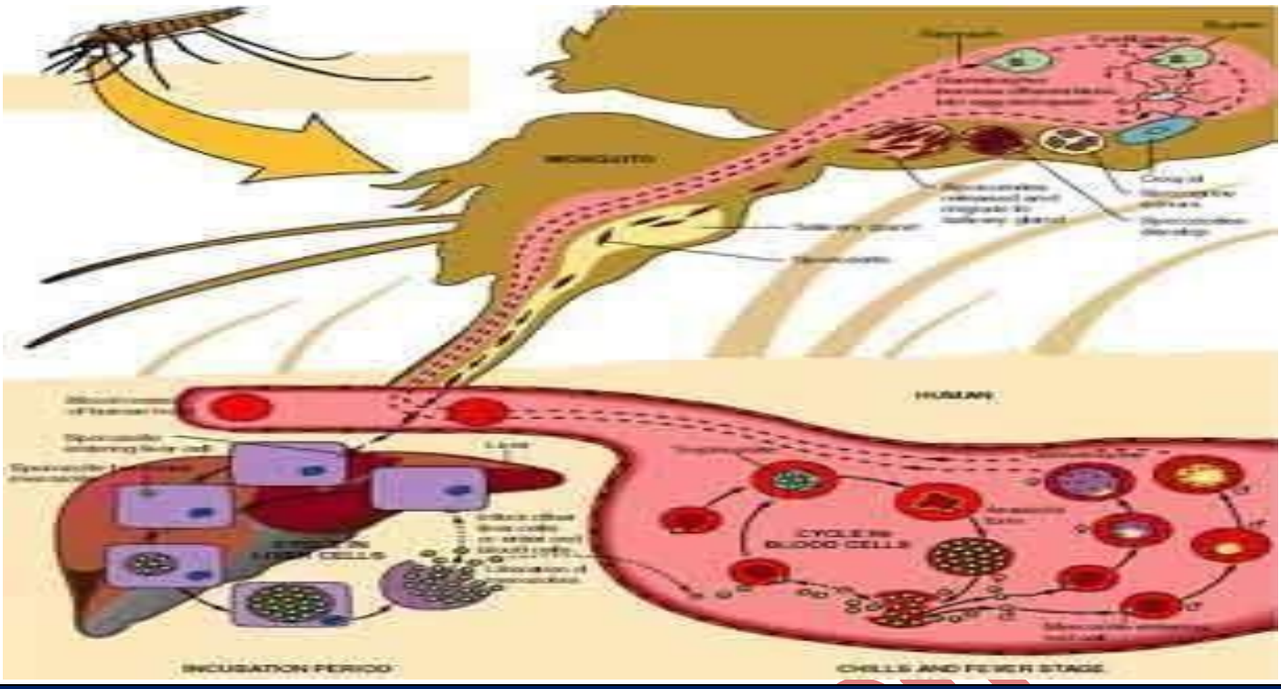
Vector-Borne Pathogens

- Typhus
  - 1996 Burundi 24K cases
  - 1945-6 Japan and Korea 30K cases
  - Historical outbreaks with millions
- Scrub typhus – Up to 20% of fever
- Chagas' Disease
  - 18M cases of 100M in endemic areas
  - Incurable chronic form
- Dengue
  - 2000 cases/100K, reported cases lower
  - 4 x increase 1975-1995
- Leishmaniasis – 57 cases/100K (350M at risk)
- Lyme disease – 31.6 cases/100K in 10 states
- WNV in US – 0.33 cases/100K in 2003

Malaria is King

- Endemic areas (41% of world population)
  - >14,228 cases/100K (350-500M/yr)
  - >4.1 deaths/100K
  - Accounts for 10.7% of childhood deaths
  - Malawi: 28% of hospital deaths

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236 Who is Responsible for This

Who is Responsible?

1. The individual
2. The local community
3. Local government
4. State government
5. National government
6. International organizations

Integrated Pest Management

- **Risk assessment**

  – What you do to find out about the problem without on-the-ground measurements

- **Surveillance**

  – Direct measurements to find the target

- **Control**

  – The suite of techniques used to render the population

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harmless – Hopefully integrated

Monitoring and Sustainability

- After success, then what?
- Our most common source of failure

Risk Assessment

- Much basic work can fit into this slot
  - Population biology
  - Genetics
  - Transmission biology



- **Global Information Systems**
  - What is adequately fine scale?
  - Integrate spatial analysis and modeling
- Products are localization, prioritization, and organization

Surveillance

- Sensitivity vs specificity
- Multiple methods often necessary
- How to handle dirty data
- Archival vs operational surveillance
- Models
  - Scale of inputs and outputs
  - Consideration of communication (=politics)

237 Control of Pathogen Vectors

How we control of pathagpone vectors?

Control

- Stop them at their source

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- Kill the population that remains
- Erect barriers against the ones you miss
- Advocate personal protection as the final layer of protection

Stop Them at Their Source

- **Household**
  - Water sources
  - Rodent harborage
  - Access into the house
  - Harborage in the house, animals
- **Community**
  - Civil engineering, particularly drainage
  - Zoning ✓
  - Economics

Wanted Dead or Dead

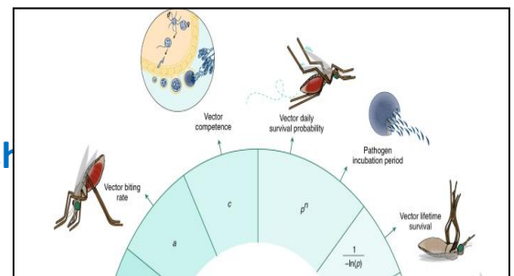
Household – Outdoors

- Larvicides
- Barrier sprays, residuals
- Traps?

– Indoors

- Residuals
- Aerosols

- **Community**
  - Organized mosquito abatement
  - Organized campaigns against other vectors
    - Triatomines



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- Black flies

Good Fences Make Good Neighbors

Household

- Structure of walls, roof
- Screens
- Doors, interior and exterior

- **Community**
  - Screens (sand flies)
  - Barrier fogging
  - Barrier spray

238 Last resort or First Line of Defence

Write Last Resort or First Line of Defense?

- **Personal**
  - Topical repellents
  - Clothing
    - Textile
    - Conformation
    - Chemical treatment
- **Household**
  - **Area repellent systems**
    - Passive chemical dispersion
    - Active chemical dispersion
  - **Excitorepellency**

Monitoring and Sustainability

How weDetect re-emergence of the problem?

- Early detection -

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Cheap to run

- Associated with other activities
- Inexpensive apparatus
- Clear interpretation

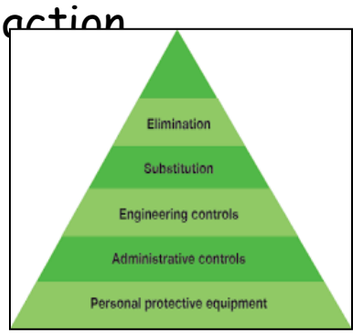
- write **Resources and methods for response ?**
  - Mobile response team
  - Avoid need for new decision process
  - Informed public
- Political motivation in absence of active damage

Sq: Integrated Disease Management

- Objective is reduction or elimination of disease
- Considers medical interventions
- Intelligently applied, actions chosen to leverage IPM and medical

Challenges

- Public Health vs. Environmental Health
- Practical application of theoretical knowledge
- Communication and good will between action agencies



Additive Measures vs. Integration

- **Lintel vs. Arch**
  - Role of each piece
  - Strength of whole

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Maintenance focus vs. expansion focus

- Prioritization of targets
- Attacking the life stages that matter most to transmission
- Using most economical control of each element of disease system to achieve

IDM ✓ Breaking the chain of transmission

239 Scale of Problems

Write The Scale of the Problems?

- **Macro: What's the point?**
  - Interventions limited (e.g., economics)
- **Meso: Getting to the point.**
  - Potentially powerful if cooperative forces unleashed
- **Micro: The point of the spear.**
  - Where interventions take place
  - Where all the action happens Where Does Research Fit In?
- Tends to form leadership positions
  - Most evidence based knowledge
  - Most scholarly knowledge
- Inherent communication gap between research and operations
- Problems solved through experience or experiments?
- Imagination and logic are rusty keys

Some Successes

1. 1890-1920: Transmission studies, mosquito abatement
2. 1942-1955: Antibiotics and pesticides

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3. 1940s: Eradication of *Anopheles gambiae* 1940-1955: Elimination of malaria in US
4. 1950-present: Eradication of screwworm
5. 1940-1960s: Eradication of *Aedes aegypti*
6. 1950-1970: Reduction in malaria
7. 1980s-present: Reduction in *Onchocerca*
8. 2000s: Roll Back Malaria and PMI

Some Failures

1. 1980s-present: Resurgence of malaria
2. 1978-present: Expansion of dengue
3. 1980s-present: Reintroduction of *Aedes aegypti*
4. 1980s-present: Expansion of Lyme disease, ehrlichioses
5. 1986-present: Introduction and expansion of *Aedes albopictus, japonicus*
6. 1999-present: West Nile virus

What Does the Future Hold?

- **Negatives**
  - Global climate change
  - Exponential increases in introductions
  - Energy and nutritional impoverishment(failure)
- **Positives**
  1. Continuing discovery of interventions
  2. Management of wild habitats
  3. Intensification of agriculture
  4. Intensification of community effort

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